

HCIA-Security Comprehensive Exam Question Bank

V3.0 Certification Preparation

Total Questions: 1070

Question Types: Single Choice, Multiple Choice, True/False

Coverage: All HCIA-Security V3.0 Exam Topics

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PART I

Single-Answer Questions

1 Network Security Concepts

1singlecol singlecolSingle Choice

Topic: Network Security Concepts

In a broad sense, network security refers to:

Physical security of devices only

Cybersecurity / cyberspace security at a national level

Only firewall deployment

Only antivirus protection

Correct Answer:

Correct Answer: B

Concept: Broadly, network security is synonymous with cybersecurity, covering national laws, regulations, and processes to protect cyberspace.

2singlecol singlecolSingle Choice

Topic: Network Security Concepts

In a narrow sense, network security focuses on:

National cyberspace policies only

Vendor devices and solutions that ensure secure running of enterprise networks

Only social engineering defense

Only data-center cooling

Correct Answer:

Correct Answer: B

Concept: In a narrow sense, network security refers to measures such as firewalls, IPS, VPNs, and policies that protect enterprise networks.

3singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which period of security history mainly used stream ciphers and physical protection?

Information security period

Communication security period

Cyberspace security period

Information assurance period

Correct Answer:

Correct Answer: B

Concept: The communication security period (1940s) relied on physical security and cryptographic communication, mainly stream ciphers.

4singlecol singlecolSingle Choice

Topic: Network Security Concepts

During the information security period, which three goals were the main focus?

Confidentiality, integrity, availability

Speed, bandwidth, latency

Power, cooling, cabling

Routing, switching, NAT

Correct Answer:

Correct Answer: A

Concept: The classic CIA triad — confidentiality, integrity, and availability — became the core objectives of information security.

5singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which two properties were added to the CIA triad during the information assurance period?

Controllability and non-repudiation

Speed and redundancy

Encryption and hashing

NAT and VLAN

Correct Answer:

Correct Answer: A

Concept: In the information assurance period, controllability (monitoring and control) and non-repudiation (proof of actions) joined the CIA triad.

6singlecol singlecolSingle Choice

Topic: Network Security Concepts

The cyberspace security period aims to protect:

Only web servers

Only user passwords

Facilities, data, users, and operations across cyberspace

Only email gateways

Correct Answer:

Correct Answer: C

Concept: Cyberspace security expands protection to the entire digital ecosystem: facilities, data, users, and operations.

7singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which term describes the practice of protecting networks and data from unauthorized access?

Cybersecurity

Marketing

Accounting

Logistics

Correct Answer:

Correct Answer: A

Concept: Cybersecurity protects networks, devices, and data from unauthorized access or attack.

8singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which of the following best defines a vulnerability?

A weakness that can be exploited

A security patch

A firewall rule

A network diagram

Correct Answer:

Correct Answer: A

Concept: A vulnerability is a weakness in a system that a threat actor can exploit.

9singlecol singlecolSingle Choice

Topic: Network Security Concepts

What is an exploit?

- A software patch
- A method to take advantage of a vulnerability
- A type of firewall
- A routing protocol

Correct Answer:

Correct Answer: B

Concept: An exploit is a technique or code that leverages a vulnerability.

10singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which term describes a potential danger to an asset?

- Risk
- Threat
- Vulnerability
- Control

Correct Answer:

Correct Answer: B

Concept: A threat is any potential danger that could exploit a vulnerability.

11singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which term describes the likelihood and impact of a threat exploiting a vulnerability?

- Threat
- Risk
- Asset
- Control

Correct Answer:

Correct Answer: B

Concept: Risk measures the probability and impact of a security incident.

12singlecol singlecolSingle Choice

Topic: Network Security Concepts

Future security architecture is moving toward:

- Single perimeter defense
- Zero-trust and dynamic trust models
- Open anonymous access
- No logging

Correct Answer:

Correct Answer: B

Concept: Zero trust assumes no implicit trust and verifies every access request dynamically.

13singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which technology is increasingly used to detect unknown threats?

- Signature-only antivirus
- Artificial intelligence and machine learning
- Hub-based networks
- Plaintext protocols

Correct Answer:

Correct Answer: B

Concept: AI/ML helps analyze behavior and detect anomalies and unknown threats.

14singlecol singlecolSingle Choice

Topic: Network Security Concepts

What does confidentiality ensure?

Only speed

Information is accessible only to authorized users

Unlimited access

Data duplication

Correct Answer:

Correct Answer: B

Concept: Confidentiality ensures that information is disclosed only to authorized individuals.

15singlecol singlecolSingle Choice

Topic: Network Security Concepts

What does integrity ensure?

Fast transmission

Information is not tampered with during transmission/storage

Free access

Unencrypted storage

Correct Answer:

Correct Answer: B

Concept: Integrity ensures data is not altered or destroyed in an unauthorized manner.

16singlecol singlecolSingle Choice

Topic: Network Security Concepts

What does availability ensure?

Authorized users can access information when needed

All users can access

Systems are offline

Data is hidden

Correct Answer:

Correct Answer: A

Concept: Availability guarantees reliable and timely access to information for authorized users.

17singlecol singlecolSingle Choice

Topic: Network Security Concepts

What does controllability mean in information security?

No monitoring

Ability to monitor and control information/systems

Unlimited downloads

No access controls

Correct Answer:

Correct Answer: B

Concept: Controllability enables organizations to monitor, manage, and protect information and systems.

18singlecol singlecolSingle Choice

Topic: Network Security Concepts

What does non-repudiation prevent?

Fast data transfer

Senders or receivers from denying their actions

Encryption

Backups

Correct Answer:

Correct Answer: B

Concept: Non-repudiation provides proof of origin and delivery so parties cannot deny participation.

19singlecol singlecolSingle Choice

Topic: Network Security Concepts

The broad definition of network security is closest to:

- Cybersecurity
- Firewall configuration
- Antivirus deployment
- Physical security

Correct Answer:

Correct Answer: A

Concept: Broadly, network security equates to cybersecurity covering national and societal cyberspace protection.

20singlecol singlecolSingle Choice

Topic: Network Security Concepts

The narrow definition of network security focuses on:

- National laws only
- Enterprise network protection solutions
- Social media policies
- Hardware warranties

Correct Answer:

Correct Answer: B

Concept: Narrowly, network security means vendor solutions and measures that keep enterprise networks secure.

21singlecol singlecolSingle Choice

Topic: Network Security Concepts

During which period did communication cipher security emerge?

- Cyberspace security period
- Communication security period
- Information assurance period
- Cloud security period

Correct Answer:

Correct Answer: B

Concept: Stream ciphers and communication confidentiality appeared during the communication security period.

22singlecol singlecolSingle Choice

Topic: Network Security Concepts

The information security period added which goal to the earlier communication security focus?

- Availability
- Confidentiality, integrity, and availability
- Non-repudiation only
- Controllability only

Correct Answer:

Correct Answer: B

Concept: The information security period formalized confidentiality, integrity, and availability as core goals.

23singlecol singlecolSingle Choice

Topic: Network Security Concepts

Controllability and non-repudiation became prominent during:

- Communication security period
- Information security period
- Information assurance period

Cloud computing period

Correct Answer:

Correct Answer: C

Concept: The information assurance period expanded security to include monitoring and proof of actions.

24singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: is synonymous with national cyberspace security?

Cyberspace security

Broad network security

Non-repudiation

Controllability

Correct Answer:

Correct Answer: B

Concept: Broad network security refers to the concept where is synonymous with national cyberspace security.

25singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: focuses on enterprise network protection solutions?

Integrity

AI and machine learning

Narrow network security

Confidentiality

Correct Answer:

Correct Answer: C

Concept: Narrow network security refers to the concept where focuses on enterprise network protection solutions.

26singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: consists of confidentiality, integrity, and availability?

Integrity

Vulnerability

Narrow network security

CIA triad

Correct Answer:

Correct Answer: D

Concept: CIA triad refers to the concept where consists of confidentiality, integrity, and availability.

27singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: added controllability and non-repudiation?

Information assurance period

AI and machine learning

Non-repudiation

Threat

Correct Answer:

Correct Answer: A

Concept: Information assurance period refers to the concept where added controllability and non-repudiation.

28singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: protects facilities, data, users, and operations?

CIA triad

Availability

Cyberspace security

Vulnerability

Correct Answer:

Correct Answer: C

Concept: Cyberspace security refers to the concept where protects facilities, data, users, and operations.

29singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: ensures information is accessible only to authorized users?

Availability

Risk

Non-repudiation

Confidentiality

Correct Answer:

Correct Answer: D

Concept: Confidentiality refers to the concept where ensures information is accessible only to authorized users.

30singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: ensures data is not tampered with during transmission or storage?

Availability

Broad network security

Future security trend

Integrity

Correct Answer:

Correct Answer: D

Concept: Integrity refers to the concept where ensures data is not tampered with during transmission or storage.

31singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: ensures authorized users can access information when needed?

Integrity

Exploit

AI and machine learning

Availability

Correct Answer:

Correct Answer: D

Concept: Availability refers to the concept where ensures authorized users can access information when needed.

32singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: enables monitoring and control of information and systems?

Non-repudiation

Broad network security

Controllability

Narrow network security

Correct Answer:

Correct Answer: C

Concept: Controllability refers to the concept where enables monitoring and control of information and systems.

33singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: prevents senders or receivers from denying actions?

Threat

Vulnerability

Non-repudiation

Controllability

Correct Answer:

Correct Answer: C

Concept: Non-repudiation refers to the concept where prevents senders or receivers from denying actions.

34singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: is the combination of threat likelihood and impact?

Exploit

Risk

Cyberspace security

Information assurance period

Correct Answer:

Correct Answer: B

Concept: Risk refers to the concept where is the combination of threat likelihood and impact.

35singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: is a weakness that can be exploited by threats?

Controllability

Risk

Confidentiality

Vulnerability

Correct Answer:

Correct Answer: D

Concept: Vulnerability refers to the concept where is a weakness that can be exploited by threats.

36singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: is a potential danger to an asset?

Confidentiality

Vulnerability

Threat

Exploit

Correct Answer:

Correct Answer: C

Concept: Threat refers to the concept where is a potential danger to an asset.

37singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: is a method that takes advantage of a vulnerability?

Non-repudiation

Confidentiality

Controllability

Exploit

Correct Answer:

Correct Answer: D

Concept: Exploit refers to the concept where is a method that takes advantage of a vulnerability.

38singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: includes zero-trust architecture?

Vulnerability

Future security trend

Broad network security

Integrity

Correct Answer:

Correct Answer: B

Concept: Future security trend refers to the concept where includes zero-trust architecture.

39singlecol singlecolSingle Choice

Topic: Network Security Concepts

Which concept is described as: help detect unknown threats and anomalies?

AI and machine learning

Availability

Broad network security

Controllability

Correct Answer:

Correct Answer: A

Concept: AI and machine learning refers to the concept where help detect unknown threats and anomalies.

2 Network Security Standards

1singlecol singlecolSingle Choice

Topic: Network Security Standards

ISO 27001 specifies requirements for:

An information security management system (ISMS)

A firewall hardware model

A routing protocol

A programming language

Correct Answer:

Correct Answer: A

Concept: ISO/IEC 27001 is the international standard for establishing, implementing, maintaining, and continually improving an ISMS.

2singlecol singlecolSingle Choice

Topic: Network Security Standards

ISO 27002 is best described as:

A certification standard

A code of practice with security controls

A firewall configuration guide

A wireless protocol

Correct Answer:

Correct Answer: B

Concept: ISO/IEC 27002 provides best-practice security controls, while ISO 27001 is the certifiable requirements standard.

3singlecol singlecolSingle Choice

Topic: Network Security Standards

Cybersecurity Classified Protection 2.0 is issued by:

Huawei

China (as a national standard/policy)

IEEE

IETF

Correct Answer:

Correct Answer: B

Concept: Classified Protection 2.0 is a Chinese national cybersecurity baseline and compliance framework.

4singlecol singlecolSingle Choice

Topic: Network Security Standards

Classified Protection 2.0 uses a protection-level scale from:

Level 1 to Level 5

Level 0 to Level 4

Level A to Level E

Level I to Level V

Correct Answer:

Correct Answer: A

Concept: The Chinese classified protection system defines five levels, with Level 5 being the highest.

5singlecol singlecolSingle Choice

Topic: Network Security Standards

Which of the following is NOT a typical Classified Protection 2.0 lifecycle phase?

Classification

Filing

Construction and rectification

Marketing and sales

Correct Answer:

Correct Answer: D

Concept: The Classified Protection lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

6singlecol singlecolSingle Choice

Topic: Network Security Standards

Which ISO standard family addresses information security management?

ISO 9000

ISO 27000

ISO 14000

ISO 20000

Correct Answer:

Correct Answer: B

Concept: The ISO/IEC 27000 family covers information security management systems and controls.

7singlecol singlecolSingle Choice

Topic: Network Security Standards

Which Chinese regulation is the basis for Classified Protection?

Cybersecurity Law

GDPR

HIPAA

PCI DSS

Correct Answer:

Correct Answer: A

Concept: China Cybersecurity Law underpins the Classified Protection compliance framework.

8singlecol singlecolSingle Choice

Topic: Network Security Standards

ISO 27001 uses which methodology for risk management?

Ignore risks

Plan-Do-Check-Act (PDCA)

Random patching

No reviews

Correct Answer:

Correct Answer: B

Concept: ISO 27001 follows the PDCA cycle for continuous improvement of the ISMS.

9singlecol singlecolSingle Choice

Topic: Network Security Standards

Which document in ISO 27000 family provides implementation guidance for controls?

ISO 27001

ISO 27002

ISO 27003

ISO 27005

Correct Answer:

Correct Answer: B

Concept: ISO 27002 lists security controls and implementation guidance.

10singlecol singlecolSingle Choice

Topic: Network Security Standards

Which is a key principle of Classified Protection 2.0?

Security synchronization with construction

Security as an afterthought

No auditing

Anonymous access

Correct Answer:

Correct Answer: A

Concept: Classified Protection requires security to be planned, constructed, and used simultaneously with information systems.

11singlecol singlecolSingle Choice

Topic: Network Security Standards

In Classified Protection 2.0, the security protection level is preliminarily determined by the network operator, but when the level reaches which level or above, expert review and approval are required?

Level 1

Level 2

Level 3

Level 4

Correct Answer:

Correct Answer: B

Concept: For protection Level 2 and above, the preliminary determination must be reviewed and approved; Level 1 can be determined by the operator.

12singlecol singlecolSingle Choice

Topic: Network Security Standards

Which standard is used to certify an ISMS?

ISO 27001

ISO 27002

ISO 9001

ISO 14001

Correct Answer:

Correct Answer: A

Concept: ISO 27001 contains the certifiable requirements for an ISMS.

13singlecol singlecolSingle Choice

Topic: Network Security Standards

ISO 27002 is primarily a:

Certification standard

Code of practice for security controls

Risk assessment methodology

Networking protocol

Correct Answer:

Correct Answer: B

Concept: ISO 27002 provides best-practice controls but is not itself a certification standard.

14singlecol singlecolSingle Choice

Topic: Network Security Standards

Classified Protection 2.0 compliance is primarily relevant in:

China

United States only

European Union only

Every country equally

Correct Answer:

Correct Answer: A

Concept: Classified Protection 2.0 is a Chinese national cybersecurity compliance framework.

15singlecol singlecolSingle Choice

Topic: Network Security Standards

Which is the highest protection level in Classified Protection 2.0?

Level 1

Level 3

Level 5

Level 10

Correct Answer:

Correct Answer: C

Concept: Classified Protection defines five levels, with Level 5 being the most stringent.

16singlecol singlecolSingle Choice

Topic: Network Security Standards

Expert review is required for Classified Protection levels starting at:

Level 1

Level 2

Level 3

Level 4

Correct Answer:

Correct Answer: B

Concept: For Level 2 and above, expert review and approval are required.

17singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: specifies certifiable ISMS requirements?

Classified Protection 2.0

ISO 27001:2022

ISO 27002

ISO 27001

Correct Answer:

Correct Answer: D

Concept: ISO 27001 refers to the concept where specifies certifiable ISMS requirements.

18singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: provides a code of practice for security controls?

PDCA

Level 2 and above

Classified Protection 2.0

ISO 27002

Correct Answer:

Correct Answer: D

Concept: ISO 27002 refers to the concept where provides a code of practice for security controls.

19singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: is the continuous improvement methodology used in ISO 27001?

Classified Protection lifecycle

Classified Protection 2.0

PDCA

Security synchronization

Correct Answer:

Correct Answer: C

Concept: PDCA refers to the concept where is the continuous improvement methodology used in ISO 27001.

20singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: is a Chinese national cybersecurity framework?

ISO 27002

ISO 27001

Classified Protection levels

Classified Protection 2.0

Correct Answer:

Correct Answer: D

Concept: Classified Protection 2.0 refers to the concept where is a Chinese national cybersecurity framework.

21singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: range from Level 1 to Level 5?

Classified Protection levels

Classified Protection lifecycle

ISO 27001

PDCA

Correct Answer:

Correct Answer: A

Concept: Classified Protection levels refers to the concept where range from Level 1 to Level 5.

22singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: require expert review and approval?

Classified Protection 2.0

ISO 27002

Classified Protection lifecycle

Level 2 and above

Correct Answer:

Correct Answer: D

Concept: Level 2 and above refers to the concept where require expert review and approval.

23singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: includes classification, filing, construction, assessment, and supervision?

Classified Protection lifecycle

Classified Protection levels

PDCA

ISO 27001

Correct Answer:

Correct Answer: A

Concept: Classified Protection lifecycle refers to the concept where includes classification, filing, construction, assessment, and supervision.

24singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: means planning security with information system construction?

Classified Protection levels

PDCA

Security synchronization

Classified Protection 2.0

Correct Answer:

Correct Answer: C

Concept: Security synchronization refers to the concept where means planning security with information system construction.

25singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: organizes controls into four themes?

ISO 27002

ISO 27001:2022

Classified Protection levels

Cybersecurity Law

Correct Answer:

Correct Answer: B

Concept: ISO 27001:2022 refers to the concept where organizes controls into four themes.

26singlecol singlecolSingle Choice

Topic: Network Security Standards

Which concept is described as: underpins Classified Protection in China?

Cybersecurity Law

ISO 27001:2022

ISO 27002

Classified Protection lifecycle

Correct Answer:

Correct Answer: A

Concept: Cybersecurity Law refers to the concept where underpins Classified Protection in China.

3 Network Fundamentals

1singlecol singlecolSingle Choice

Topic: Network Fundamentals

What is the default subnet mask for a Class C IPv4 address?

255.0.0.0

255.255.0.0

255.255.255.0

255.255.255.255

Correct Answer:

Correct Answer: C

Concept: Class C networks use a default /24 subnet mask: 255.255.255.0.

2singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol is used to automatically assign IPv6 addresses using MAC addresses?

DHCPv6

SLAAC

ARP

DNS

Correct Answer:

Correct Answer: B

Concept: SLAAC can generate IPv6 addresses from MAC addresses.

3singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which layer-2 protocol prevents loops in redundant Ethernet topologies?

VTP

STP

RSTP

Both STP and RSTP

Correct Answer:

Correct Answer: D

Concept: Spanning Tree Protocol and Rapid STP prevent Layer 2 loops.

4singlecol singlecolSingle Choice

Topic: Network Fundamentals

What does VLAN stand for?

Virtual Local Area Network

Virtual Large Area Network

Verified Local Access Node

Virtual Link Aggregation Network

Correct Answer:

Correct Answer: A

Concept: VLANs logically segment a switch into separate broadcast domains.

5singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which device interconnects VLANs at Layer 3?

Layer 2 switch

Router or Layer 3 switch

Hub

Bridge

Correct Answer:

Correct Answer: B

Concept: Routers or Layer 3 switches perform inter-VLAN routing.

6singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which OSI layer is responsible for reliable data transfer between hosts?

Network

Transport

Data link

Physical

Correct Answer:

Correct Answer: B

Concept: The transport layer manages end-to-end reliability, flow control, and error recovery.

7singlecol singlecolSingle Choice

Topic: Network Fundamentals

In the TCP/IP model, which layer corresponds to the OSI network layer?

Internet layer

Network interface layer

Transport layer

Application layer

Correct Answer:

Correct Answer: A

Concept: The TCP/IP internet layer maps to the OSI network layer and handles IP routing.

8singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which layer handles encryption and data format conversion?

Application

Presentation

Session

Transport

Correct Answer:

Correct Answer: B

Concept: The presentation layer handles syntax, encryption, compression, and character encoding.

9singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which layer establishes, manages, and terminates application dialogs?

Application

Presentation

Session

Transport

Correct Answer:

Correct Answer: C

Concept: The session layer controls dialog establishment, maintenance, and termination.

10singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol is used to send email between servers?

HTTP

SMTP

POP3

IMAP

Correct Answer:

Correct Answer: B

Concept: SMTP is used to send and relay email.

11singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol is used by clients to retrieve email from a server while keeping messages on the server?

POP3

IMAP

SMTP

SNMP

Correct Answer:

Correct Answer: B

Concept: IMAP allows clients to access and manage email while messages remain on the server.

12singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol dynamically assigns IP addresses to hosts?

DNS

DHCP

ARP

FTP

Correct Answer:

Correct Answer: B

Concept: DHCP dynamically assigns IP addresses and network configuration parameters.

13singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol maps IP addresses to MAC addresses on a local network?

DNS

ARP

DHCP

ICMP

Correct Answer:

Correct Answer: B

Concept: ARP resolves IP addresses to MAC addresses.

14singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which protocol is used for network management and monitoring?

SNMP

SMTP

FTP

Telnet

Correct Answer:

Correct Answer: A

Concept: SNMP manages and monitors network devices.

15singlecol singlecolSingle Choice

Topic: Network Fundamentals

What does ICMP primarily provide?

File transfer

Error reporting and diagnostics

Email delivery

Web browsing

Correct Answer:

Correct Answer: B

Concept: ICMP reports errors and provides diagnostic functions such as ping and traceroute.

16singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which port does HTTP use by default?

20

21

80

443

Correct Answer:

Correct Answer: C

Concept: HTTP uses TCP port 80 by default.

17singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which port does HTTPS use by default?

80

443

22

25

Correct Answer:

Correct Answer: B

Concept: HTTPS uses TCP port 443 by default, providing encrypted web traffic.

18singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which port does SSH use by default?

21

22

23

25

Correct Answer:

Correct Answer: B

Concept: SSH uses TCP port 22 for secure remote access.

19singlecol singlecolSingle Choice

Topic: Network Fundamentals

Telnet uses which port and is considered insecure because it transmits data in plaintext?

21
22
23
25

Correct Answer:

Correct Answer: C

Concept: Telnet uses TCP port 23 and sends data, including credentials, in plaintext.

20singlecol singlecolSingle Choice
Which address is a Layer 2 address?

Topic: Network Fundamentals

IP address
MAC address
Port number
Domain name

Correct Answer:

Correct Answer: B

Concept: A MAC address is a hardware address used at the data link layer.

21singlecol singlecolSingle Choice
A router forwards packets based on:

Topic: Network Fundamentals

MAC address
IP address
Port number
VLAN ID

Correct Answer:

Correct Answer: B

Concept: Routers use IP addresses to make forwarding decisions.

22singlecol singlecolSingle Choice
What is a subnet mask used for?

Topic: Network Fundamentals

MAC address
IP address class
Network and host portions of an IP address
DNS server

Correct Answer:

Correct Answer: C

Concept: A subnet mask divides an IP address into network and host portions.

23singlecol singlecolSingle Choice
The OSI model layer closest to the physical medium is:

Topic: Network Fundamentals

Data link
Physical
Network
Transport

Correct Answer:

Correct Answer: B

Concept: The physical layer deals with bits, cables, and signals.

24singlecol singlecolSingle Choice
TCP is considered reliable because it uses:

Topic: Network Fundamentals

Sequence numbers and acknowledgments

Broadcasting

No connection setup

Fixed ports only

Correct Answer:

Correct Answer: A

Concept: TCP uses sequence numbers, acknowledgments, and retransmissions for reliability.

25singlecol singlecolSingle Choice

Topic: Network Fundamentals

UDP is preferred for:

Reliable file transfer

Real-time streaming and low-latency applications

Email delivery

Web browsing requiring guaranteed delivery

Correct Answer:

Correct Answer: B

Concept: UDP low overhead and lack of retransmission make it suitable for real-time traffic.

26singlecol singlecolSingle Choice

Topic: Network Fundamentals

DNS primarily uses:

TCP port 53

UDP port 53

TCP port 80

UDP port 161

Correct Answer:

Correct Answer: B

Concept: DNS queries primarily use UDP port 53.

27singlecol singlecolSingle Choice

Topic: Network Fundamentals

A Layer 3 switch combines functions of a:

Hub and repeater

Switch and router

Firewall and IDS

Modem and gateway

Correct Answer:

Correct Answer: B

Concept: Layer 3 switches perform both Layer 2 switching and Layer 3 routing.

28singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: has seven layers?

Hub

OSI model

SSH

TCP/IP equivalent model

Correct Answer:

Correct Answer: B

Concept: OSI model refers to the concept where has seven layers.

29singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: has four layers?

Physical layer

TCP/IP standard model

RFC 1918

SMTP

Correct Answer:

Correct Answer: B

Concept: TCP/IP standard model refers to the concept where has four layers.

30singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: has five layers?

MAC address

HTTP

TCP/IP equivalent model

STP

Correct Answer:

Correct Answer: C

Concept: TCP/IP equivalent model refers to the concept where has five layers.

31singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: transmits raw bits over a medium?

Data link layer

SMTP

Physical layer

MAC address

Correct Answer:

Correct Answer: C

Concept: Physical layer refers to the concept where transmits raw bits over a medium.

32singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: handles framing and MAC addressing?

TCP/IP equivalent model

Switch

Data link layer

Application layer

Correct Answer:

Correct Answer: C

Concept: Data link layer refers to the concept where handles framing and MAC addressing.

33singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: handles logical addressing and routing?

ICMP

Network layer

Transport layer

Application layer

Correct Answer:

Correct Answer: B

Concept: Network layer refers to the concept where handles logical addressing and routing.

34singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: provides end-to-end communication and reliability?

SMTP

Transport layer

Application layer

Hub

Correct Answer:

Correct Answer: B

Concept: Transport layer refers to the concept where provides end-to-end communication and reliability.

35singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: manages application dialogs?

MAC address

ARP

Session layer

DNS

Correct Answer:

Correct Answer: C

Concept: Session layer refers to the concept where manages application dialogs.

36singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: handles encryption and data format conversion?

ICMP

HTTPS

Presentation layer

Hub

Correct Answer:

Correct Answer: C

Concept: Presentation layer refers to the concept where handles encryption and data format conversion.

37singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: provides network services to applications?

SMTP

Application layer

FTP

UDP

Correct Answer:

Correct Answer: B

Concept: Application layer refers to the concept where provides network services to applications.

38singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: is connection-oriented and reliable?

Switch

TCP

UDP

DNS

Correct Answer:

Correct Answer: B

Concept: TCP refers to the concept where is connection-oriented and reliable.

39singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: is connectionless and has low overhead?

Switch

UDP

STP

SSH

Correct Answer:

Correct Answer: B

Concept: UDP refers to the concept where is connectionless and has low overhead.

40singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP ports 20 and 21?

HTTPS

Presentation layer

OSI model

FTP

Correct Answer:

Correct Answer: D

Concept: FTP refers to the concept where uses TCP ports 20 and 21.

41singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP port 80?

ARP

ICMP

FTP

HTTP

Correct Answer:

Correct Answer: D

Concept: HTTP refers to the concept where uses TCP port 80.

42singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP port 443?

STP

VLAN

SSH

HTTPS

Correct Answer:

Correct Answer: D

Concept: HTTPS refers to the concept where uses TCP port 443.

43singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP port 22?

FTP

Network layer

SSH

DHCP

Correct Answer:

Correct Answer: C

Concept: SSH refers to the concept where uses TCP port 22.

44singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP port 23 and is insecure?

Telnet

Application layer

SMTP

IPv4 address

Correct Answer:

Correct Answer: A

Concept: Telnet refers to the concept where uses TCP port 23 and is insecure.

45singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: primarily uses UDP port 53?

Switch

DNS

Router

SMTP

Correct Answer:

Correct Answer: B

Concept: DNS refers to the concept where primarily uses UDP port 53.

46singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: uses TCP port 25?

TCP/IP equivalent model

UDP

Presentation layer

SMTP

Correct Answer:

Correct Answer: D

Concept: SMTP refers to the concept where uses TCP port 25.

47singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: dynamically assigns IP addresses?

RFC 1918

FTP

VLAN

DHCP

Correct Answer:

Correct Answer: D

Concept: DHCP refers to the concept where dynamically assigns IP addresses.

48singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: maps IP addresses to MAC addresses?

SSH

Presentation layer

MAC address

ARP

Correct Answer:

Correct Answer: D

Concept: ARP refers to the concept where maps IP addresses to MAC addresses.

49singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: provides error reporting and diagnostics?

ICMP

STP

ARP

Session layer

Correct Answer:

Correct Answer: A

Concept: ICMP refers to the concept where provides error reporting and diagnostics.

50singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: is 32 bits long?

IPv4 address

OSI model

MAC address

Router

Correct Answer:

Correct Answer: A

Concept: IPv4 address refers to the concept where is 32 bits long.

51singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: is 128 bits long?

RFC 1918

IPv6 address

Telnet

STP

Correct Answer:

Correct Answer: B

Concept: IPv6 address refers to the concept where is 128 bits long.

52singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: is a Layer 2 hardware address?

VLAN

MAC address

Network layer

OSI model

Correct Answer:

Correct Answer: B

Concept: MAC address refers to the concept where is a Layer 2 hardware address.

53singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: operates at Layer 2 and forwards frames by MAC?

Switch

STP

IPv6 address

Transport layer

Correct Answer:

Correct Answer: A

Concept: Switch refers to the concept where operates at Layer 2 and forwards frames by MAC.

54singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: operates at Layer 3 and forwards packets by IP?

FTP

VLAN

SSH

Router

Correct Answer:

Correct Answer: D

Concept: Router refers to the concept where operates at Layer 3 and forwards packets by IP.

55singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: operates at Layer 1 and repeats signals?

DHCP

TCP

Hub

DNS

Correct Answer:

Correct Answer: C

Concept: Hub refers to the concept where operates at Layer 1 and repeats signals.

56singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: logically segments a network into broadcast domains?

UDP

SSH

VLAN

DNS

Correct Answer:

Correct Answer: C

Concept: VLAN refers to the concept where logically segments a network into broadcast domains.

57singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: prevents Layer 2 loops?

STP

SMTP

VLAN

MAC address

Correct Answer:

Correct Answer: A

Concept: STP refers to the concept where prevents Layer 2 loops.

58singlecol singlecolSingle Choice

Topic: Network Fundamentals

Which concept is described as: defines private IPv4 address ranges?

RFC 1918

TCP/IP standard model

DHCP

Physical layer

Correct Answer:

Correct Answer: A

Concept: RFC 1918 refers to the concept where defines private IPv4 address ranges.

4 OSI and TCP/IP Models

1singlecol singlecolSingle Choice

Topic: OSI and TCP/IP Models

How many layers does the OSI reference model have?

4

5

7

9

Correct Answer:

Correct Answer: C

Concept: The OSI model has seven layers: physical, data link, network, transport, session, presentation, and application.

2singlecol singlecolSingle Choice

Topic: OSI and TCP/IP Models

How many layers does the TCP/IP standard model have?

4

5

6

7

Correct Answer:

Correct Answer: A

Concept: The TCP/IP standard model has four layers: network interface, internet, transport, and application.

3singlecol singlecolSingle Choice

Topic: OSI and TCP/IP Models

Which layer is responsible for end-to-end communication and flow control?

Network layer

Transport layer

Data link layer

Physical layer

Correct Answer:

Correct Answer: B

Concept: The transport layer provides end-to-end communication, reliability, and flow control (e.g., TCP).

4singlecol singlecolSingle Choice

Topic: OSI and TCP/IP Models

Which layer handles logical addressing and routing?

Transport layer

Network layer

Session layer

Presentation layer

Correct Answer:

Correct Answer: B

Concept: The network layer uses logical addresses (IP) and routes packets between networks.

single choice

Topic: OSI and TCP/IP Models

Which layer is responsible for framing, MAC addressing, and error detection on a local link?

Network layer

Transport layer

Data link layer

Application layer

Correct Answer:

Correct Answer: C

Concept: The data link layer handles framing, MAC addresses, and local link error detection.

5 Network Protocols

single choice

Topic: Network Protocols

Which protocol uses ports 20 and 21 for file transfer?

HTTP

FTP

SMTP

DNS

Correct Answer:

Correct Answer: B

Concept: FTP uses TCP port 21 for control and port 20 for data transfer.

single choice

Topic: Network Protocols

Which protocol resolves domain names to IP addresses?

DHCP

DNS

SNMP

NTP

Correct Answer:

Correct Answer: B

Concept: DNS (Domain Name System) maps human-readable domain names to IP addresses.

single choice

Topic: Network Protocols

Which transport protocol is connection-oriented and reliable?

UDP

TCP

ICMP

IP

Correct Answer:

Correct Answer: B

Concept: TCP provides connection-oriented, reliable, ordered delivery with flow and congestion control.

single choice

Topic: Network Protocols

Which transport protocol is connectionless and has low overhead?

TCP
UDP
HTTP
FTP

Correct Answer:

Correct Answer: B

Concept: UDP is connectionless, offers low latency and overhead, but does not guarantee delivery.

6 Network Devices

1singlecol singlecolSingle Choice

Topic: Network Devices

Which device operates at Layer 2 and forwards frames based on MAC addresses?

Router
Switch
Hub
Firewall

Correct Answer:

Correct Answer: B

Concept: A switch operates at the data link layer and forwards frames using MAC address tables.

2singlecol singlecolSingle Choice

Topic: Network Devices

Which device operates at Layer 3 and forwards packets based on IP addresses?

Switch
Router
Hub
Bridge

Correct Answer:

Correct Answer: B

Concept: A router operates at the network layer and makes forwarding decisions based on IP routing tables.

3singlecol singlecolSingle Choice

Topic: Network Devices

Which device simply repeats signals to all connected ports?

Switch
Router
Hub
Firewall

Correct Answer:

Correct Answer: C

Concept: A hub is a Layer 1 device that broadcasts signals to all connected devices.

7 Enterprise Security Threats

1singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack sends an overwhelming amount of traffic from many sources?

DoS
DDoS

Phishing

Spoofing

Correct Answer:

Correct Answer: B

Concept: DDoS uses multiple distributed sources to flood a target.

2singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack falsifies the source address of packets?

Phishing

Spoofing

Sniffing

Spamming

Correct Answer:

Correct Answer: B

Concept: Spoofing impersonates a trusted source by falsifying addresses or identities.

3singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which type of malware replicates itself across networks without user action?

Virus

Worm

Trojan

Spyware

Correct Answer:

Correct Answer: B

Concept: Worms can self-replicate and spread across networks, often without user interaction.

4singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which malware disguises itself as legitimate software?

Virus

Worm

Trojan

Adware

Correct Answer:

Correct Answer: C

Concept: A Trojan horse masquerades as legitimate software to trick users into executing it.

5singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Ransomware primarily does what?

Steals credentials

Encrypts data and demands payment

Deletes logs

Monitors traffic

Correct Answer:

Correct Answer: B

Concept: Ransomware encrypts files and demands a ransom for decryption.

6singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack intercepts communications between two parties without their knowledge?

Man-in-the-middle

DoS

Phishing

Tailgating

Correct Answer:

Correct Answer: A

Concept: A man-in-the-middle attack intercepts and possibly alters communication between parties.

7singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which protocol can help prevent man-in-the-middle attacks by verifying server identity?

HTTP

HTTPS

FTP

Telnet

Correct Answer:

Correct Answer: B

Concept: HTTPS uses TLS/SSL to encrypt traffic and authenticate the server.

8singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which wireless security protocol is more secure than WEP?

WPA2/WPA3

WEP

WPA (TKIP only)

Open network

Correct Answer:

Correct Answer: A

Concept: WPA2 and WPA3 provide stronger encryption and authentication than WEP.

9singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

What is a DMZ typically used for?

Hosting internal databases only

Hosting public-facing servers

User workstations

Printer sharing

Correct Answer:

Correct Answer: B

Concept: A DMZ hosts public-facing services between the Internet and internal network.

10singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which security zone usually has the highest priority in Huawei firewalls?

Untrust

Trust

DMZ

Local

Correct Answer:

Correct Answer: D

Concept: The Local zone (firewall itself) has priority 100, the highest default priority.

11singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which practice limits the damage from compromised accounts?

Least privilege
Sharing admin passwords
Disabling logs
Open shares

Correct Answer:

Correct Answer: A

Concept: Least privilege gives users only the access they need, limiting blast radius.

12singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

What is the purpose of host-based intrusion prevention?

Route packets
Detect and block malicious host activity
Assign IP addresses
Synchronize time

Correct Answer:

Correct Answer: B

Concept: Host-based IPS monitors and blocks suspicious activity on an endpoint.

13singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which technology collects and analyzes security logs centrally?

SIEM
SNMP
DHCP
ARP

Correct Answer:

Correct Answer: A

Concept: SIEM centralizes log collection and analysis.

14singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

What is a key benefit of centralized security management?

Reduced visibility
Unified policy and monitoring
More complex passwords
Faster malware spread

Correct Answer:

Correct Answer: B

Concept: Centralized management provides unified visibility, policy control, and monitoring.

15singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

What is an insider threat?

An attack from outside the network
A security risk originating from within the organization
A hardware vendor
A natural disaster

Correct Answer:

Correct Answer: B

Concept: Insider threats come from employees, contractors, or partners with internal access.

16singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack sends oversized ICMP packets to crash a target?

Ping of death

Ping sweep

ARP spoofing

DNS hijacking

Correct Answer:

Correct Answer: A

Concept: Ping of death sends ICMP packets larger than the maximum allowed size.

17singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack uses broadcast pings to amplify traffic?

Smurf attack

SYN flood

Fraggle

Session hijacking

Correct Answer:

Correct Answer: A

Concept: Smurf attacks send ICMP echo requests to broadcast addresses with a spoofed source.

18singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which attack intercepts ARP messages to redirect traffic?

ARP spoofing

DNS spoofing

IP spoofing

MAC flooding

Correct Answer:

Correct Answer: A

Concept: ARP spoofing sends forged ARP replies to associate an attacker MAC with another IP.

19singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

A SYN flood is a type of:

Denial of Service attack

Phishing attack

Data exfiltration

Physical attack

Correct Answer:

Correct Answer: A

Concept: SYN floods exhaust server connection resources with half-open TCP connections.

20singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

ARP spoofing is used to:

Intercept traffic on a local network

Encrypt files

Speed up routing

Authenticate users

Correct Answer:

Correct Answer: A

Concept: Attackers use forged ARP messages to redirect local traffic through themselves.

21singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

The principle of least privilege aims to:

Give everyone admin rights

Limit user access to the minimum necessary

Disable all logging

Remove firewalls

Correct Answer:

Correct Answer: B

Concept: Least privilege reduces risk by restricting access to only what is needed.

22singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

A SIEM system is primarily used for:

Centralized log analysis and threat detection

Assigning IP addresses

Switching frames

Manufacturing cables

Correct Answer:

Correct Answer: A

Concept: SIEM aggregates and analyzes security events and logs.

23singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: monitors traffic without altering it?

SIEM

Passive attack

Worm

SYN flood

Correct Answer:

Correct Answer: B

Concept: Passive attack refers to the concept where monitors traffic without altering it.

24singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: modifies or disrupts systems or data?

Ransomware

Passive attack

Patch management

Active attack

Correct Answer:

Correct Answer: D

Concept: Active attack refers to the concept where modifies or disrupts systems or data.

25singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: includes viruses, worms, Trojans, and ransomware?

Smurf attack

Passive attack

Malware

Worm

Correct Answer:

Correct Answer: C

Concept: Malware refers to the concept where includes viruses, worms, Trojans, and ransomware.

26singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: requires a host program to spread?

Worm

DDoS attack

Virus

Insider threat

Correct Answer:

Correct Answer: C

Concept: Virus refers to the concept where requires a host program to spread.

27singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: can self-replicate across networks?

Worm

Ping of death

Virus

Passive attack

Correct Answer:

Correct Answer: A

Concept: Worm refers to the concept where can self-replicate across networks.

28singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: disguises itself as legitimate software?

Trojan

Active attack

Worm

Man-in-the-middle attack

Correct Answer:

Correct Answer: A

Concept: Trojan refers to the concept where disguises itself as legitimate software.

29singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: encrypts data and demands payment?

SIEM

Smurf attack

Social engineering

Ransomware

Correct Answer:

Correct Answer: D

Concept: Ransomware refers to the concept where encrypts data and demands payment.

30singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: uses deception to steal information?

Worm

Patch management

Phishing

Virus

Correct Answer:

Correct Answer: C

Concept: Phishing refers to the concept where uses deception to steal information.

31singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: manipulates people into breaking security rules?

Patch management

Social engineering

Worm

ARP spoofing

Correct Answer:

Correct Answer: B

Concept: Social engineering refers to the concept where manipulates people into breaking security rules.

32singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: aims to make a service unavailable?

SIEM

Worm

Ransomware

DoS attack

Correct Answer:

Correct Answer: D

Concept: DoS attack refers to the concept where aims to make a service unavailable.

33singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: uses many distributed sources to flood a target?

Virus

DDoS attack

ARP spoofing

Man-in-the-middle attack

Correct Answer:

Correct Answer: B

Concept: DDoS attack refers to the concept where uses many distributed sources to flood a target.

34singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: intercepts communication between two parties?

ARP spoofing

Patch management

Man-in-the-middle attack

Social engineering

Correct Answer:

Correct Answer: C

Concept: Man-in-the-middle attack refers to the concept where intercepts communication between two parties.

35singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: forges ARP replies to redirect traffic?

ARP spoofing
Least privilege
DDoS attack
Malware

Correct Answer:

Correct Answer: A

Concept: ARP spoofing refers to the concept where forges ARP replies to redirect traffic.

36singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: sends many half-open TCP connections?

Ransomware
DMZ
Trojan
SYN flood

Correct Answer:

Correct Answer: D

Concept: SYN flood refers to the concept where sends many half-open TCP connections.

37singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: sends oversized ICMP packets?

Ping of death
Phishing
SIEM
Malware

Correct Answer:

Correct Answer: A

Concept: Ping of death refers to the concept where sends oversized ICMP packets.

38singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: uses broadcast pings to amplify traffic?

Social engineering
ARP spoofing
Trojan
Smurf attack

Correct Answer:

Correct Answer: D

Concept: Smurf attack refers to the concept where uses broadcast pings to amplify traffic.

39singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: limits user access to the minimum necessary?

Passive attack
Social engineering
Worm
Least privilege

Correct Answer:

Correct Answer: D

Concept: Least privilege refers to the concept where limits user access to the minimum necessary.

40singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: eliminates known vulnerabilities?

Phishing
 SYN flood
 Patch management
 Trojan

Correct Answer:

Correct Answer: C

Concept: Patch management refers to the concept where eliminates known vulnerabilities.

41singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: centralizes log collection and analysis?

Malware
 Active attack
 Phishing
 SIEM

Correct Answer:

Correct Answer: D

Concept: SIEM refers to the concept where centralizes log collection and analysis.

42singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: hosts public-facing servers between Internet and internal network?

DMZ
 DDoS attack
 DoS attack
 Patch management

Correct Answer:

Correct Answer: A

Concept: DMZ refers to the concept where hosts public-facing servers between Internet and internal network.

43singlecol singlecolSingle Choice

Topic: Enterprise Security Threats

Which concept is described as: originates from within the organization?

Insider threat
 Worm
 Man-in-the-middle attack
 Active attack

Correct Answer:

Correct Answer: A

Concept: Insider threat refers to the concept where originates from within the organization.

8 Communication Network Security

1singlecol singlecolSingle Choice

Topic: Communication Network Security

Which threat involves intercepting data traveling across a network?

Eavesdropping
 DoS
 Phishing

Tailgating

Correct Answer:

Correct Answer: A

Concept: Eavesdropping intercepts network communications to steal information.

9 Zone Border Security

1singlecol singlecolSingle Choice

Topic: Zone Border Security

What is the main purpose of a security zone?

Increase Internet speed

Group interfaces with similar security levels

Replace antivirus

Disable routing

Correct Answer:

Correct Answer: B

Concept: Security zones group network interfaces with similar trust levels to enforce inter-zone policies.

2singlecol singlecolSingle Choice

Topic: Zone Border Security

Which device is typically deployed at a zone border to enforce access control?

Hub

Firewall

Repeater

Bridge

Correct Answer:

Correct Answer: B

Concept: Firewalls are deployed at network boundaries to enforce security policies between zones.

10 Computing Environment Security

1singlecol singlecolSingle Choice

Topic: Computing Environment Security

Which control helps protect endpoints from malware?

Antivirus/EDR

Physical locks only

Hub replacement

Disabling DNS

Correct Answer:

Correct Answer: A

Concept: Antivirus and endpoint detection and response (EDR) protect hosts from malware and suspicious behavior.

11 Firewall Basics

1singlecol singlecolSingle Choice

Topic: Firewall Basics

Which firewall feature inspects application-layer data to detect attacks?

Packet filtering

Application inspection

Hub forwarding

NAT

Correct Answer:

Correct Answer: B

Concept: Application inspection examines payload content for malicious patterns.

2singlecol singlecolSingle Choice

Topic: Firewall Basics

Which Huawei firewall command mode is used for system-level configuration?

User view

System view

Interface view

Policy view

Correct Answer:

Correct Answer: B

Concept: System view allows global-level configuration on Huawei devices.

3singlecol singlecolSingle Choice

Topic: Firewall Basics

Which view is entered after the system-view command?

User view

System view

Interface view

None

Correct Answer:

Correct Answer: B

Concept: The system-view command enters system view from user view.

4singlecol singlecolSingle Choice

Topic: Firewall Basics

What is the purpose of a security profile in NGFW?

Physical installation

Applying advanced security features such as IPS/AV/URL filtering

Layer 1 forwarding

DHCP relay

Correct Answer:

Correct Answer: B

Concept: Security profiles group advanced inspection features applied to policies.

5singlecol singlecolSingle Choice

Topic: Firewall Basics

Which firewall generation combines traditional firewall functions with application awareness and intrusion prevention?

Packet-filtering firewall

Next-generation firewall

Stateful firewall

Proxy firewall

Correct Answer:

Correct Answer: B

Concept: Next-generation firewalls integrate application awareness, IPS, antivirus, and user identity.

6singlecol singlecolSingle Choice

Topic: Firewall Basics

What is the purpose of a default security policy deny action?

Improve performance
 Ensure least-privilege access
 Enable logging
 Allow management

Correct Answer:

Correct Answer: B

Concept: Default deny enforces least privilege by blocking unspecified traffic.

7singlecol singlecolSingle Choice

Topic: Firewall Basics

Which zone would typically contain public web servers?

Trust
 Untrust
 DMZ
 Local

Correct Answer:

Correct Answer: C

Concept: Public-facing servers are placed in the DMZ for isolation.

8singlecol singlecolSingle Choice

Topic: Firewall Basics

What does policy hit counting help with?

Routing optimization
 Verifying policy usage and troubleshooting
 MAC learning
 DHCP assignment

Correct Answer:

Correct Answer: B

Concept: Hit counts show which policies are triggered, aiding auditing and troubleshooting.

9singlecol singlecolSingle Choice

Topic: Firewall Basics

Which of the following is NOT a firewall deployment mode?

Routing mode
 Transparent mode
 Hybrid mode
 Broadcast mode

Correct Answer:

Correct Answer: D

Concept: Firewalls deploy in routing, transparent (bridge), or hybrid modes; broadcast mode is not standard.

10singlecol singlecolSingle Choice

Topic: Firewall Basics

What happens when traffic moves from a higher-priority zone to a lower-priority zone?

Inbound traffic
 Outbound traffic
 No traffic allowed
 Always dropped

Correct Answer:

Correct Answer: B

Concept: Traffic from a higher-priority zone to a lower-priority zone is considered outbound.

11singlecol singlecolSingle Choice

Topic: Firewall Basics

What is the priority range for custom security zones on Huawei firewalls?

0-50

1-100

1-99 excluding defaults

0-100

Correct Answer:

Correct Answer: C

Concept: Custom zone priorities range from 1 to 99, avoiding reserved default values.

12singlecol singlecolSingle Choice

Topic: Firewall Basics

Which action in a security policy can log permitted or denied traffic?

Permit

Deny

Both permit and deny

Neither

Correct Answer:

Correct Answer: C

Concept: Security policies can enable logging for both permit and deny actions.

13singlecol singlecolSingle Choice

Topic: Firewall Basics

Which security policy action is typically applied to suspicious traffic for further inspection?

Permit

Deny

Block

Redirect

Correct Answer:

Correct Answer: B

Concept: Suspicious traffic is commonly denied or blocked by policy.

14singlecol singlecolSingle Choice

Topic: Firewall Basics

When a return packet arrives, what does a stateful firewall check first?

Routing table

Session table

MAC table

ARP table

Correct Answer:

Correct Answer: B

Concept: A stateful firewall first checks the session table to find a matching connection.

15singlecol singlecolSingle Choice

Topic: Firewall Basics

What happens if a return packet does not match any session table entry?

Forwarded normally

Dropped or inspected against policies

Always permitted

Sent to DMZ

Correct Answer:

Correct Answer: B

Concept: Without a matching session, the firewall treats the packet as a new connection and applies policies.

16singlecol singlecolSingle Choice

Topic: Firewall Basics

Which protocol commonly requires ASPF support due to dynamic port negotiation?

HTTP

FTP

DNS

NTP

Correct Answer:

Correct Answer: B

Concept: FTP negotiates dynamic data ports in its control channel, requiring ASPF.

17singlecol singlecolSingle Choice

Topic: Firewall Basics

ASPF primarily inspects traffic at which layer?

Network layer only

Application layer

Physical layer

Data link layer

Correct Answer:

Correct Answer: B

Concept: ASPF performs application-layer inspection to identify negotiated connections.

18singlecol singlecolSingle Choice

Topic: Firewall Basics

In Huawei firewalls, the default priority of Trust is:

5

50

85

100

Correct Answer:

Correct Answer: C

Concept: Trust zone default priority is 85.

19singlecol singlecolSingle Choice

Topic: Firewall Basics

Traffic from a low-priority zone to a high-priority zone is considered:

Outbound

Inbound

Blocked by default

Always permitted

Correct Answer:

Correct Answer: B

Concept: Traffic entering a higher-priority zone from a lower-priority zone is inbound.

20singlecol singlecolSingle Choice

Topic: Firewall Basics

The default security policy action is:

Permit

Deny

Log

Redirect

Correct Answer:

Correct Answer: B

Concept: Huawei firewalls deny traffic by default unless explicitly permitted.

21singlecol singlecolSingle Choice
Stateful inspection relies on:

Topic: Firewall Basics

Session tables

MAC tables only

DNS records

DHCP leases

Correct Answer:

Correct Answer: A

Concept: Session tables track connection state for stateful inspection.

22singlecol singlecolSingle Choice
ASPF helps firewalls handle:

Topic: Firewall Basics

Multi-channel protocols

Static routing

MAC learning

DHCP relay

Correct Answer:

Correct Answer: A

Concept: ASPF inspects protocols that negotiate dynamic secondary channels.

23singlecol singlecolSingle Choice
Which concept is described as: is 85 by default?

Topic: Firewall Basics

Stateful firewall

Hybrid mode

Untrust zone priority

Trust zone priority

Correct Answer:

Correct Answer: D

Concept: Trust zone priority refers to the concept where is 85 by default.

24singlecol singlecolSingle Choice
Which concept is described as: is 50 by default?

Topic: Firewall Basics

DMZ zone priority

Trust zone priority

Security zones

Policy hit count

Correct Answer:

Correct Answer: A

Concept: DMZ zone priority refers to the concept where is 50 by default.

25singlecol singlecolSingle Choice
Which concept is described as: is 5 by default?

Topic: Firewall Basics

Next-generation firewall

Stateful firewall

Security zones

Untrust zone priority

Correct Answer:

Correct Answer: D

Concept: Untrust zone priority refers to the concept where is 5 by default.

26singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: is 100 by default?

Transparent mode

Untrust zone priority

Default security policy action

Local zone priority

Correct Answer:

Correct Answer: D

Concept: Local zone priority refers to the concept where is 100 by default.

27singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: is deny?

Intra-zone traffic

Routing mode

Default security policy action

Hybrid mode

Correct Answer:

Correct Answer: C

Concept: Default security policy action refers to the concept where is deny.

28singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: tracks connections in a session table?

Next-generation firewall

Security policy logging

Untrust zone priority

Stateful firewall

Correct Answer:

Correct Answer: D

Concept: Stateful firewall refers to the concept where tracks connections in a session table.

29singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: inspects packets individually without state?

Packet-filtering firewall

Hybrid mode

DMZ zone priority

Untrust zone priority

Correct Answer:

Correct Answer: A

Concept: Packet-filtering firewall refers to the concept where inspects packets individually without state.

30singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: inspects multi-channel protocols like FTP?

Intra-zone traffic

ASPF

Routing mode

Local zone priority

Correct Answer:

Correct Answer: B

Concept: ASPF refers to the concept where inspects multi-channel protocols like FTP.

31singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: includes application awareness and IPS?

Hybrid mode

Security policy logging

Next-generation firewall

Stateful firewall

Correct Answer:

Correct Answer: C

Concept: Next-generation firewall refers to the concept where includes application awareness and IPS.

32singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: group interfaces with similar trust levels?

Next-generation firewall

Security zones

Default security policy action

Untrust zone priority

Correct Answer:

Correct Answer: B

Concept: Security zones refers to the concept where group interfaces with similar trust levels.

33singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: can be controlled by policies?

Security policy logging

Intra-zone traffic

Untrust zone priority

Next-generation firewall

Correct Answer:

Correct Answer: B

Concept: Intra-zone traffic refers to the concept where can be controlled by policies.

34singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: helps verify which rules are used?

Next-generation firewall

Policy hit count

Intra-zone traffic

Routing mode

Correct Answer:

Correct Answer: B

Concept: Policy hit count refers to the concept where helps verify which rules are used.

35singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: can be enabled for permit and deny actions?

Security policy logging
Policy hit count
DMZ zone priority
Packet-filtering firewall

Correct Answer:

Correct Answer: A

Concept: Security policy logging refers to the concept where can be enabled for permit and deny actions.

36singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

Trust zone priority
Routing mode
DMZ zone priority
Untrust zone priority

Correct Answer:

Correct Answer: B

Concept: Routing mode refers to the concept where is a firewall deployment mode.

37singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

Intra-zone traffic
ASPF
Transparent mode
Next-generation firewall

Correct Answer:

Correct Answer: C

Concept: Transparent mode refers to the concept where is a firewall deployment mode.

38singlecol singlecolSingle Choice

Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

Hybrid mode
DMZ zone priority
ASPF
Default security policy action

Correct Answer:

Correct Answer: A

Concept: Hybrid mode refers to the concept where is a firewall deployment mode.

12 Security Zones

1singlecol singlecolSingle Choice

Topic: Security Zones

What is the default priority of the Trust zone on a Huawei firewall?

5
50
85

100

Correct Answer:

Correct Answer: C

Concept: The Trust zone has a default priority of 85.

2singlecol singlecolSingle Choice

Topic: Security Zones

What is the default priority of the DMZ zone on a Huawei firewall?

5

50

85

100

Correct Answer:

Correct Answer: B

Concept: The DMZ zone has a default priority of 50.

3singlecol singlecolSingle Choice

Topic: Security Zones

What is the default priority of the Untrust zone on a Huawei firewall?

5

50

85

100

Correct Answer:

Correct Answer: A

Concept: The Untrust zone has a default priority of 5.

4singlecol singlecolSingle Choice

Topic: Security Zones

What is the default priority of the Local zone on a Huawei firewall?

5

50

85

100

Correct Answer:

Correct Answer: D

Concept: The Local zone represents the firewall itself and has the highest default priority of 100.

13 Security Policies

1singlecol singlecolSingle Choice

Topic: Security Policies

What is the default action of a Huawei firewall security policy if no rule matches?

Permit

Deny

Log only

Redirect

Correct Answer:

Correct Answer: B

Concept: By default, unmatched traffic is denied.

2singlecol singlecolSingle Choice

Topic: Security Policies

Which components can be used as matching conditions in a security policy? (single best answer)

Source/destination zone, address, service, application, user, time

Only source IP

Only destination port

Only protocol

Correct Answer:

Correct Answer: A

Concept: Security policies can match on zones, addresses, users, services, applications, and time.

14 Stateful Inspection

1singlecol singlecolSingle Choice

Topic: Stateful Inspection

What is a session table used for in a stateful firewall?

Storing user passwords

Tracking established connections

Routing packets

Encrypting traffic

Correct Answer:

Correct Answer: B

Concept: The session table tracks connection state so return traffic can be permitted without a new policy lookup.

2singlecol singlecolSingle Choice

Topic: Stateful Inspection

Which type of firewall inspects packets individually without connection state?

Stateful firewall

Packet-filtering firewall

Proxy firewall

Next-generation firewall

Correct Answer:

Correct Answer: B

Concept: Packet-filtering firewalls inspect each packet independently based on header information.

15 ASPF

1singlecol singlecolSingle Choice

Topic: ASPF

What does ASPF stand for?

Advanced Stateful Packet Filtering

Application Specific Packet Filtering

Automatic Security Policy Forwarding

Advanced Stateful Policy Firewall

Correct Answer:

Correct Answer: B

Concept: ASPF is Application Specific Packet Filtering, used to inspect multi-channel protocols.

2singlecol singlecolSingle Choice

Topic: ASPF

Why is ASPF needed for protocols like FTP?

FTP only uses one port

FTP opens dynamic data channels

FTP encrypts all traffic

FTP does not use TCP

Correct Answer:

Correct Answer: B

Concept: FTP opens a dynamic data channel, and ASPF inspects the control channel to permit the negotiated data channel.

16 NAT

1singlecol singlecolSingle Choice

Topic: NAT

Which NAT type is also called many-to-one NAT?

Static NAT

Dynamic NAT

NAPT

NAT Server

Correct Answer:

Correct Answer: C

Concept: NAPT maps multiple private addresses to one public IP using port numbers.

2singlecol singlecolSingle Choice

Topic: NAT

Which NAT type preserves a one-to-one mapping between public and private addresses?

Static NAT

NAPT

Easy-IP

Dynamic PAT

Correct Answer:

Correct Answer: A

Concept: Static NAT maps one public IP to one private IP consistently.

3singlecol singlecolSingle Choice

Topic: NAT

Which NAT type uses the outbound interface IP as the public address?

NAPT

NAT No-PAT

Easy-IP

Destination NAT

Correct Answer:

Correct Answer: C

Concept: Easy-IP uses the firewall egress interface IP for translation.

4singlecol singlecolSingle Choice

Topic: NAT

Which source NAT mode maps one private address to one public address without port translation?

NAPT

NAT No-PAT

Easy-IP

NAT Server

Correct Answer:

Correct Answer: B

Concept: NAT No-PAT performs one-to-one address translation without changing port numbers.

5singlecol singlecolSingle Choice

Topic: NAT

When many internal hosts access the Internet using one public IP, which NAT is used?

NAT No-PAT

NAPT

NAT Server

Static NAT

Correct Answer:

Correct Answer: B

Concept: NAPT allows many-to-one address translation using port numbers.

6singlecol singlecolSingle Choice

Topic: NAT

Destination NAT is also known as:

Static NAT

Port forwarding / virtual IP

PAT

Proxy ARP

Correct Answer:

Correct Answer: B

Concept: Destination NAT maps external destinations to internal hosts, commonly called port forwarding or virtual IP.

7singlecol singlecolSingle Choice

Topic: NAT

Which NAT type maps a public IP/port to a private server IP/port?

Source NAT

NAT Server

NAPT

Bidirectional NAT

Correct Answer:

Correct Answer: B

Concept: NAT Server publishes internal services to external networks.

8singlecol singlecolSingle Choice

Topic: NAT

Bidirectional NAT is useful when:

Only source address translation is needed

Both source and destination must be rewritten

No NAT is needed

Only DNS traffic is involved

Correct Answer:

Correct Answer: B

Concept: It translates both source and destination addresses in a single flow.

9singlecol singlecolSingle Choice

Topic: NAT

Which problem can occur if NAT ALG is disabled for FTP?

FTP control channel fails

FTP data channel may fail because dynamic ports are not opened

FTP becomes encrypted

FTP speeds up

Correct Answer:

Correct Answer: B

Concept: Without ALG, the firewall cannot open the negotiated FTP data channel.

10singlecol singlecolSingle Choice

Topic: NAT

Which address type is preserved by NAT No-PAT?

Source port

Destination port

Both source and destination ports

No ports

Correct Answer:

Correct Answer: C

Concept: NAT No-PAT translates only IP addresses, preserving port numbers.

11singlecol singlecolSingle Choice

Topic: NAT

What is a major benefit of NAT?

One-to-one address mapping

Conserves public IP addresses

Full protocol transparency

No ALG needed

Correct Answer:

Correct Answer: B

Concept: NAT conserves public IPs by sharing one address across many internal hosts.

12singlecol singlecolSingle Choice

Topic: NAT

A NAT Server entry is typically:

Bidirectional and persistent

Temporary and random

Encrypted

Only for outbound traffic

Correct Answer:

Correct Answer: A

Concept: NAT Server entries are configured statically and persist for inbound access.

13singlecol singlecolSingle Choice

Topic: NAT

NAT ALG is often enabled together with:

ASPF

Routing

DHCP

DNS

Correct Answer:

Correct Answer: A

Concept: ALG and ASPF work together to inspect dynamic application channels.

14singlecol singlecolSingle Choice

Topic: NAT

Which RFC defines private IPv4 address space?

RFC 1918
RFC 2328
RFC 793
RFC 959

Correct Answer:

Correct Answer: A

Concept: RFC 1918 defines the private IPv4 address ranges.

15singlecol singlecolSingle Choice

Topic: NAT

Hairpin NAT allows internal users to access an internal server using:

The server private IP only
The server public NAT address
A VPN tunnel
Broadcast address

Correct Answer:

Correct Answer: B

Concept: Hairpin NAT lets internal hosts reach an internal server via its public NAT address.

16singlecol singlecolSingle Choice

Topic: NAT

NAPT is also known as:

PAT
Static NAT
NAT Server
Bridge NAT

Correct Answer:

Correct Answer: A

Concept: Port Address Translation (PAT) is another name for NAPT.

17singlecol singlecolSingle Choice

Topic: NAT

Static NAT is most often used when:

A server must always use the same public IP
Many hosts share one IP
Ports must be preserved
Outbound only traffic exists

Correct Answer:

Correct Answer: A

Concept: Static NAT provides a consistent one-to-one public-to-private mapping.

18singlecol singlecolSingle Choice

Topic: NAT

Destination NAT is commonly used to:

Allow internal hosts to browse the web
Publish internal services to external users
Encrypt VPN traffic
Assign IP addresses

Correct Answer:

Correct Answer: B

Concept: Destination NAT maps external destinations to internal servers.

19singlecol singlecolSingle Choice

Topic: NAT

Which feature rewrites IP addresses inside application payloads?

NAT ALG

NAPT

Static NAT

Route lookup

Correct Answer:

Correct Answer: A

Concept: NAT ALG handles protocols that embed address information in payloads.

20singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: translates IP addresses between private and public?

NAT ALG

NAT Server

Easy-IP

NAT

Correct Answer:

Correct Answer: D

Concept: NAT refers to the concept where translates IP addresses between private and public.

21singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: translates the source IP of outbound packets?

Bidirectional NAT

Source NAT

NAT No-PAT

Easy-IP

Correct Answer:

Correct Answer: B

Concept: Source NAT refers to the concept where translates the source IP of outbound packets.

22singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: translates the destination IP of inbound packets?

NAT

NAT No-PAT

Easy-IP

Destination NAT

Correct Answer:

Correct Answer: D

Concept: Destination NAT refers to the concept where translates the destination IP of inbound packets.

23singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: publishes internal services to the Internet?

NAT Server

Bidirectional NAT

NAPT

Source NAT

Correct Answer:

Correct Answer: A

Concept: NAT Server refers to the concept where publishes internal services to the Internet.

24singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: translates both source and destination addresses?

Bidirectional NAT

RFC 1918

Static NAT

Destination NAT

Correct Answer:

Correct Answer: A

Concept: Bidirectional NAT refers to the concept where translates both source and destination addresses.

25singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: translates many private addresses to one public IP using ports?

Static NAT

NAT No-PAT

NAPT

NAT Server

Correct Answer:

Correct Answer: C

Concept: NAPT refers to the concept where translates many private addresses to one public IP using ports.

26singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: performs one-to-one address translation preserving ports?

Hairpin NAT

NAT No-PAT

Source NAT

NAT ALG

Correct Answer:

Correct Answer: B

Concept: NAT No-PAT refers to the concept where performs one-to-one address translation preserving ports.

27singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: uses the outbound interface IP for translation?

Hairpin NAT

Easy-IP

Static NAT

Destination NAT

Correct Answer:

Correct Answer: B

Concept: Easy-IP refers to the concept where uses the outbound interface IP for translation.

28singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: rewrites addresses inside application payloads?

NAT ALG

Bidirectional NAT

Hairpin NAT

NAPT

Correct Answer:

Correct Answer: A

Concept: NAT ALG refers to the concept where rewrites addresses inside application payloads.

29singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: defines private IPv4 address ranges?

Destination NAT

RFC 1918

Source NAT

NAT

Correct Answer:

Correct Answer: B

Concept: RFC 1918 refers to the concept where defines private IPv4 address ranges.

30singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: lets internal users access servers via public NAT address?

Hairpin NAT

Easy-IP

NAT ALG

Source NAT

Correct Answer:

Correct Answer: A

Concept: Hairpin NAT refers to the concept where lets internal users access servers via public NAT address.

31singlecol singlecolSingle Choice

Topic: NAT

Which concept is described as: provides a persistent one-to-one mapping?

Static NAT

NAT No-PAT

NAT ALG

RFC 1918

Correct Answer:

Correct Answer: A

Concept: Static NAT refers to the concept where provides a persistent one-to-one mapping.

17 NAT Basics

1singlecol singlecolSingle Choice

Topic: NAT Basics

What does NAT primarily translate?

MAC addresses

IP addresses

Port numbers only

Domain names

Correct Answer:

Correct Answer: B

Concept: NAT translates IP addresses, typically between private and public addresses.

18 Source NAT

1singlecol singlecolSingle Choice

Topic: Source NAT

Which NAT type translates the source IP address of outgoing packets?

Source NAT

Destination NAT

Bidirectional NAT

NAT Server

Correct Answer:

Correct Answer: A

Concept: Source NAT changes the source IP so internal hosts can use public addresses to access the Internet.

2singlecol singlecolSingle Choice

Topic: Source NAT

What is the main purpose of NAT?

Translate many private addresses to one public IP using port multiplexing

Hide the destination IP

Encrypt traffic

Provide VPN access

Correct Answer:

Correct Answer: A

Concept: NAT (Network Address and Port Translation) maps multiple private addresses to a single public IP using different ports.

19 Destination NAT

1singlecol singlecolSingle Choice

Topic: Destination NAT

Which NAT type translates the destination IP address of incoming packets?

Source NAT

Destination NAT

Bidirectional NAT

NAT Server

Correct Answer:

Correct Answer: B

Concept: Destination NAT rewrites the destination IP, often to expose internal servers to external users.

20 NAT Server

1singlecol singlecolSingle Choice

Topic: NAT Server

What is a NAT Server used for?

Hiding internal server IP addresses

Publishing internal services to the Internet

Encrypting server traffic

Load balancing only

Correct Answer:

Correct Answer: B

Concept: NAT Server maps public IPs/ports to private server IPs/ports so external users can access internal services.

21 NAT ALG

1singlecol singlecolSingle Choice

Topic: NAT ALG

What is the purpose of NAT ALG?

Encrypt NAT traffic

Translate application-layer addresses embedded in protocol payloads

Replace firewalls

Assign IP addresses

Correct Answer:

Correct Answer: B

Concept: NAT Application Level Gateway inspects and rewrites IP addresses inside application-layer protocols (e.g., FTP, SIP).

22 Firewall Hot Standby

1singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

In active/standby mode, how many firewalls forward traffic at the same time?

One

Two

All

None

Correct Answer:

Correct Answer: A

Concept: In active/standby mode, one firewall actively forwards while the other remains standby.

2singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

In load-sharing mode, how many firewalls forward traffic?

One

Two or more

None

Only the backup

Correct Answer:

Correct Answer: B

Concept: Load-sharing mode allows multiple firewalls to forward traffic simultaneously.

3singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

What is the default VRRP advertisement interval?

1 second

3 seconds

10 seconds

30 seconds

Correct Answer:

Correct Answer: A

Concept: VRRP master sends advertisements approximately every 1 second by default.

4singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which VRRP priority value is highest configurable?

1

100

254

255

Correct Answer:

Correct Answer: C

Concept: VRRP priority ranges from 1 to 254 configurable; 255 is reserved for the IP owner.

5singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

What happens if a backup router stops receiving VRRP advertisements?

It remains backup

It preempts and becomes master

It shuts down

It sends ARP requests

Correct Answer:

Correct Answer: B

Concept: Missing advertisements trigger a backup to assume the master role.

6singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

What does VGMP group priority determine?

NAT pool size

Which firewall becomes active

Number of interfaces

DHCP lease time

Correct Answer:

Correct Answer: B

Concept: VGMP priority determines the active/standby role among redundant firewalls.

7singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which HRP state indicates the firewall is currently forwarding traffic?

Standby

Active

Initialize

Backup

Correct Answer:

Correct Answer: B

Concept: The active HRP peer forwards traffic and synchronizes state to the standby.

8singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which HRP command enables automatic configuration synchronization?

hrp enable

hrp auto-sync config

hrp mirror session

hrp vrrp enable

Correct Answer:

Correct Answer: B

Concept: hrp auto-sync config enables automatic configuration synchronization between peers.

9singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which deployment places both firewalls inline with the same traffic path?

Dual-hot standby
Active/standby inline
Route backup only
Transparent bridge

Correct Answer:

Correct Answer: B

Concept: Inline active/standby deployment places firewalls in the same forwarding path.

10singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

What is a common requirement for HRP heartbeat links?

Use the same interface as user traffic
Use a dedicated high-reliability link
Use wireless link only
No link required

Correct Answer:

Correct Answer: B

Concept: Heartbeat links should be dedicated and reliable to avoid false failovers.

11singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which mechanism reduces unnecessary VRRP state changes when interface flaps briefly?

Preempt delay
Track interface
Authentication
Priority 255

Correct Answer:

Correct Answer: A

Concept: Preempt delay prevents rapid master/backup flapping.

12singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which HRP feature ensures sessions survive a failover?

Session backup
Route backup
NAT ALG
URL filtering

Correct Answer:

Correct Answer: A

Concept: HRP session backup keeps session state synchronized so connections continue after failover.

13singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

VRRP master election is based primarily on:

Priority value
Router hostname
Number of interfaces
Uptime only

Correct Answer:

Correct Answer: A

Concept: The router with the highest priority becomes the VRRP master.

14singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

VGMP group management prevents:

Split-brain VRRP states

NAT exhaustion

DNS failures

DHCP conflicts

Correct Answer:

Correct Answer: A

Concept: VGMP ensures VRRP groups fail over consistently.

15singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

HRP is used to synchronize:

Session and configuration data

Email messages

Web pages

Printer drivers

Correct Answer:

Correct Answer: A

Concept: HRP synchronizes sessions, NAT, server-map entries, and configurations.

16singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

In active/standby mode, the standby firewall:

Does not forward traffic until failover

Forwards half the traffic

Processes all traffic

Acts as a switch

Correct Answer:

Correct Answer: A

Concept: The standby firewall remains ready but inactive until failover.

17singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: provides gateway redundancy with a virtual IP?

Load-sharing mode

VGMP

VRRP

Preempt delay

Correct Answer:

Correct Answer: C

Concept: VRRP refers to the concept where provides gateway redundancy with a virtual IP.

18singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: forwards traffic for the virtual IP?

Heartbeat link

VGMP

HRP

VRRP master

Correct Answer:

Correct Answer: D

Concept: VRRP master refers to the concept where forwards traffic for the virtual IP.

19singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: takes over if the master fails?

VGMP

VRRP backup

HRP session backup

Load-sharing mode

Correct Answer:

Correct Answer: B

Concept: VRRP backup refers to the concept where takes over if the master fails.

20singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: manages VRRP group states consistently?

VGMP

HRP

VRRP

VRRP master

Correct Answer:

Correct Answer: A

Concept: VGMP refers to the concept where manages VRRP group states consistently.

21singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: synchronizes sessions and configurations?

VRRP master

Heartbeat link

VRRP

HRP

Correct Answer:

Correct Answer: D

Concept: HRP refers to the concept where synchronizes sessions and configurations.

22singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: has one firewall forwarding traffic?

Preempt delay

HRP session backup

Active/standby mode

Load-sharing mode

Correct Answer:

Correct Answer: C

Concept: Active/standby mode refers to the concept where has one firewall forwarding traffic.

23singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: has multiple firewalls forwarding traffic?

Load-sharing mode

VRRP backup

Heartbeat link

HRP

Correct Answer:

Correct Answer: A

Concept: Load-sharing mode refers to the concept where has multiple firewalls forwarding traffic.

24singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: should be dedicated and reliable?

Heartbeat link

Preempt delay

VRRP

VRRP backup

Correct Answer:

Correct Answer: A

Concept: Heartbeat link refers to the concept where should be dedicated and reliable.

25singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: ensures connections survive failover?

Load-sharing mode

VGMP

HRP session backup

Heartbeat link

Correct Answer:

Correct Answer: C

Concept: HRP session backup refers to the concept where ensures connections survive failover.

26singlecol singlecolSingle Choice

Topic: Firewall Hot Standby

Which concept is described as: reduces rapid master-backup flapping?

Preempt delay

VRRP master

Heartbeat link

Active/standby mode

Correct Answer:

Correct Answer: A

Concept: Preempt delay refers to the concept where reduces rapid master-backup flapping.

23 VRRP

1singlecol singlecolSingle Choice

Topic: VRRP

What does VRRP stand for?

Virtual Router Redundancy Protocol

Virtual Routing Resolution Protocol

Virtual Redundant Routing Path

Virtual Routing Relay Protocol

Correct Answer:

Correct Answer: A

Concept: VRRP provides gateway redundancy by allowing multiple routers to share a virtual IP address.

2singlecol singlecolSingle Choice

Topic: VRRP

In VRRP, which router forwards traffic for the virtual IP?

Backup router

Master router

All routers simultaneously

Only the physical IP owner

Correct Answer:

Correct Answer: B

Concept: The VRRP master router owns the virtual IP and forwards traffic; backups take over if it fails.

24 VGMP

1singlecol singlecolSingle Choice

Topic: VGMP

What does VGMP manage on Huawei firewalls?

VRRP groups

NAT pools

Routing tables

DHCP scopes

Correct Answer:

Correct Answer: A

Concept: VGMP (VRRP Group Management Protocol) manages the state of multiple VRRP groups consistently.

25 HRP

1singlecol singlecolSingle Choice

Topic: HRP

What is the primary purpose of HRP?

Encrypt traffic

Synchronize sessions and configurations between firewalls

Assign IP addresses

Filter URLs

Correct Answer:

Correct Answer: B

Concept: HRP (Huawei Redundancy Protocol) synchronizes sessions, configurations, and state between hot-standby firewalls.

26 Intrusion Prevention and Antivirus

1singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which phase of an attack involves identifying vulnerable services?

Exploitation

Reconnaissance

Installation

Exfiltration

Correct Answer:

Correct Answer: B

Concept: Reconnaissance gathers information about targets before exploitation.

2singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which phase involves gaining unauthorized access?

Reconnaissance

Exploitation

Covering tracks

Reporting

Correct Answer:

Correct Answer: B

Concept: Exploitation uses vulnerabilities to gain access to the target.

3singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which deployment mode allows IDS to monitor traffic without affecting it?

Inline

Tap/SPAN

Bridge

Gateway

Correct Answer:

Correct Answer: B

Concept: IDS commonly connects to a tap or switch SPAN port for passive monitoring.

4singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which signature action drops the offending packet?

Alert

Block

Permit

Mirror

Correct Answer:

Correct Answer: B

Concept: Block actions discard malicious packets.

5singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

What is a false positive in IPS?

Attack missed

Legitimate traffic blocked as malicious

Correct detection

Signature update failure

Correct Answer:

Correct Answer: B

Concept: False positives incorrectly classify legitimate traffic as malicious.

6singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

What is a false negative in IPS?

Legitimate traffic blocked

Attack missed by IPS

Correct block

Logging error

Correct Answer:

Correct Answer: B

Concept: False negatives occur when an actual attack is not detected.

7singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which malware type encrypts user files and demands ransom?

Virus

Worm

Trojan

Ransomware

Correct Answer:

Correct Answer: D

Concept: Ransomware encrypts files and demands a ransom for decryption.

8singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which approach only allows known-good applications to run?

Blacklisting

Whitelisting

Heuristics

Sandboxing

Correct Answer:

Correct Answer: B

Concept: Whitelisting permits only approved applications.

9singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which technique detects malware by examining code behavior rather than signatures?

Signature scanning

Heuristic analysis

Whitelisting

Patching

Correct Answer:

Correct Answer: B

Concept: Heuristics analyze code behavior to identify potentially malicious activity.

10singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which protocol is commonly targeted by brute-force attacks against remote access?

HTTP

SSH

DNS

NTP

Correct Answer:

Correct Answer: B

Concept: SSH and similar remote-access protocols are frequent brute-force targets.

11singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

What does an IPS signature database contain?

User passwords

Known attack patterns

Routing tables

DNS records

Correct Answer:

Correct Answer: B

Concept: Signature databases contain patterns used to identify known attacks.

12singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

What is the role of the antivirus engine on a firewall?

Route packets

Scan files and traffic for malware

Assign IP addresses

Authenticate users

Correct Answer:

Correct Answer: B

Concept: The antivirus engine inspects files and traffic to detect and block malware.

13singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

An IPS differs from an IDS mainly because an IPS can:

Block traffic

Only log events

Operate out-of-band only

Never generate alerts

Correct Answer:

Correct Answer: A

Concept: IPS can take active blocking actions, while IDS primarily detects.

14singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Signature-based detection cannot easily detect:

Known attacks

Zero-day attacks

Old malware

Common exploits

Correct Answer:

Correct Answer: B

Concept: Zero-day attacks have no known signatures.

15singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Sandboxing helps detect malware by:

Observing behavior in an isolated environment

Matching signatures

Comparing file sizes

Checking file names

Correct Answer:

Correct Answer: A

Concept: Sandboxing executes suspicious files safely to observe malicious behavior.

16singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Antivirus engines require regular updates to:

Recognize new malware variants
 Improve routing
 Increase bandwidth
 Disable firewalls

Correct Answer:

Correct Answer: A

Concept: Signature and detection databases must be updated to detect emerging threats.

17singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects intrusions and generates alerts?

Antivirus signatures
 False negative
 Signature detection
 IDS

Correct Answer:

Correct Answer: D

Concept: IDS refers to the concept where detects intrusions and generates alerts.

18singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects and blocks intrusions inline?

False positive
 IPS
 Anomaly detection
 NIDS

Correct Answer:

Correct Answer: B

Concept: IPS refers to the concept where detects and blocks intrusions inline.

19singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: matches known attack patterns?

Signature detection
 Whitelisting
 IPS
 Exploitation

Correct Answer:

Correct Answer: A

Concept: Signature detection refers to the concept where matches known attack patterns.

20singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: flags deviations from normal behavior?

NIDS
 Anomaly detection
 Exploitation
 Whitelisting

Correct Answer:

Correct Answer: B

Concept: Anomaly detection refers to the concept where flags deviations from normal behavior.

21singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: incorrectly blocks legitimate traffic?

Heuristic analysis
Anomaly detection
Exploitation
False positive

Correct Answer:

Correct Answer: D

Concept: False positive refers to the concept where incorrectly blocks legitimate traffic.

22singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: misses an actual attack?

Sandboxing
Anomaly detection
Whitelisting
False negative

Correct Answer:

Correct Answer: D

Concept: False negative refers to the concept where misses an actual attack.

23singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: observes file behavior in an isolated environment?

False negative
Sandboxing
Heuristic analysis
NIDS

Correct Answer:

Correct Answer: B

Concept: Sandboxing refers to the concept where observes file behavior in an isolated environment.

24singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects malware by examining behavior?

Reconnaissance
Heuristic analysis
Whitelisting
Signature detection

Correct Answer:

Correct Answer: B

Concept: Heuristic analysis refers to the concept where detects malware by examining behavior.

25singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: allows only approved applications?

IDS
HIDS
Whitelisting
Sandboxing

Correct Answer:

Correct Answer: C

Concept: Whitelisting refers to the concept where allows only approved applications.

26singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: must be updated regularly?

IDS

Antivirus signatures

False positive

NIDS

Correct Answer:

Correct Answer: B

Concept: Antivirus signatures refers to the concept where must be updated regularly.

27singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: monitors network traffic for intrusions?

NIDS

HIDS

Whitelisting

Anomaly detection

Correct Answer:

Correct Answer: A

Concept: NIDS refers to the concept where monitors network traffic for intrusions.

28singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: monitors host activity for intrusions?

Antivirus signatures

HIDS

IDS

False negative

Correct Answer:

Correct Answer: B

Concept: HIDS refers to the concept where monitors host activity for intrusions.

29singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: is an early attack phase?

IDS

Reconnaissance

False positive

Heuristic analysis

Correct Answer:

Correct Answer: B

Concept: Reconnaissance refers to the concept where is an early attack phase.

30singlecol singlecolSingle Choice

Topic: Intrusion Prevention and Antivirus

Which concept is described as: gains unauthorized access using vulnerabilities?

Antivirus signatures

Exploitation

Sandboxing

HIDS

Correct Answer:

Correct Answer: B

Concept: Exploitation refers to the concept where gains unauthorized access using vulnerabilities.

27 Intrusion Overview

1singlecol singlecolSingle Choice

Topic: Intrusion Overview

What is an intrusion?

Authorized access

Unauthorized access or attack on a network/system

A software patch

A backup operation

Correct Answer:

Correct Answer: B

Concept: An intrusion is any unauthorized attempt to access, manipulate, or harm a system or network.

28 Intrusion Prevention

1singlecol singlecolSingle Choice

Topic: Intrusion Prevention

What is the main difference between IDS and IPS?

IDS only detects; IPS detects and blocks

IDS is faster

IPS cannot log

IDS operates inline

Correct Answer:

Correct Answer: A

Concept: IDS detects intrusions and alerts; IPS can also take action to block malicious traffic.

2singlecol singlecolSingle Choice

Topic: Intrusion Prevention

Which detection method compares traffic against known attack signatures?

Anomaly detection

Signature detection

Behavioral analysis

Heuristic detection

Correct Answer:

Correct Answer: B

Concept: Signature-based detection matches traffic patterns against a database of known attack signatures.

3singlecol singlecolSingle Choice

Topic: Intrusion Prevention

Which detection method builds a baseline of normal behavior and flags deviations?

Signature detection

Anomaly detection

Static analysis

Port scanning

Correct Answer:

Correct Answer: B

Concept: Anomaly detection identifies traffic that deviates from a learned baseline.

4singlecol singlecolSingle Choice

Topic: Intrusion Prevention

Where is an IPS typically deployed?

Out-of-band only

Inline in the traffic path

On user desktops only

Inside a printer

Correct Answer:

Correct Answer: B

Concept: IPS is deployed inline so it can block malicious traffic in real time.

29 Antivirus

1singlecol singlecolSingle Choice

Topic: Antivirus

What does antivirus software primarily detect and remove?

Routing loops

Malware

Cable faults

DNS errors

Correct Answer:

Correct Answer: B

Concept: Antivirus detects and removes malware such as viruses, worms, and Trojans.

2singlecol singlecolSingle Choice

Topic: Antivirus

Which antivirus technique looks for known byte sequences of malware?

Heuristic analysis

Signature scanning

Sandboxing

Whitelisting

Correct Answer:

Correct Answer: B

Concept: Signature scanning matches files/traffic against known malware signatures.

3singlecol singlecolSingle Choice

Topic: Antivirus

Which method executes suspicious files in an isolated environment to observe behavior?

Signature scanning

Heuristic analysis

Sandboxing

Whitelisting

Correct Answer:

Correct Answer: C

Concept: Sandboxing runs files in an isolated environment to detect malicious behavior.

30 AAA and User Authentication

1singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which AAA protocol commonly uses UDP ports 1812 and 1813?

TACACS+

RADIUS

LDAP

Kerberos

Correct Answer:

Correct Answer: B

Concept: RADIUS commonly uses UDP ports 1812/1813 (or 1645/1646 legacy).

2singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which AAA protocol uses TCP port 49 by default?

RADIUS

TACACS+

LDAP

DHCP

Correct Answer:

Correct Answer: B

Concept: TACACS+ uses TCP port 49 by default.

3singlecol singlecolSingle Choice

Topic: AAA and User Authentication

In AAA, which process maps users to privileges?

Authentication

Authorization

Accounting

Admission

Correct Answer:

Correct Answer: B

Concept: Authorization maps authenticated identities to permitted actions/resources.

4singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which authentication type identifies users based on source IP?

User authentication

Access user authentication

Device authentication

IP/MAC binding

Correct Answer:

Correct Answer: B

Concept: Access user authentication can identify users by IP/MAC for transparent access control.

5singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which user type is created directly on the firewall?

RADIUS user

LDAP user

Local user

Domain user

Correct Answer:

Correct Answer: C

Concept: Local users are defined directly in the firewall local user database.

6singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which component is queried when using LDAP authentication?

DNS server

LDAP directory server

NTP server

SNMP server

Correct Answer:

Correct Answer: B

Concept: The firewall queries an LDAP directory server to validate credentials.

7singlecol singlecolSingle Choice

Topic: AAA and User Authentication

What is the purpose of a user group?

Increase Internet speed

Apply policies to multiple users collectively

Replace firewalls

Encrypt traffic

Correct Answer:

Correct Answer: B

Concept: User groups simplify policy management by grouping users with similar access needs.

8singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which authentication method is often used for wired/wireless network access control?

Portal

802.1X

Local

RADIUS accounting

Correct Answer:

Correct Answer: B

Concept: 802.1X is commonly used for port-based network access control on wired and wireless networks.

9singlecol singlecolSingle Choice

Topic: AAA and User Authentication

What does single sign-on (SSO) provide?

Multiple passwords

One authentication for multiple services

No authentication

Local only access

Correct Answer:

Correct Answer: B

Concept: SSO allows users to authenticate once and access multiple services.

10singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which user authentication policy condition can match traffic?

Source zone

Time range

Source/destination address

All of the above

Correct Answer:

Correct Answer: D

Concept: Authentication policies can match zones, addresses, time ranges, and other conditions.

11singlecol singlecolSingle Choice

Topic: AAA and User Authentication

The first A in AAA stands for:

Authentication

Authorization

Accounting

Auditing

Correct Answer:

Correct Answer: A

Concept: AAA stands for Authentication, Authorization, and Accounting.

12singlecol singlecolSingle Choice

Topic: AAA and User Authentication

RADIUS commonly runs over:

UDP

TCP

SCTP

ICMP

Correct Answer:

Correct Answer: A

Concept: RADIUS typically uses UDP ports 1812/1813.

13singlecol singlecolSingle Choice

Topic: AAA and User Authentication

TACACS+ commonly runs over:

UDP

TCP

HTTP

FTP

Correct Answer:

Correct Answer: B

Concept: TACACS+ uses TCP port 49.

14singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Portal authentication is typically used for:

Web-based guest access

Cable modem registration

Switch MAC learning

DNS resolution

Correct Answer:

Correct Answer: A

Concept: Portal authentication presents a web login page.

15singlecol singlecolSingle Choice

Topic: AAA and User Authentication

802.1X authentication occurs at:

Layer 2 port level

Application layer only

Physical layer only

Transport layer only

Correct Answer:

Correct Answer: A

Concept: 802.1X provides port-based network access control at Layer 2.

16singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: verifies user identity?

Accounting
Authorization
RADIUS
Authentication

Correct Answer:

Correct Answer: D

Concept: Authentication refers to the concept where verifies user identity.

17singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: determines what a user can access?

Accounting
Portal authentication
Local authentication
Authorization

Correct Answer:

Correct Answer: D

Concept: Authorization refers to the concept where determines what a user can access.

18singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: records user activities and resource usage?

Accounting
Authorization
User group
Portal authentication

Correct Answer:

Correct Answer: A

Concept: Accounting refers to the concept where records user activities and resource usage.

19singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: uses UDP ports 1812 and 1813?

Portal authentication
RADIUS
SSO
Local authentication

Correct Answer:

Correct Answer: B

Concept: RADIUS refers to the concept where uses UDP ports 1812 and 1813.

20singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: uses TCP port 49?

RADIUS
TACACS+
802.1X
Authentication

Correct Answer:

Correct Answer: B

Concept: TACACS+ refers to the concept where uses TCP port 49.

21singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: uses a web login page?

Authorization

User group

Local authentication

Portal authentication

Correct Answer:

Correct Answer: D

Concept: Portal authentication refers to the concept where uses a web login page.

22singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: provides port-based network access control?

User group

Accounting

SSO

802.1X

Correct Answer:

Correct Answer: D

Concept: 802.1X refers to the concept where provides port-based network access control.

23singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: uses the device user database?

Authentication

LDAP

Accounting

Local authentication

Correct Answer:

Correct Answer: D

Concept: Local authentication refers to the concept where uses the device user database.

24singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: queries a directory server for authentication?

802.1X

RADIUS

LDAP

Authentication

Correct Answer:

Correct Answer: C

Concept: LDAP refers to the concept where queries a directory server for authentication.

25singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: applies policies to multiple users collectively?

Portal authentication

User group

802.1X

TACACS+

Correct Answer:

Correct Answer: B

Concept: User group refers to the concept where applies policies to multiple users collectively.

26singlecol singlecolSingle Choice

Topic: AAA and User Authentication

Which concept is described as: allows one authentication for multiple services?

802.1X

LDAP

SSO

RADIUS

Correct Answer:

Correct Answer: C

Concept: SSO refers to the concept where allows one authentication for multiple services.

31 AAA Basics

1singlecol singlecolSingle Choice

Topic: AAA Basics

What does AAA stand for?

Authentication, Authorization, Accounting

Access, Audit, Alert

Attack, Analysis, Action

Application, Authorization, Audit

Correct Answer:

Correct Answer: A

Concept: AAA provides authentication (who are you), authorization (what can you do), and accounting (what did you do).

2singlecol singlecolSingle Choice

Topic: AAA Basics

Which AAA component verifies a user identity?

Authentication

Authorization

Accounting

Auditing

Correct Answer:

Correct Answer: A

Concept: Authentication verifies identity, for example, by username/password or certificate.

3singlecol singlecolSingle Choice

Topic: AAA Basics

Which AAA component determines what resources a user can access?

Authentication

Authorization

Accounting

Auditing

Correct Answer:

Correct Answer: B

Concept: Authorization defines permissions and access rights after authentication.

4singlecol singlecolSingle Choice

Topic: AAA Basics

Which AAA component records user activities for billing or auditing?

Authentication

Authorization

Accounting

Auditing

Correct Answer:

Correct Answer: C

Concept: Accounting logs user activities such as login time, data usage, and commands executed.

32 Firewall User Authentication

1singlecol singlecolSingle Choice

Topic: Firewall User Authentication

Which authentication method redirects a user to a web portal for login?

Local authentication

RADIUS authentication

Portal authentication

LDAP authentication

Correct Answer:

Correct Answer: C

Concept: Portal authentication presents a web page to users for credential entry.

2singlecol singlecolSingle Choice

Topic: Firewall User Authentication

Which authentication method uses a client-side agent to authenticate users before network access?

Portal authentication

802.1X authentication

Local authentication

Certificate authentication

Correct Answer:

Correct Answer: B

Concept: 802.1X uses a supplicant client to authenticate devices before granting network access.

3singlecol singlecolSingle Choice

Topic: Firewall User Authentication

Which protocol is commonly used for centralized authentication and accounting?

RADIUS

SMTP

SNMP

NTP

Correct Answer:

Correct Answer: A

Concept: RADIUS is a widely used AAA protocol for centralized authentication, authorization, and accounting.

4singlecol singlecolSingle Choice

Topic: Firewall User Authentication

Which protocol provides authentication with stronger security and uses TCP?

RADIUS

TACACS+

LDAP

DHCP

Correct Answer:

Correct Answer: B

Concept: TACACS+ uses TCP and separates authentication, authorization, and accounting functions.

33 Cryptography and PKI

1singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which symmetric algorithm uses variable key lengths of 128, 192, or 256 bits?

DES

AES

RSA

MD5

Correct Answer:

Correct Answer: B

Concept: AES supports 128, 192, and 256-bit keys.

2singlecol singlecolSingle Choice

Topic: Cryptography and PKI

How many bits is a DES key?

56

64

128

256

Correct Answer:

Correct Answer: A

Concept: DES uses a 56-bit effective key length.

3singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which asymmetric algorithm is commonly used for key exchange in TLS?

AES

RSA

SHA-256

HMAC

Correct Answer:

Correct Answer: B

Concept: RSA or Diffie-Hellman are commonly used for TLS key exchange.

4singlecol singlecolSingle Choice

Topic: Cryptography and PKI

What is the main disadvantage of symmetric encryption?

Slow speed

Key distribution challenge

Large key sizes

No confidentiality

Correct Answer:

Correct Answer: B

Concept: Distributing the shared secret key securely is the main challenge.

5singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which property ensures a small input change produces a very different hash?

Determinism

Avalanche effect

Fixed output

Reversibility

Correct Answer:

Correct Answer: B

Concept: The avalanche effect means tiny input changes cause large output changes.

6singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which hash algorithm produces a 128-bit digest?

MD5

SHA-1

SHA-256

SHA-512

Correct Answer:

Correct Answer: A

Concept: MD5 produces a 128-bit digest.

7singlecol singlecolSingle Choice

Topic: Cryptography and PKI

What is HMAC?

A hash-based message authentication code

A symmetric cipher

A key exchange protocol

A firewall feature

Correct Answer:

Correct Answer: A

Concept: HMAC combines a hash function with a secret key to provide message authentication.

8singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which key is used to create a digital signature?

Public key

Private key

Shared secret

Session key

Correct Answer:

Correct Answer: B

Concept: The signer uses their private key to create a digital signature.

9singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which key is used to verify a digital signature?

Public key

Private key

Shared secret

Master key

Correct Answer:

Correct Answer: A

Concept: The verifier uses the signer public key to verify the signature.

10singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which standard defines the format of digital certificates?

X.509

X.500

LDAP

DNS

Correct Answer:

Correct Answer: A

Concept: X.509 is the standard format for public key certificates.

11singlecol singlecolSingle Choice

Topic: Cryptography and PKI

What does OCSP provide?

Online certificate status checking

Certificate issuance

Key generation

Email encryption

Correct Answer:

Correct Answer: A

Concept: OCSP checks certificate revocation status in real time.

12singlecol singlecolSingle Choice

Topic: Cryptography and PKI

In a PKI hierarchy, which certificate is self-signed?

End-entity certificate

Root CA certificate

Intermediate CA certificate

Server certificate

Correct Answer:

Correct Answer: B

Concept: Root CA certificates are self-signed because they are the trust anchor.

13singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which entity stores issued certificates for retrieval?

CA

RA

Repository

CRL

Correct Answer:

Correct Answer: C

Concept: A repository distributes certificates and CRLs to relying parties.

14singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which cipher mode provides both confidentiality and authentication?

ECB

CBC

GCM

CTR

Correct Answer:

Correct Answer: C

Concept: Galois/Counter Mode (GCM) provides authenticated encryption.

15singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which mode should be avoided because identical plaintext blocks produce identical ciphertext?

ECB

CBC

GCM

CFB

Correct Answer:

Correct Answer: A

Concept: ECB reveals patterns in plaintext and is generally insecure.

16singlecol singlecolSingle Choice

AES is an example of:

Topic: Cryptography and PKI

Symmetric encryption

Asymmetric encryption

Hashing

Steganography

Correct Answer:

Correct Answer: A

Concept: AES uses a single shared key for encryption and decryption.

17singlecol singlecolSingle Choice

RSA is an example of:

Topic: Cryptography and PKI

Asymmetric encryption

Symmetric encryption

Hash algorithm

Compression algorithm

Correct Answer:

Correct Answer: A

Concept: RSA uses public and private key pairs.

18singlecol singlecolSingle Choice

A hash function produces:

Topic: Cryptography and PKI

A fixed-length digest

The original input

An encrypted file

A random key

Correct Answer:

Correct Answer: A

Concept: Hash functions map variable input to a fixed-size digest.

19singlecol singlecolSingle Choice

MD5 is considered insecure because:

Topic: Cryptography and PKI

Collision vulnerabilities exist

It is too slow

It produces too long output

It requires a public key

Correct Answer:

Correct Answer: A

Concept: MD5 collision attacks make it unsuitable for security purposes.

20singlecol singlecolSingle Choice

A digital certificate binds a public key to:

Topic: Cryptography and PKI

An identity
A password
A MAC address
A VLAN

Correct Answer:

Correct Answer: A

Concept: Certificates associate a public key with an entity identity.

21singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Certificate revocation status can be checked using:

CRL or OCSP
DNS only
DHCP only
SNMP only

Correct Answer:

Correct Answer: A

Concept: CRLs and OCSP provide revocation status information.

22singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: uses the same key for encryption and decryption?

Root CA certificate
Public key
Private key
Symmetric encryption

Correct Answer:

Correct Answer: D

Concept: Symmetric encryption refers to the concept where uses the same key for encryption and decryption.

23singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: uses public and private key pairs?

Asymmetric encryption
X.509
Root CA certificate
Hash function

Correct Answer:

Correct Answer: A

Concept: Asymmetric encryption refers to the concept where uses public and private key pairs.

24singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is a symmetric encryption standard?

GCM
SHA-256
AES
RA

Correct Answer:

Correct Answer: C

Concept: AES refers to the concept where is a symmetric encryption standard.

25singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is an asymmetric encryption algorithm?

PKI

Asymmetric encryption

RSA

HMAC

Correct Answer:

Correct Answer: C

Concept: RSA refers to the concept where is an asymmetric encryption algorithm.

26singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: uses a 56-bit effective key?

CA

OCSP

DES

SHA-256

Correct Answer:

Correct Answer: C

Concept: DES refers to the concept where uses a 56-bit effective key.

27singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: produces a fixed-length digest?

Root CA certificate

Hash function

CA

RA

Correct Answer:

Correct Answer: B

Concept: Hash function refers to the concept where produces a fixed-length digest.

28singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is considered broken due to collision attacks?

Hash function

MD5

AES

CRL

Correct Answer:

Correct Answer: B

Concept: MD5 refers to the concept where is considered broken due to collision attacks.

29singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: produces a 256-bit digest?

Digital signature

RA

Hash function

SHA-256

Correct Answer:

Correct Answer: D

Concept: SHA-256 refers to the concept where produces a 256-bit digest.

30singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: provides message authentication using a hash and key?

PKI
CRL
HMAC
Hash function

Correct Answer:

Correct Answer: C

Concept: HMAC refers to the concept where provides message authentication using a hash and key.

31singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: verifies authenticity and integrity?

Public key
OCSP
Digital signature
Asymmetric encryption

Correct Answer:

Correct Answer: C

Concept: Digital signature refers to the concept where verifies authenticity and integrity.

32singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is used to verify a digital signature?

Root CA certificate
Public key
HMAC
Hash function

Correct Answer:

Correct Answer: B

Concept: Public key refers to the concept where is used to verify a digital signature.

33singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is used to create a digital signature?

Asymmetric encryption
CA
Private key
OCSP

Correct Answer:

Correct Answer: C

Concept: Private key refers to the concept where is used to create a digital signature.

34singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: manages public keys and certificates?

PKI
HMAC
Public key
OCSP

Correct Answer:

Correct Answer: A

Concept: PKI refers to the concept where manages public keys and certificates.

35singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: issues and signs digital certificates?

CA
Symmetric encryption
CRL
SHA-256

Correct Answer:

Correct Answer: A

Concept: CA refers to the concept where issues and signs digital certificates.

36singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: validates identity before certificate issuance?

Public key
RA
Private key
PKI

Correct Answer:

Correct Answer: B

Concept: RA refers to the concept where validates identity before certificate issuance.

37singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: lists revoked certificates?

HMAC
CRL
Digital signature
ECB

Correct Answer:

Correct Answer: B

Concept: CRL refers to the concept where lists revoked certificates.

38singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: provides online certificate status checking?

X.509
OCSP
Hash function
HMAC

Correct Answer:

Correct Answer: B

Concept: OCSP refers to the concept where provides online certificate status checking.

39singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is the standard certificate format?

AES
X.509
Symmetric encryption
Digital signature

Correct Answer:

Correct Answer: B

Concept: X.509 refers to the concept where is the standard certificate format.

40singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: is self-signed and acts as a trust anchor?

Root CA certificate

Digital signature

ECB

CA

Correct Answer:

Correct Answer: A

Concept: Root CA certificate refers to the concept where is self-signed and acts as a trust anchor.

41singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: provides authenticated encryption?

GCM

Root CA certificate

X.509

OCSP

Correct Answer:

Correct Answer: A

Concept: GCM refers to the concept where provides authenticated encryption.

42singlecol singlecolSingle Choice

Topic: Cryptography and PKI

Which concept is described as: should be avoided because it reveals plaintext patterns?

RA

Root CA certificate

DES

ECB

Correct Answer:

Correct Answer: D

Concept: ECB refers to the concept where should be avoided because it reveals plaintext patterns.

34 Cryptography Basics

1singlecol singlecolSingle Choice

Topic: Cryptography Basics

Which encryption type uses the same key for encryption and decryption?

Symmetric encryption

Asymmetric encryption

Hashing

Steganography

Correct Answer:

Correct Answer: A

Concept: Symmetric encryption uses a single shared key for both encryption and decryption.

2singlecol singlecolSingle Choice

Topic: Cryptography Basics

Which encryption type uses a key pair: public key and private key?

Symmetric encryption

Asymmetric encryption

Hashing

Encoding

Correct Answer:

Correct Answer: B

Concept: Asymmetric encryption uses a public key to encrypt and a private key to decrypt (or vice versa).

3singlecol singlecolSingle Choice

Topic: Cryptography Basics

Which algorithm is a symmetric encryption standard?

RSA

AES

ECC

DSA

Correct Answer:

Correct Answer: B

Concept: AES (Advanced Encryption Standard) is a widely used symmetric encryption algorithm.

4singlecol singlecolSingle Choice

Topic: Cryptography Basics

Which algorithm is an asymmetric encryption algorithm?

AES

DES

RSA

3DES

Correct Answer:

Correct Answer: C

Concept: RSA is an asymmetric algorithm used for encryption, digital signatures, and key exchange.

35 Hash Algorithms

1singlecol singlecolSingle Choice

Topic: Hash Algorithms

What is a key property of a cryptographic hash function?

Reversible output

Fixed-length output regardless of input size

Requires a public key

Uses the same key for hashing and verification

Correct Answer:

Correct Answer: B

Concept: Hash functions produce a fixed-length digest for any input size.

2singlecol singlecolSingle Choice

Topic: Hash Algorithms

Which hash algorithm produces a 256-bit digest?

MD5

SHA-1

SHA-256

SHA-512

Correct Answer:

Correct Answer: C

Concept: SHA-256 produces a 256-bit hash digest.

3singlecol singlecolSingle Choice

Topic: Hash Algorithms

Which hash algorithm is considered broken and unsuitable for security use?

SHA-256

MD5

SHA-3

BLAKE2

Correct Answer:**Correct Answer: B****Concept:** MD5 is vulnerable to collision attacks and should not be used for security-sensitive applications.

36 Digital Signatures

1singlecol singlecolSingle Choice

Topic: Digital Signatures

What is the purpose of a digital signature?

Encrypt all data

Verify authenticity and integrity

Compress files

Hide sender identity

Correct Answer:**Correct Answer: B****Concept:** Digital signatures verify the sender identity and ensure data has not been altered.

37 PKI

1singlecol singlecolSingle Choice

Topic: PKI

What does PKI stand for?

Public Key Infrastructure

Private Key Interface

Packet Knowledge Index

Policy Key Identifier

Correct Answer:**Correct Answer: A****Concept:** PKI provides a framework for managing public keys, certificates, and trust relationships.

2singlecol singlecolSingle Choice

Topic: PKI

Which entity issues digital certificates in a PKI?

Registration Authority

Certificate Authority

End user

Repository

Correct Answer:**Correct Answer: B****Concept:** A Certificate Authority (CA) issues and signs digital certificates.

3singlecol singlecolSingle Choice

Topic: PKI

Which PKI component validates an applicant identity before certificate issuance?

Certificate Authority

Registration Authority

Repository

CRL

Correct Answer:

Correct Answer: B

Concept: A Registration Authority (RA) verifies identities and forwards requests to the CA.

4singlecol singlecolSingle Choice

Topic: PKI

What is a CRL used for?

Storing public keys

Listing revoked certificates

Encrypting emails

Generating hash values

Correct Answer:

Correct Answer: B

Concept: A Certificate Revocation List (CRL) contains certificates that have been revoked before expiration.

38 VPN

1singlecol singlecolSingle Choice

Topic: VPN

Which VPN deployment connects two fixed sites?

Remote access VPN

Site-to-site VPN

Clientless VPN

Host-to-host VPN

Correct Answer:

Correct Answer: B

Concept: Site-to-site VPNs connect entire networks at different locations.

2singlecol singlecolSingle Choice

Topic: VPN

What is a key advantage of GRE?

Built-in encryption

Can encapsulate multicast and non-IP protocols

Uses TCP

Automatic key exchange

Correct Answer:

Correct Answer: B

Concept: GRE can tunnel multicast and non-IP traffic, making it useful for routing protocols.

3singlecol singlecolSingle Choice

Topic: VPN

Which protocol number is assigned to GRE?

47

50

51

89

Correct Answer:

Correct Answer: A

Concept: GRE is IP protocol 47.

4singlecol singlecolSingle Choice
What does IKE Phase 1 establish?

Topic: VPN

- IPsec SA
- IKE SA
- Routing table
- DHCP lease

Correct Answer:

Correct Answer: B

Concept: IKE Phase 1 establishes a secure IKE SA for protected negotiation.

5singlecol singlecolSingle Choice
What does IKE Phase 2 establish?

Topic: VPN

- IKE SA
- IPsec SA
- SSL session
- ARP entry

Correct Answer:

Correct Answer: B

Concept: IKE Phase 2 negotiates the IPsec SA that protects actual data traffic.

6singlecol singlecolSingle Choice
Which authentication method does IKE support? (single best answer)

Topic: VPN

- Pre-shared key
- Digital signatures
- Both pre-shared key and digital signatures
- None

Correct Answer:

Correct Answer: C

Concept: IKE supports both pre-shared key and digital signature authentication.

7singlecol singlecolSingle Choice
Which Diffie-Hellman group provides stronger key exchange?

Topic: VPN

- Group 1
- Group 2
- Group 5
- Group 14 or higher

Correct Answer:

Correct Answer: D

Concept: Higher Diffie-Hellman groups such as 14 or above provide stronger security.

8singlecol singlecolSingle Choice
Which IKE version provides faster SA setup and improved reliability?

Topic: VPN

- IKEv1
- IKEv2
- IKEv3

ISAKMP

Correct Answer:

Correct Answer: B

Concept: IKEv2 simplifies negotiation and improves resilience compared to IKEv1.

9singlecol singlecolSingle Choice

Topic: VPN

Which UDP ports does L2TP use?

500 and 4500

1701

443

50 and 51

Correct Answer:

Correct Answer: B

Concept: L2TP uses UDP port 1701.

10singlecol singlecolSingle Choice

Topic: VPN

Why is L2TP often combined with IPsec?

To provide encryption

To increase speed

To reduce overhead

To avoid tunnels

Correct Answer:

Correct Answer: A

Concept: L2TP lacks encryption, so IPsec is added for confidentiality and integrity.

11singlecol singlecolSingle Choice

Topic: VPN

Which SSL VPN mode grants the remote host a virtual IP on the internal network?

Web proxy

Network extension

File sharing

Port forwarding

Correct Answer:

Correct Answer: B

Concept: Network extension mode assigns a virtual IP and routes traffic into the corporate network.

12singlecol singlecolSingle Choice

Topic: VPN

Which SSL VPN mode is best for accessing internal web apps without installing a client?

Web proxy

Network extension

IPsec

L2TP

Correct Answer:

Correct Answer: A

Concept: Web proxy mode requires only a browser.

13singlecol singlecolSingle Choice

Topic: VPN

Which encapsulation mode is typically used for site-to-site VPNs?

Transport mode

Tunnel mode

Bridge mode

Trunk mode

Correct Answer:

Correct Answer: B

Concept: Site-to-site VPNs typically use tunnel mode to protect the entire original packet.

14singlecol singlecolSingle Choice

Topic: VPN

Which tunneling protocol is considered a Layer 2 VPN technology?

GRE

IPsec

L2TP

SSL VPN

Correct Answer:

Correct Answer: C

Concept: L2TP operates at Layer 2, while GRE, IPsec, and SSL VPN operate at higher layers.

15singlecol singlecolSingle Choice

Topic: VPN

Which component defines the security parameters for an IPsec connection?

SPD

SA

NAT

ARP

Correct Answer:

Correct Answer: B

Concept: A Security Association (SA) defines algorithms, keys, and lifetimes for IPsec.

16singlecol singlecolSingle Choice

Topic: VPN

The main goal of a VPN is to provide:

Secure communication over public networks

Faster Internet speed

Free public Wi-Fi

Unencrypted remote access

Correct Answer:

Correct Answer: A

Concept: VPNs create secure encrypted tunnels over untrusted networks.

17singlecol singlecolSingle Choice

Topic: VPN

GRE protocol number is:

47

50

51

89

Correct Answer:

Correct Answer: A

Concept: GRE is encapsulated using IP protocol 47.

18singlecol singlecolSingle Choice

Topic: VPN

IPsec ESP provides:

Encryption and authentication

Only routing

Only NAT

Only compression

Correct Answer:

Correct Answer: A

Concept: ESP provides confidentiality, integrity, and authentication.

19singlecol singlecolSingle Choice

Topic: VPN

IPsec AH provides:

Authentication and integrity without encryption

Encryption only

Compression only

Key exchange

Correct Answer:

Correct Answer: A

Concept: AH authenticates and protects integrity but does not encrypt.

20singlecol singlecolSingle Choice

Topic: VPN

IKE negotiates:

Security associations and keys for IPsec

DHCP leases

DNS records

MAC addresses

Correct Answer:

Correct Answer: A

Concept: IKE handles key exchange and SA negotiation.

21singlecol singlecolSingle Choice

Topic: VPN

SSL VPN commonly uses port:

80

443

22

1701

Correct Answer:

Correct Answer: B

Concept: SSL VPN uses HTTPS/TCP 443.

22singlecol singlecolSingle Choice

Topic: VPN

L2TP is commonly paired with which protocol for encryption?

IPsec

GRE

OSPF

BGP

Correct Answer:

Correct Answer: A

Concept: L2TP provides tunneling and is paired with IPsec for encryption.

23singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: provides secure communication over public networks?

Remote-access VPN

AH

VPN

SSL VPN

Correct Answer:

Correct Answer: C

Concept: VPN refers to the concept where provides secure communication over public networks.

24singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: connects entire networks at different locations?

Site-to-site VPN

IPsec

Tunnel mode

GRE

Correct Answer:

Correct Answer: A

Concept: Site-to-site VPN refers to the concept where connects entire networks at different locations.

25singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: connects individual users to a corporate network?

Remote-access VPN

IKEv2

Tunnel mode

Network extension mode

Correct Answer:

Correct Answer: A

Concept: Remote-access VPN refers to the concept where connects individual users to a corporate network.

26singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: is IP protocol 47 and lacks built-in encryption?

ESP

GRE

IKE

IPsec

Correct Answer:

Correct Answer: B

Concept: GRE refers to the concept where is IP protocol 47 and lacks built-in encryption.

27singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: can tunnel multicast and non-IP protocols?

Tunnel mode

Transport mode

GRE

Remote-access VPN

Correct Answer:

Correct Answer: C

Concept: GRE refers to the concept where can tunnel multicast and non-IP protocols.

28singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: provides confidentiality, integrity, and authentication?

Network extension mode

Remote-access VPN

IPsec

IKEv2

Correct Answer:

Correct Answer: C

Concept: IPsec refers to the concept where provides confidentiality, integrity, and authentication.

29singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: provides encryption and authentication?

ESP

IKEv2

SSL VPN

GRE

Correct Answer:

Correct Answer: A

Concept: ESP refers to the concept where provides encryption and authentication.

30singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: provides authentication without encryption?

ESP

GRE

AH

IPsec

Correct Answer:

Correct Answer: C

Concept: AH refers to the concept where provides authentication without encryption.

31singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: negotiates IPsec keys and security associations?

GRE

Tunnel mode

IKEv2

IKE

Correct Answer:

Correct Answer: D

Concept: IKE refers to the concept where negotiates IPsec keys and security associations.

32singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: encrypts the entire original IP packet?

IKEv2

L2TP

IKE

Tunnel mode

Correct Answer:

Correct Answer: D

Concept: Tunnel mode refers to the concept where encrypts the entire original IP packet.

33singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: encrypts only the IP payload?

Remote-access VPN

Web proxy mode

Transport mode

GRE

Correct Answer:

Correct Answer: C

Concept: Transport mode refers to the concept where encrypts only the IP payload.

34singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: uses UDP port 1701 and lacks encryption?

IKEv2

VPN

L2TP over IPsec

L2TP

Correct Answer:

Correct Answer: D

Concept: L2TP refers to the concept where uses UDP port 1701 and lacks encryption.

35singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: combines L2TP tunneling with IPsec encryption?

Network extension mode

L2TP

L2TP over IPsec

Site-to-site VPN

Correct Answer:

Correct Answer: C

Concept: L2TP over IPsec refers to the concept where combines L2TP tunneling with IPsec encryption.

36singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: commonly uses TCP port 443?

SSL VPN

IKE

IPsec

Web proxy mode

Correct Answer:

Correct Answer: A

Concept: SSL VPN refers to the concept where commonly uses TCP port 443.

37singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: provides browser-based access to web apps?

Network extension mode

Web proxy mode

IKEv2

GRE

Correct Answer:

Correct Answer: B

Concept: Web proxy mode refers to the concept where provides browser-based access to web apps.

38singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: assigns a virtual IP to the remote host?

Network extension mode

Site-to-site VPN

Web proxy mode

VPN

Correct Answer:

Correct Answer: A

Concept: Network extension mode refers to the concept where assigns a virtual IP to the remote host.

39singlecol singlecolSingle Choice

Topic: VPN

Which concept is described as: is faster and more reliable than IKEv1?

IKEv2

Web proxy mode

IPsec

Site-to-site VPN

Correct Answer:

Correct Answer: A

Concept: IKEv2 refers to the concept where is faster and more reliable than IKEv1.

39 VPN Overview

1singlecol singlecolSingle Choice

Topic: VPN Overview

What does VPN stand for?

Virtual Private Network

Virtual Public Network

Verified Private Node

Virtual Protocol Network

Correct Answer:

Correct Answer: A

Concept: A VPN extends a private network across a public network using encryption and tunneling.

2singlecol singlecolSingle Choice

Topic: VPN Overview

Which benefit does a VPN provide over the public Internet?

Higher physical speed

Confidentiality and secure remote access

Free bandwidth

No encryption needed

Correct Answer:

Correct Answer: B

Concept: VPNs provide confidentiality, integrity, and secure access over untrusted networks.

3singlecol singlecolSingle Choice

Topic: VPN Overview

Which VPN type connects individual remote users to a corporate network?

Site-to-site VPN

Remote-access VPN

Intranet VPN

Extranet VPN

Correct Answer:

Correct Answer: B

Concept: Remote-access VPNs allow individual users to securely connect to a corporate network.

40 GRE VPN

1singlecol singlecolSingle Choice

Topic: GRE VPN

Which protocol does GRE use as its transport protocol?

TCP

UDP

IP protocol 47

IP protocol 50

Correct Answer:

Correct Answer: C

Concept: GRE encapsulates packets using IP protocol number 47.

2singlecol singlecolSingle Choice

Topic: GRE VPN

Which statement about GRE is correct?

GRE provides encryption by default

GRE is a simple tunneling protocol without built-in encryption

GRE cannot tunnel multicast

GRE uses UDP port 500

Correct Answer:

Correct Answer: B

Concept: GRE provides encapsulation and tunneling but does not provide encryption or authentication by itself.

41 IPsec VPN

1singlecol singlecolSingle Choice

Topic: IPsec VPN

Which IPsec protocol provides encryption and confidentiality?

AH

ESP

IKE

GRE

Correct Answer:

Correct Answer: B

Concept: ESP (Encapsulating Security Payload) provides encryption, authentication, and integrity.

2singlecol singlecolSingle Choice

Topic: IPsec VPN

Which IPsec protocol provides authentication and integrity but not encryption?

AH

ESP

IKE

L2TP

Correct Answer:**Correct Answer: A****Concept:** AH (Authentication Header) provides source authentication and integrity but does not encrypt payloads.

3singlecol singlecolSingle Choice

Topic: IPsec VPN

Which protocol is used to establish and manage IPsec security associations?

AH

ESP

IKE

GRE

Correct Answer:**Correct Answer: C****Concept:** IKE (Internet Key Exchange) negotiates keys and Security Associations for IPsec.

4singlecol singlecolSingle Choice

Topic: IPsec VPN

Which IPsec mode encrypts the entire original IP packet and adds a new IP header?

Transport mode

Tunnel mode

Gre mode

Bridge mode

Correct Answer:**Correct Answer: B****Concept:** Tunnel mode encapsulates and encrypts the entire original packet, adding a new IP header.

5singlecol singlecolSingle Choice

Topic: IPsec VPN

Which IPsec mode encrypts only the payload and is used for host-to-host communications?

Transport mode

Tunnel mode

Bridge mode

Trunk mode

Correct Answer:**Correct Answer: A****Concept:** Transport mode encrypts only the IP payload, leaving the original IP header intact.

42 L2TP VPN

1singlecol singlecolSingle Choice

Topic: L2TP VPN

Which layer-2 tunneling protocol is often combined with IPsec for encryption?

PPTP

L2TP

GRE

MPLS

Correct Answer:**Correct Answer: B****Concept:** L2TP provides tunneling and is commonly paired with IPsec to provide encryption.

43 SSL VPN

1singlecol singlecolSingle Choice

Topic: SSL VPN

Which VPN type is commonly accessed through a web browser without a dedicated client?

IPsec VPN

SSL VPN

GRE VPN

L2TP VPN

Correct Answer:

Correct Answer: B

Concept: SSL VPN can be accessed via a web browser, making it convenient for remote users.

2singlecol singlecolSingle Choice

Topic: SSL VPN

Which SSL VPN mode provides access to specific web applications through a browser?

Web proxy mode

Network extension mode

File sharing mode

Full tunnel mode

Correct Answer:

Correct Answer: A

Concept: Web proxy mode allows browser-based access to internal web applications.

PART II

Multiple-Answer Questions

44 Network Security Concepts

1multicol multicolMultiple Choice

Topic: Network Security Concepts

Which of the following are characteristics of network security? (Choose all that apply)

Confidentiality

Integrity

Availability

Controllability

Non-repudiation

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: Network security commonly encompasses confidentiality, integrity, availability, controllability, and non-repudiation.

2multicol multicolMultiple Choice

Topic: Network Security Concepts

Which are stages in the development history of network security? (Choose all that apply)

Communication security period

Information security period

Information assurance period

Cyberspace security period

Correct Answer:

Correct Answer: A,B,C,D

Concept: Network security has evolved through communication security, information security, information assurance, and cyberspace security periods.

3multicol multicolMultiple Choice

Topic: Network Security Concepts

Which are core information security properties? (Choose all that apply)

Confidentiality

Integrity

Availability

Controllability

Non-repudiation

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: Network security includes confidentiality, integrity, availability, controllability, and non-repudiation.

45 Network Security Standards

1multicol multicolMultiple Choice

Topic: Network Security Standards

Which are typical Classified Protection 2.0 lifecycle phases? (Choose all that apply)

Classification

Filing

Construction and rectification

Level assessment

Supervision and inspection

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

2multicol multicolMultiple Choice

Topic: Network Security Standards

Which are benefits of implementing ISO 27001? (Choose all that apply)

Systematic risk management

Improved stakeholder confidence

Legal/regulatory compliance

Guaranteed zero attacks

Correct Answer:

Correct Answer: A,B,C

Concept: ISO 27001 helps manage risk, build confidence, and comply with regulations, but cannot guarantee zero attacks.

3multicol multicolMultiple Choice

Topic: Network Security Standards

Which are typical Classified Protection 2.0 security extension requirements? (Choose all that apply)

Cloud computing

Mobile Internet

Internet of Things

Industrial control systems

Correct Answer:

Correct Answer: A,B,C,D

Concept: Classified Protection 2.0 extends coverage to cloud, mobile Internet, IoT, and industrial control scenarios.

4multicol multicolMultiple Choice

Topic: Network Security Standards

Which belong to the Classified Protection lifecycle? (Choose all that apply)

Classification

Filing

Construction and rectification

Level assessment

Supervision and inspection

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction, level assessment, and supervision.

5multicol multicolMultiple Choice

Topic: Network Security Standards

Which are ISO 27000 family standards? (Choose all that apply)

ISO 27001

ISO 27002

ISO 27005

ISO 9001

Correct Answer:

Correct Answer: A,B,C

Concept: ISO 27001, 27002, and 27005 are information security standards; ISO 9001 is quality management.

6multicol multicolMultiple Choice

Topic: Network Security Standards

Which of the following relate to ISO? (Choose all that apply)

ISO 27001

ISO 27002

ISO 27001:2022

Correct Answer:

Correct Answer: A,B,C

Concept: These concepts all relate to ISO in Network Security Standards.

7multicol multicolMultiple Choice

Topic: Network Security Standards

Which of the following relate to Classified? (Choose all that apply)

Classified Protection 2.0

Classified Protection levels

Classified Protection lifecycle

Correct Answer:

Correct Answer: A,B,C

Concept: These concepts all relate to Classified in Network Security Standards.

46 Network Fundamentals

1multicol multicolMultiple Choice

Topic: Network Fundamentals

Which layers are included in the TCP/IP equivalent model? (Choose all that apply)

Application

Transport

Network

Data link

Physical

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes application, transport, network, data link, and physical layers.

2multicol multicolMultiple Choice

Topic: Network Fundamentals

Which are connectionless protocols? (Choose all that apply)

UDP

IP

ICMP

TCP

Correct Answer:

Correct Answer: A,B,C

Concept: UDP, IP, and ICMP are connectionless; TCP is connection-oriented.

3multicol multicolMultiple Choice

Topic: Network Fundamentals

Which fields are in a TCP header? (Choose all that apply)

Source port

Destination port

Sequence number

Acknowledgment number

Correct Answer:

Correct Answer: A,B,C,D

Concept: TCP headers include source/destination ports, sequence and acknowledgment numbers, flags, window size, etc.

4multicol multicolMultiple Choice

Topic: Network Fundamentals

Which devices operate primarily at Layer 1? (Choose all that apply)

Hub

Repeater

Switch

Router

Correct Answer:

Correct Answer: A,B

Concept: Hubs and repeaters operate at the physical layer.

5multicol multicolMultiple Choice

Topic: Network Fundamentals

Which protocols use TCP? (Choose all that apply)

HTTP

FTP

SMTP

DNS (sometimes)

Correct Answer:

Correct Answer: A,B,C,D

Concept: HTTP, FTP, and SMTP use TCP; DNS primarily uses UDP but can use TCP for zone transfers and large responses.

6multicol multicolMultiple Choice

Topic: Network Fundamentals

Which protocols use UDP? (Choose all that apply)

DNS

TFTP

SNMP

DHCP

Correct Answer:

Correct Answer: A,B,C,D

Concept: DNS, TFTP, SNMP, and DHCP commonly use UDP.

7multicol multicolMultiple Choice

Topic: Network Fundamentals

Which are valid IPv4 private address ranges? (Choose all that apply)

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

224.0.0.0/4

Correct Answer:

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

8multicol multicolMultiple Choice

Topic: Network Fundamentals

Which are OSI layers? (Choose all that apply)

Physical

Data link

Network

Transport

Application

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: The OSI model includes these layers plus session and presentation.

9multicol multicolMultiple Choice

Topic: Network Fundamentals

Which are TCP/IP equivalent model layers? (Choose all that apply)

Application

Transport

Network

Data link

Physical

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes all five layers.

10multicol multicolMultiple Choice

Topic: Network Fundamentals

Which protocols use TCP? (Choose all that apply)

HTTP

FTP

SMTP

SSH

Correct Answer:**Correct Answer: A,B,C,D****Concept:** HTTP, FTP, SMTP, and SSH all use TCP.

11multicol multicolMultiple Choice

Topic: Network Fundamentals

Which devices operate at Layer 2? (Choose all that apply)

Switch

Bridge

Router

Hub

Correct Answer:**Correct Answer: A,B****Concept:** Switches and bridges operate at the data link layer.

12multicol multicolMultiple Choice

Topic: Network Fundamentals

Which of the following relate to TCP/IP? (Choose all that apply)

TCP/IP standard model

TCP/IP equivalent model

Correct Answer:**Correct Answer: A,B****Concept:** These concepts all relate to TCP/IP in Network Fundamentals.

47 OSI and TCP/IP Models

1multicol multicolMultiple Choice

Topic: OSI and TCP/IP Models

Which layers are part of the OSI model? (Choose all that apply)

Physical

Data link

Network

Transport

Application

Correct Answer:**Correct Answer: A,B,C,D,E****Concept:** The OSI model includes physical, data link, network, transport, session, presentation, and application layers.

48 Network Protocols

1multicol multicolMultiple Choice

Topic: Network Protocols

Which protocols operate at the application layer? (Choose all that apply)

HTTP

FTP
DNS
SMTP

Correct Answer:

Correct Answer: A,B,C,D

Concept: HTTP, FTP, DNS, and SMTP are all application-layer protocols.

49 Network Devices

1multicol multicolMultiple Choice

Topic: Network Devices

Which devices can operate at Layer 2? (Choose all that apply)

Switch
Bridge
Router
Hub

Correct Answer:

Correct Answer: A,B

Concept: Switches and bridges operate at the data link layer; routers are Layer 3 and hubs are Layer 1.

50 Enterprise Security Threats

1multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are examples of malware? (Choose all that apply)

Virus
Worm
Trojan
Ransomware
Spyware

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: All listed items are types of malware.

2multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are social engineering attacks? (Choose all that apply)

Phishing
Pretexting
Baiting
Tailgating

Correct Answer:

Correct Answer: A,B,C,D

Concept: Social engineering manipulates people through phishing, pretexting, baiting, tailgating, etc.

3multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which protect communication confidentiality? (Choose all that apply)

Encryption
VPN

TLS/SSL

MAC filtering

Correct Answer:

Correct Answer: A,B,C

Concept: Encryption, VPNs, and TLS/SSL protect confidentiality; MAC filtering controls device access.

4multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are typical security zones? (Choose all that apply)

Trust

Untrust

DMZ

Local

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls define Trust, Untrust, DMZ, and Local security zones by default.

5multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are endpoint security controls? (Choose all that apply)

Antivirus

Host firewall

Application whitelisting

Full-disk encryption

Correct Answer:

Correct Answer: A,B,C,D

Concept: Endpoint security includes antivirus, host firewalls, whitelisting, and encryption.

6multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are malware types? (Choose all that apply)

Virus

Worm

Trojan

Ransomware

Correct Answer:

Correct Answer: A,B,C,D

Concept: All listed are categories of malware.

7multicol multicolMultiple Choice

Topic: Enterprise Security Threats

Which are denial-of-service attacks? (Choose all that apply)

SYN flood

DDoS

Ping of death

Smurf attack

Correct Answer:

Correct Answer: A,B,C,D

Concept: These attacks aim to disrupt service availability.

51 Zone Border Security

1multicol multicolMultiple Choice

Topic: Zone Border Security

Which controls are typically applied at zone borders? (Choose all that apply)

Firewall policies

NAT

IPS/IDS

VPN termination

Correct Answer:

Correct Answer: A,B,C,D

Concept: Zone borders commonly use firewalls, NAT, IPS/IDS, and VPN gateways for protection.

52 Firewall Basics

1multicol multicolMultiple Choice

Topic: Firewall Basics

Which are Huawei firewall default security zones? (Choose all that apply)

Trust

Untrust

DMZ

Local

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

2multicol multicolMultiple Choice

Topic: Firewall Basics

Which actions can a security policy perform? (Choose all that apply)

Permit

Deny

Log

Redirect

Correct Answer:

Correct Answer: A,B,C

Concept: Policies can permit, deny, and log traffic; redirect may be available in specific features.

3multicol multicolMultiple Choice

Topic: Firewall Basics

Which are common firewall deployment modes? (Choose all that apply)

Routing mode

Transparent mode

Hybrid mode

Bridge mode

Correct Answer:

Correct Answer: A,B,C

Concept: Firewalls commonly deploy in routing, transparent (bridge), and hybrid modes.

4multicol multicolMultiple Choice

Topic: Firewall Basics

Which conditions can be matched in a security policy? (Choose all that apply)

Source zone

Destination zone

Source IP

Destination port

User

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, IPs, ports, users, applications, services, and time.

5multicol multicolMultiple Choice

Topic: Firewall Basics

Which are default Huawei security zones? (Choose all that apply)

Trust

Untrust

DMZ

Local

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

6multicol multicolMultiple Choice

Topic: Firewall Basics

Which can be matched in security policies? (Choose all that apply)

Source zone

Destination zone

Source IP

Destination port

User

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, addresses, ports, users, services, applications, and time.

7multicol multicolMultiple Choice

Topic: Firewall Basics

Which of the following relate to Security? (Choose all that apply)

Security zones

Security policy logging

Correct Answer:

Correct Answer: A,B

Concept: These concepts all relate to Security in Firewall Basics.

53 Security Zones

1multicol multicolMultiple Choice

Topic: Security Zones

Which are default security zones on Huawei firewalls? (Choose all that apply)

Trust

Untrust

DMZ

Local

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide four default zones: Trust, Untrust, DMZ, and Local.

54 Security Policies

1multicol multicolMultiple Choice

Topic: Security Policies

Which elements can be referenced in a security policy? (Choose all that apply)

Security zones

IP addresses/address groups

Services

Users and user groups

Time ranges

Correct Answer:

Correct Answer: A,B,C,D,E

Concept: Security policies can reference zones, addresses, services, users, applications, and time ranges.

55 Stateful Inspection

1multicol multicolMultiple Choice

Topic: Stateful Inspection

Which are advantages of stateful inspection over simple packet filtering? (Choose all that apply)

Tracks connection state

Allows return traffic automatically

Better performance for established sessions

Can inspect application-layer payloads

Correct Answer:

Correct Answer: A,B,C

Concept: Stateful inspection tracks connections, permits return traffic, and improves performance; deep payload inspection is more advanced.

56 NAT

1multicol multicolMultiple Choice

Topic: NAT

Which are private IPv4 address ranges? (Choose all that apply)

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

127.0.0.0/8

Correct Answer:

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10/8, 172.16/12, and 192.168/16.

2multicol multicolMultiple Choice

Topic: NAT

Which are valid uses of NAT Server? (Choose all that apply)

Publishing a web server

Publishing an email server

Remote access to an internal host

Hiding internal topology

Correct Answer:

Correct Answer: A,B,C,D

Concept: NAT Server can publish services, enable remote access, and obscure internal addressing.

3multicol multicolMultiple Choice

Topic: NAT

Which are NAT types? (Choose all that apply)

Source NAT

Destination NAT

Bidirectional NAT

NAT Server

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls support all four NAT types.

4multicol multicolMultiple Choice

Topic: NAT

Which source NAT modes exist? (Choose all that apply)

NAT No-PAT

NAPT

Easy-IP

NAT Server

Correct Answer:

Correct Answer: A,B,C

Concept: NAT Server is not a source NAT mode.

5multicol multicolMultiple Choice

Topic: NAT

Which of the following relate to NAT? (Choose all that apply)

NAT

NAT Server

NAT No-PAT

NAT ALG

Correct Answer:

Correct Answer: A,B,C,D

Concept: These concepts all relate to NAT in NAT.

57 NAT Basics

1multicol multicolMultiple Choice

Topic: NAT Basics

Which are types of NAT supported by Huawei firewalls? (Choose all that apply)

Source NAT

Destination NAT

Bidirectional NAT

NAT Server

Correct Answer:

Correct Answer: A,B,C,D

Concept: Huawei firewalls support source NAT, destination NAT, bidirectional NAT, and NAT server.

58 Source NAT

1multicol multicolMultiple Choice

Topic: Source NAT

Which are common source NAT modes? (Choose all that apply)

NAT No-PAT

NAPT

Easy-IP

NAT Server

Correct Answer:

Correct Answer: A,B,C

Concept: Source NAT includes NAT No-PAT (one-to-one), NAPT (many-to-one), and Easy-IP (interface-based).

59 NAT ALG

1multicol multicolMultiple Choice

Topic: NAT ALG

Which protocols may require NAT ALG? (Choose all that apply)

FTP

SIP

H.323

DNS

Correct Answer:

Correct Answer: A,B,C

Concept: Protocols embedding address/port information in payloads (FTP, SIP, H.323) need NAT ALG.

60 Firewall Hot Standby

1multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which are VRRP components? (Choose all that apply)

Virtual IP

Virtual MAC

Master router

Backup router

Correct Answer:

Correct Answer: A,B,C,D

Concept: VRRP uses a virtual IP, virtual MAC, master, and backup routers.

2multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which are HRP synchronization modes? (Choose all that apply)

Real-time

Fast

Batch

Auto

Correct Answer:

Correct Answer: A,B,C

Concept: HRP supports real-time, fast, and batch synchronization modes.

3multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which are valid hot-standby modes? (Choose all that apply)

Active/standby

Load sharing

Cluster

Standalone

Correct Answer:

Correct Answer: A,B

Concept: Huawei firewall hot standby supports active/standby and load-sharing modes.

4multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which items should be checked when HRP is not synchronizing? (Choose all that apply)

Heartbeat link status

HRP enable status

VGMP priority

Configuration consistency

Correct Answer:

Correct Answer: A,B,C,D

Concept: Check heartbeat links, HRP status, priorities, and configuration consistency.

5multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which are HRP synchronization contents? (Choose all that apply)

Session table

NAT entries

Server-map entries

Configuration

Correct Answer:

Correct Answer: A,B,C,D

Concept: HRP synchronizes all these items between peers.

6multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which are VRRP states? (Choose all that apply)

Master

Backup

Initialize

Failover

Correct Answer:

Correct Answer: A,B,C

Concept: VRRP states include master, backup, and initialize.

7multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which of the following relate to VRRP? (Choose all that apply)

VRRP

VRRP master

VRRP backup

Correct Answer:

Correct Answer: A,B,C

Concept: These concepts all relate to VRRP in Firewall Hot Standby.

8multicol multicolMultiple Choice

Topic: Firewall Hot Standby

Which of the following relate to HRP? (Choose all that apply)

HRP

HRP session backup

Correct Answer:**Correct Answer: A,B****Concept:** These concepts all relate to HRP in Firewall Hot Standby.

61 VRRP

1multicol multicolMultiple Choice

Topic: VRRP

Which are VRRP roles? (Choose all that apply)

Master

Backup

Initialize

Failover

Correct Answer:**Correct Answer: A,B,C****Concept:** VRRP devices can be in master, backup, or initialize states.

62 HRP

1multicol multicolMultiple Choice

Topic: HRP

Which data does HRP synchronize? (Choose all that apply)

Session table

NAT entries

Server-map entries

Configurations

Correct Answer:**Correct Answer: A,B,C,D****Concept:** HRP synchronizes sessions, NAT entries, server-map entries, and configurations.

63 Intrusion Prevention and Antivirus

1multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which are common attack phases? (Choose all that apply)

Reconnaissance

Exploitation

Lateral movement

Exfiltration

Correct Answer:**Correct Answer: A,B,C,D****Concept:** Cyber kill chain phases include reconnaissance, exploitation, lateral movement, and exfiltration.

2multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which are types of intrusion detection/prevention systems? (Choose all that apply)

NIDS
NIPS
HIDS
HIPS

Correct Answer:

Correct Answer: A,B,C,D

Concept: Network and host-based IDS/IPS exist for different monitoring scopes.

3multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which factors should be considered when tuning IPS signatures? (Choose all that apply)

False positive rate
False negative rate
Performance impact
Business context

Correct Answer:

Correct Answer: A,B,C,D

Concept: Effective tuning balances detection accuracy, performance, and business needs.

4multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which are methods to improve malware detection? (Choose all that apply)

Keep signatures updated
Use heuristics
Deploy sandboxing
User education

Correct Answer:

Correct Answer: A,B,C,D

Concept: Updated signatures, heuristics, sandboxing, and user awareness improve detection.

5multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which are IPS deployment modes? (Choose all that apply)

Inline
Tap/SPAN
Out-of-band
Bridge

Correct Answer:

Correct Answer: A,B,C

Concept: IPS can be inline, tapped, or out-of-band.

6multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which are antivirus detection methods? (Choose all that apply)

Signature-based
Heuristic
Sandboxing
Behavioral analysis

Correct Answer:

Correct Answer: A,B,C,D

Concept: Modern antivirus uses all these techniques.

7multicol multicolMultiple Choice

Topic: Intrusion Prevention and Antivirus

Which of the following relate to False? (Choose all that apply)

False positive

False negative

Correct Answer:

Correct Answer: A,B

Concept: These concepts all relate to False in Intrusion Prevention and Antivirus.

64 Intrusion Prevention

1multicol multicolMultiple Choice

Topic: Intrusion Prevention

Which are common IPS actions? (Choose all that apply)

Block

Permit

Reset connection

Alert/log

Correct Answer:

Correct Answer: A,C,D

Concept: IPS can block malicious traffic, reset connections, and alert/log events.

65 Antivirus

1multicol multicolMultiple Choice

Topic: Antivirus

Which are antivirus detection techniques? (Choose all that apply)

Signature-based

Heuristic

Sandboxing

Behavioral analysis

Correct Answer:

Correct Answer: A,B,C,D

Concept: Modern antivirus uses signatures, heuristics, sandboxing, and behavioral analysis.

66 AAA and User Authentication

1multicol multicolMultiple Choice

Topic: AAA and User Authentication

Which are AAA functions? (Choose all that apply)

Authentication

Authorization

Accounting

Alerting

Correct Answer:

Correct Answer: A,B,C

Concept: AAA provides authentication, authorization, and accounting.

2multicol multicolMultiple Choice

Topic: AAA and User Authentication

Which can be authentication sources on Huawei firewalls? (Choose all that apply)

Local database

RADIUS server
HWTACACS server
LDAP server

Correct Answer:

Correct Answer: A,B,C,D

Concept: Firewalls can authenticate against local, RADIUS, HWTACACS, and LDAP sources.

3multicol multicolMultiple Choice

Topic: AAA and User Authentication

Which are differences between RADIUS and TACACS+? (Choose all that apply)

RADIUS uses UDP; TACACS+ uses TCP
RADIUS combines auth/authz; TACACS+ separates them
RADIUS encrypts only password; TACACS+ encrypts entire payload
RADIUS is Cisco proprietary

Correct Answer:

Correct Answer: A,B,C

Concept: RADIUS uses UDP and combines auth/authz; TACACS+ uses TCP and separates AAA; only the password is encrypted in RADIUS.

4multicol multicolMultiple Choice

Topic: AAA and User Authentication

Which are AAA components? (Choose all that apply)

Authentication
Authorization
Accounting
Auditing

Correct Answer:

Correct Answer: A,B,C

Concept: AAA consists of authentication, authorization, and accounting.

5multicol multicolMultiple Choice

Topic: AAA and User Authentication

Which can authenticate users on a Huawei firewall? (Choose all that apply)

Local database
RADIUS
HWTACACS
LDAP

Correct Answer:

Correct Answer: A,B,C,D

Concept: Firewalls support all these authentication sources.

67 AAA Basics

1multicol multicolMultiple Choice

Topic: AAA Basics

Which are components of AAA? (Choose all that apply)

Authentication
Authorization
Accounting

Auditing

Correct Answer:**Correct Answer: A,B,C****Concept:** AAA consists of authentication, authorization, and accounting.

68 Firewall User Authentication

1multicol multicolMultiple Choice

Topic: Firewall User Authentication

Which are common firewall authentication methods? (Choose all that apply)

Local authentication

RADIUS

LDAP

HWTACACS

Correct Answer:**Correct Answer: A,B,C,D****Concept:** Firewalls support local, RADIUS, LDAP, and HWTACACS authentication.

69 Cryptography and PKI

1multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are asymmetric algorithms? (Choose all that apply)

RSA

ECC

DSA

AES

Correct Answer:**Correct Answer: A,B,C****Concept:** RSA, ECC, and DSA are asymmetric; AES is symmetric.

2multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are block cipher modes? (Choose all that apply)

ECB

CBC

CTR

GCM

Correct Answer:**Correct Answer: A,B,C,D****Concept:** ECB, CBC, CTR, and GCM are all AES block cipher modes.

3multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are secure hash algorithms? (Choose all that apply)

SHA-256

SHA-3

BLAKE2

MD5

Correct Answer:

Correct Answer: A,B,C

Concept: SHA-256, SHA-3, and BLAKE2 are considered secure; MD5 is broken.

4multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are PKI trust models? (Choose all that apply)

Single CA

Hierarchical CA

Cross-certification

Bridge CA

Correct Answer:

Correct Answer: A,B,C,D

Concept: PKI can use single, hierarchical, cross-certified, or bridge CA models.

5multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which fields are typically found in an X.509 certificate? (Choose all that apply)

Subject

Issuer

Public key

Validity period

Correct Answer:

Correct Answer: A,B,C,D

Concept: X.509 certificates contain subject, issuer, public key, validity, serial number, and signature.

6multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are symmetric algorithms? (Choose all that apply)

AES

DES

3DES

RSA

Correct Answer:

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

7multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are hash algorithms? (Choose all that apply)

MD5

SHA-256

SHA-1

SHA-3

Correct Answer:

Correct Answer: A,B,C,D

Concept: These are all hash algorithms, though MD5 and SHA-1 are deprecated for security use.

8multicol multicolMultiple Choice

Topic: Cryptography and PKI

Which are PKI components? (Choose all that apply)

CA

RA

Repository
CRL/OCSP

Correct Answer:

Correct Answer: A,B,C,D

Concept: PKI includes certificate authorities, registration authorities, repositories, and revocation systems.

70 Cryptography Basics

1multicol multicolMultiple Choice

Topic: Cryptography Basics

Which are symmetric encryption algorithms? (Choose all that apply)

AES
DES
3DES
RSA

Correct Answer:

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

71 PKI

1multicol multicolMultiple Choice

Topic: PKI

Which are core PKI components? (Choose all that apply)

Certificate Authority
Registration Authority
Certificate repository
CRL/OCSP

Correct Answer:

Correct Answer: A,B,C,D

Concept: PKI includes CAs, RAs, repositories, and revocation mechanisms such as CRL and OCSP.

72 VPN

1multicol multicolMultiple Choice

Topic: VPN

Which are benefits of using a VPN? (Choose all that apply)

Confidentiality
Data integrity
Authentication
Secure remote access

Correct Answer:

Correct Answer: A,B,C,D

Concept: VPNs provide confidentiality, integrity, authentication, and secure remote access.

2multicol multicolMultiple Choice

Topic: VPN

Which are IPsec security services? (Choose all that apply)

Confidentiality

Integrity

Authentication

Anti-replay

Correct Answer:

Correct Answer: A,B,C,D

Concept: IPsec provides confidentiality, integrity, authentication, and anti-replay protection.

3multicol multicolMultiple Choice

Topic: VPN

Which are phases of IKEv1? (Choose all that apply)

Phase 1

Phase 2

Phase 3

Phase 1.5

Correct Answer:

Correct Answer: A,B

Concept: IKEv1 has two phases: Phase 1 (IKE SA) and Phase 2 (IPsec SA).

4multicol multicolMultiple Choice

Topic: VPN

Which protocols operate at the IP layer? (Choose all that apply)

AH

ESP

GRE

L2TP

Correct Answer:

Correct Answer: A,B

Concept: AH and ESP are IP-layer protocols; GRE is also IP-layer but L2TP is not.

5multicol multicolMultiple Choice

Topic: VPN

Which are VPN types? (Choose all that apply)

Site-to-site

Remote-access

Intranet

Extranet

Correct Answer:

Correct Answer: A,B,C,D

Concept: VPNs can be categorized by these deployment models.

6multicol multicolMultiple Choice

Topic: VPN

Which are IPsec protocols? (Choose all that apply)

AH

ESP

IKE

GRE

Correct Answer:

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are IPsec protocols; GRE is separate.

7multicol multicolMultiple Choice

Topic: VPN

Which are SSL VPN access modes? (Choose all that apply)

Web proxy
File sharing
Network extension
Port forwarding

Correct Answer:

Correct Answer: A,B,C,D

Concept: SSL VPN commonly supports all these modes.

8multicol multicolMultiple Choice

Topic: VPN

Which of the following relate to GRE? (Choose all that apply)

GRE

GRE

Correct Answer:

Correct Answer: A,B

Concept: These concepts all relate to GRE in VPN.

9multicol multicolMultiple Choice

Topic: VPN

Which of the following relate to L2TP? (Choose all that apply)

L2TP

L2TP over IPsec

Correct Answer:

Correct Answer: A,B

Concept: These concepts all relate to L2TP in VPN.

73 VPN Overview

1multicol multicolMultiple Choice

Topic: VPN Overview

Which are common VPN types? (Choose all that apply)

Site-to-site VPN

Remote-access VPN

Intranet VPN

Extranet VPN

Correct Answer:

Correct Answer: A,B,C,D

Concept: VPNs can be site-to-site, remote-access, intranet, or extranet based on deployment.

74 IPsec VPN

1multicol multicolMultiple Choice

Topic: IPsec VPN

Which are IPsec protocols? (Choose all that apply)

AH

ESP

IKE

GRE

Correct Answer:

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are part of IPsec; GRE is a separate tunneling protocol.

75 SSL VPN

1multicolMultiple Choice

Topic: SSL VPN

Which are common SSL VPN access modes? (Choose all that apply)

Web proxy

File sharing

Network extension

Port forwarding

Correct Answer:

Correct Answer: A,B,C,D

Concept: SSL VPN can offer web proxy, file sharing, network extension, and port forwarding.

PART III

True / False Questions

76 Network Security Concepts

1tfcolTrue / False

Topic: Network Security Concepts

Network security in the narrow sense only refers to national cybersecurity laws.

True

False

Correct Answer:

Correct Answer: False

Concept: Narrow-sense network security covers vendor solutions and enterprise network protection, not only national laws.

2tfcolTrue / False

Topic: Network Security Concepts

The CIA triad includes confidentiality, integrity, and availability.

True

False

Correct Answer:

Correct Answer: True

Concept: Confidentiality, integrity, and availability are the three foundational goals of information security.

3tfcolTrue / False

Topic: Network Security Concepts

Building network security helps enterprises reduce losses from attacks.

True

False

Correct Answer:

Correct Answer: True

Concept: Security controls reduce the impact of breaches, downtime, and data loss, thereby lowering financial losses.

4tfcolTrue / False

Topic: Network Security Concepts

Non-repudiation ensures that senders or receivers cannot deny their actions.

True

False

Correct Answer:

Correct Answer: True

Concept: Non-repudiation provides evidence to prevent denial of participation.

5tfcol tfcolTrue / False

Topic: Network Security Concepts

Availability means anyone can access the system at any time.

True

False

Correct Answer:

Correct Answer: False

Concept: Availability means authorized users can access information when required, not unlimited access.

6tfcol tfcolTrue / False

Topic: Network Security Concepts

Zero trust means everything inside the network is trusted by default.

True

False

Correct Answer:

Correct Answer: False

Concept: Zero trust never assumes trust based on network location and verifies every request.

7tfcol tfcolTrue / False

Topic: Network Security Concepts

Integrity ensures that data is not tampered with during transmission.

True

False

Correct Answer:

Correct Answer: True

Concept: Integrity protects against unauthorized modification or destruction of data.

8tfcol tfcolTrue / False

Topic: Network Security Concepts

Network security broadly refers to cyberspace security at a national level.

True

False

Correct Answer:

Correct Answer: True

Concept: Broad network security covers national cyberspace protection.

9tfcol tfcolTrue / False

Topic: Network Security Concepts

The CIA triad excludes availability.

True

False

Correct Answer:

Correct Answer: False

Concept: The CIA triad includes confidentiality, integrity, and availability.

10tfcol tfcolTrue / False

Topic: Network Security Concepts

Risk is the combination of threat likelihood and impact.

True

False

Correct Answer:

Correct Answer: True

Concept: Risk assesses probability and consequence of security incidents.

11tfcol tfcolTrue / False

Topic: Network Security Concepts

Broad network security means that is synonymous with national cyberspace security.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Broad network security.

12tfcol tfcolTrue / False

Topic: Network Security Concepts

Broad network security is unrelated to is synonymous with national cyberspace security.

True

False

Correct Answer:

Correct Answer: False

Concept: Broad network security is directly related to is synonymous with national cyberspace security.

13tfcol tfcolTrue / False

Topic: Network Security Concepts

Narrow network security means that focuses on enterprise network protection solutions.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Narrow network security.

14tfcol tfcolTrue / False

Topic: Network Security Concepts

Narrow network security is unrelated to focuses on enterprise network protection solutions.

True

False

Correct Answer:

Correct Answer: False

Concept: Narrow network security is directly related to focuses on enterprise network protection solutions.

15tfcol tfcolTrue / False

Topic: Network Security Concepts

CIA triad means that consists of confidentiality, integrity, and availability.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes CIA triad.

16tfcol tfcolTrue / False

Topic: Network Security Concepts

CIA triad is unrelated to consists of confidentiality, integrity, and availability.

True

False

Correct Answer:

Correct Answer: False

Concept: CIA triad is directly related to consists of confidentiality, integrity, and availability.

17tfcol tfcolTrue / False

Topic: Network Security Concepts

Information assurance period means that added controllability and non-repudiation.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Information assurance period.

18tfcol tfcolTrue / False

Topic: Network Security Concepts

Information assurance period is unrelated to added controllability and non-repudiation.

True

False

Correct Answer:

Correct Answer: False

Concept: Information assurance period is directly related to added controllability and non-repudiation.

19tfcol tfcolTrue / False

Topic: Network Security Concepts

Cyberspace security means that protects facilities, data, users, and operations.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Cyberspace security.

20tfcol tfcolTrue / False

Topic: Network Security Concepts

Cyberspace security is unrelated to protects facilities, data, users, and operations.

True

False

Correct Answer:

Correct Answer: False

Concept: Cyberspace security is directly related to protects facilities, data, users, and operations.

21tfcol tfcolTrue / False

Topic: Network Security Concepts

Confidentiality means that ensures information is accessible only to authorized users.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Confidentiality.

22tfcol tfcolTrue / False

Topic: Network Security Concepts

Confidentiality is unrelated to ensures information is accessible only to authorized users.

True

False

Correct Answer:

Correct Answer: False

Concept: Confidentiality is directly related to ensures information is accessible only to authorized users.

23tfcot tfcolTrue / False

Topic: Network Security Concepts

Integrity means that ensures data is not tampered with during transmission or storage.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Integrity.

24tfcot tfcolTrue / False

Topic: Network Security Concepts

Integrity is unrelated to ensures data is not tampered with during transmission or storage.

True

False

Correct Answer:

Correct Answer: False

Concept: Integrity is directly related to ensures data is not tampered with during transmission or storage.

25tfcot tfcolTrue / False

Topic: Network Security Concepts

Availability means that ensures authorized users can access information when needed.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Availability.

26tfcot tfcolTrue / False

Topic: Network Security Concepts

Availability is unrelated to ensures authorized users can access information when needed.

True

False

Correct Answer:

Correct Answer: False

Concept: Availability is directly related to ensures authorized users can access information when needed.

27tfcot tfcolTrue / False

Topic: Network Security Concepts

Controllability means that enables monitoring and control of information and systems.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Controllability.

28tfcot tfcolTrue / False

Topic: Network Security Concepts

Controllability is unrelated to enables monitoring and control of information and systems.

True

False

Correct Answer:

Correct Answer: False

Concept: Controllability is directly related to enables monitoring and control of information and systems.

29tfcot tfcolTrue / False

Topic: Network Security Concepts

Non-repudiation means that prevents senders or receivers from denying actions.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Non-repudiation.

30tfcot tfcolTrue / False

Topic: Network Security Concepts

Non-repudiation is unrelated to prevents senders or receivers from denying actions.

True

False

Correct Answer:

Correct Answer: False

Concept: Non-repudiation is directly related to prevents senders or receivers from denying actions.

31tfcot tfcolTrue / False

Topic: Network Security Concepts

Risk means that is the combination of threat likelihood and impact.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Risk.

32tfcot tfcolTrue / False

Topic: Network Security Concepts

Risk is unrelated to is the combination of threat likelihood and impact.

True

False

Correct Answer:

Correct Answer: False

Concept: Risk is directly related to is the combination of threat likelihood and impact.

33tfcot tfcolTrue / False

Topic: Network Security Concepts

Vulnerability means that is a weakness that can be exploited by threats.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Vulnerability.

34tfcot tfcolTrue / False

Topic: Network Security Concepts

Vulnerability is unrelated to is a weakness that can be exploited by threats.

True

False

Correct Answer:

Correct Answer: False

Concept: Vulnerability is directly related to is a weakness that can be exploited by threats.

35tfcot tfcolTrue / False

Topic: Network Security Concepts

Threat means that is a potential danger to an asset.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Threat.

36tfcot tfcolTrue / False

Topic: Network Security Concepts

Threat is unrelated to is a potential danger to an asset.

True

False

Correct Answer:

Correct Answer: False

Concept: Threat is directly related to is a potential danger to an asset.

37tfcot tfcolTrue / False

Topic: Network Security Concepts

Exploit means that is a method that takes advantage of a vulnerability.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Exploit.

38tfcot tfcolTrue / False

Topic: Network Security Concepts

Exploit is unrelated to is a method that takes advantage of a vulnerability.

True

False

Correct Answer:

Correct Answer: False

Concept: Exploit is directly related to is a method that takes advantage of a vulnerability.

39tfcot tfcolTrue / False

Topic: Network Security Concepts

Future security trend means that includes zero-trust architecture.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Future security trend.

40tfcot tfcolTrue / False

Topic: Network Security Concepts

Future security trend is unrelated to includes zero-trust architecture.

True

False

Correct Answer:

Correct Answer: False

Concept: Future security trend is directly related to includes zero-trust architecture.

41tfcot tfcolTrue / False

Topic: Network Security Concepts

AI and machine learning means that help detect unknown threats and anomalies.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes AI and machine learning.

42tfcot tfcolTrue / False

Topic: Network Security Concepts

AI and machine learning is unrelated to help detect unknown threats and anomalies.

True

False

Correct Answer:

Correct Answer: False

Concept: AI and machine learning is directly related to help detect unknown threats and anomalies.

77 Network Security Standards

1tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27002 can be used to certify an enterprise information security system.

True

False

Correct Answer:

Correct Answer: False

Concept: ISO 27001 is the certification standard; ISO 27002 is a code of practice with controls.

2tfcot tfcolTrue / False

Topic: Network Security Standards

Classified Protection 2.0 defines protection levels from Level 1 to Level 5.

True

False

Correct Answer:

Correct Answer: True

Concept: Classified Protection 2.0 uses five levels, with higher levels requiring stronger controls and oversight.

3tfcot tfcolTrue / False

Topic: Network Security Standards

Network operators can always determine the final protection level on their own without review.

True

False

Correct Answer:

Correct Answer: False

Concept: For Level 2 and above, expert review and approval are required; only Level 1 can be determined by the operator.

4tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27001 provides a list of security controls but is not certifiable.

True

False

Correct Answer:

Correct Answer: False

Concept: ISO 27001 is the certifiable standard specifying ISMS requirements; ISO 27002 provides controls.

5tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection 2.0 requires security and informatization to be planned together.

True

False

Correct Answer:

Correct Answer: True

Concept: The framework emphasizes synchronized planning, construction, and use of security.

6tfcol tfcolTrue / False

Topic: Network Security Standards

Level assessment is part of the Classified Protection lifecycle.

True

False

Correct Answer:

Correct Answer: True

Concept: After construction, a level assessment verifies that protection measures meet the required level.

7tfcol tfcolTrue / False

Topic: Network Security Standards

ISO 27001 is certifiable.

True

False

Correct Answer:

Correct Answer: True

Concept: Organizations can be certified against ISO 27001 requirements.

8tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection 2.0 has only three levels.

True

False

Correct Answer:

Correct Answer: False

Concept: Classified Protection 2.0 defines five levels.

9tfcol tfcolTrue / False

Topic: Network Security Standards

ISO 27001 means that specifies certifiable ISMS requirements.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ISO 27001.

10tfcol tfcolTrue / False

Topic: Network Security Standards

ISO 27001 is unrelated to specifies certifiable ISMS requirements.

True

False

Correct Answer:

Correct Answer: False

Concept: ISO 27001 is directly related to specifies certifiable ISMS requirements.

11tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27002 means that provides a code of practice for security controls.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ISO 27002.

12tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27002 is unrelated to provides a code of practice for security controls.

True

False

Correct Answer:

Correct Answer: False

Concept: ISO 27002 is directly related to provides a code of practice for security controls.

13tfcot tfcolTrue / False

Topic: Network Security Standards

PDCA means that is the continuous improvement methodology used in ISO 27001.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes PDCA.

14tfcot tfcolTrue / False

Topic: Network Security Standards

PDCA is unrelated to is the continuous improvement methodology used in ISO 27001.

True

False

Correct Answer:

Correct Answer: False

Concept: PDCA is directly related to is the continuous improvement methodology used in ISO 27001.

15tfcot tfcolTrue / False

Topic: Network Security Standards

Classified Protection 2.0 means that is a Chinese national cybersecurity framework.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Classified Protection 2.0.

16tfcot tfcolTrue / False

Topic: Network Security Standards

Classified Protection 2.0 is unrelated to is a Chinese national cybersecurity framework.

True

False

Correct Answer:

Correct Answer: False

Concept: Classified Protection 2.0 is directly related to is a Chinese national cybersecurity framework.

17tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection levels means that range from Level 1 to Level 5.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Classified Protection levels.

18tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection levels is unrelated to range from Level 1 to Level 5.

True

False

Correct Answer:

Correct Answer: False

Concept: Classified Protection levels is directly related to range from Level 1 to Level 5.

19tfcol tfcolTrue / False

Topic: Network Security Standards

Level 2 and above means that require expert review and approval.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Level 2 and above.

20tfcol tfcolTrue / False

Topic: Network Security Standards

Level 2 and above is unrelated to require expert review and approval.

True

False

Correct Answer:

Correct Answer: False

Concept: Level 2 and above is directly related to require expert review and approval.

21tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection lifecycle means that includes classification, filing, construction, assessment, and supervision.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Classified Protection lifecycle.

22tfcol tfcolTrue / False

Topic: Network Security Standards

Classified Protection lifecycle is unrelated to includes classification, filing, construction, assessment, and supervision.

True

False

Correct Answer:

Correct Answer: False

Concept: Classified Protection lifecycle is directly related to includes classification, filing, construction, assessment, and supervision.

23tfcot tfcolTrue / False

Topic: Network Security Standards

Security synchronization means that means planning security with information system construction.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Security synchronization.

24tfcot tfcolTrue / False

Topic: Network Security Standards

Security synchronization is unrelated to means planning security with information system construction.

True

False

Correct Answer:

Correct Answer: False

Concept: Security synchronization is directly related to means planning security with information system construction.

25tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27001:2022 means that organizes controls into four themes.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ISO 27001:2022.

26tfcot tfcolTrue / False

Topic: Network Security Standards

ISO 27001:2022 is unrelated to organizes controls into four themes.

True

False

Correct Answer:

Correct Answer: False

Concept: ISO 27001:2022 is directly related to organizes controls into four themes.

27tfcot tfcolTrue / False

Topic: Network Security Standards

Cybersecurity Law means that underpins Classified Protection in China.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Cybersecurity Law.

28tfcot tfcolTrue / False

Topic: Network Security Standards

Cybersecurity Law is unrelated to underpins Classified Protection in China.

True

False

Correct Answer:

Correct Answer: False

Concept: Cybersecurity Law is directly related to underpins Classified Protection in China.

78 Network Fundamentals

1tfcol tfcolTrue / False

Topic: Network Fundamentals

The OSI model was developed before the TCP/IP model.

True

False

Correct Answer:

Correct Answer: False

Concept: TCP/IP was developed earlier and became the de facto standard; OSI was later as a reference model.

2tfcol tfcolTrue / False

Topic: Network Fundamentals

DNS always uses TCP port 53.

True

False

Correct Answer:

Correct Answer: False

Concept: DNS primarily uses UDP port 53; TCP port 53 is used for zone transfers and large responses.

3tfcol tfcolTrue / False

Topic: Network Fundamentals

A switch forwards packets based on IP addresses.

True

False

Correct Answer:

Correct Answer: False

Concept: A switch forwards frames based on MAC addresses; routers forward packets based on IP addresses.

4tfcol tfcolTrue / False

Topic: Network Fundamentals

IPv4 addresses are 32 bits long.

True

False

Correct Answer:

Correct Answer: True

Concept: IPv4 addresses are 32-bit binary numbers, usually written in dotted-decimal notation.

5tfcol tfcolTrue / False

Topic: Network Fundamentals

ICMP is a transport-layer protocol.

True

False

Correct Answer:

Correct Answer: False

Concept: ICMP is a network-layer protocol used for diagnostics and error reporting.

6tfcol tfcolTrue / False

Topic: Network Fundamentals

TCP provides connection-oriented reliable delivery.

True

False

Correct Answer:

Correct Answer: True

Concept: TCP establishes connections and ensures reliable, ordered delivery.

7tfcol tfcolTrue / False

Topic: Network Fundamentals

A hub creates separate collision domains per port.

True

False

Correct Answer:

Correct Answer: False

Concept: A hub shares a single collision domain; switches create separate collision domains.

8tfcol tfcolTrue / False

Topic: Network Fundamentals

IPv6 addresses are 128 bits long.

True

False

Correct Answer:

Correct Answer: True

Concept: IPv6 addresses are 128-bit binary numbers.

9tfcol tfcolTrue / False

Topic: Network Fundamentals

OSI model means that has seven layers.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes OSI model.

10tfcol tfcolTrue / False

Topic: Network Fundamentals

OSI model is unrelated to has seven layers.

True

False

Correct Answer:

Correct Answer: False

Concept: OSI model is directly related to has seven layers.

11tfcol tfcolTrue / False

Topic: Network Fundamentals

TCP/IP standard model means that has four layers.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes TCP/IP standard model.

12tfcol tfcolTrue / False

Topic: Network Fundamentals

TCP/IP standard model is unrelated to has four layers.

True

False

Correct Answer:

Correct Answer: False

Concept: TCP/IP standard model is directly related to has four layers.

13tfcot tfcolTrue / False

TCP/IP equivalent model means that has five layers.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes TCP/IP equivalent model.

14tfcot tfcolTrue / False

TCP/IP equivalent model is unrelated to has five layers.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: TCP/IP equivalent model is directly related to has five layers.

15tfcot tfcolTrue / False

Physical layer means that transmits raw bits over a medium.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Physical layer.

16tfcot tfcolTrue / False

Physical layer is unrelated to transmits raw bits over a medium.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: Physical layer is directly related to transmits raw bits over a medium.

17tfcot tfcolTrue / False

Data link layer means that handles framing and MAC addressing.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Data link layer.

18tfcot tfcolTrue / False

Data link layer is unrelated to handles framing and MAC addressing.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: Data link layer is directly related to handles framing and MAC addressing.

19tfcot tfcolTrue / False

Topic: Network Fundamentals

Network layer means that handles logical addressing and routing.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Network layer.

20tfcot tfcolTrue / False

Topic: Network Fundamentals

Network layer is unrelated to handles logical addressing and routing.

True

False

Correct Answer:

Correct Answer: False

Concept: Network layer is directly related to handles logical addressing and routing.

21tfcot tfcolTrue / False

Topic: Network Fundamentals

Transport layer means that provides end-to-end communication and reliability.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Transport layer.

22tfcot tfcolTrue / False

Topic: Network Fundamentals

Transport layer is unrelated to provides end-to-end communication and reliability.

True

False

Correct Answer:

Correct Answer: False

Concept: Transport layer is directly related to provides end-to-end communication and reliability.

23tfcot tfcolTrue / False

Topic: Network Fundamentals

Session layer means that manages application dialogs.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Session layer.

24tfcot tfcolTrue / False

Topic: Network Fundamentals

Session layer is unrelated to manages application dialogs.

True

False

Correct Answer:

Correct Answer: False

Concept: Session layer is directly related to manages application dialogs.

25tfcol tfcolTrue / False

Topic: Network Fundamentals

Presentation layer means that handles encryption and data format conversion.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Presentation layer.

26tfcol tfcolTrue / False

Topic: Network Fundamentals

Presentation layer is unrelated to handles encryption and data format conversion.

True

False

Correct Answer:

Correct Answer: False

Concept: Presentation layer is directly related to handles encryption and data format conversion.

27tfcol tfcolTrue / False

Topic: Network Fundamentals

Application layer means that provides network services to applications.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Application layer.

28tfcol tfcolTrue / False

Topic: Network Fundamentals

Application layer is unrelated to provides network services to applications.

True

False

Correct Answer:

Correct Answer: False

Concept: Application layer is directly related to provides network services to applications.

29tfcol tfcolTrue / False

Topic: Network Fundamentals

TCP means that is connection-oriented and reliable.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes TCP.

30tfcol tfcolTrue / False

Topic: Network Fundamentals

TCP is unrelated to is connection-oriented and reliable.

True

False

Correct Answer:

Correct Answer: False

Concept: TCP is directly related to is connection-oriented and reliable.

31tfcol tfcolTrue / False

Topic: Network Fundamentals

UDP means that is connectionless and has low overhead.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes UDP.

32tfcol tfcolTrue / False

Topic: Network Fundamentals

UDP is unrelated to is connectionless and has low overhead.

True

False

Correct Answer:

Correct Answer: False

Concept: UDP is directly related to is connectionless and has low overhead.

33tfcol tfcolTrue / False

Topic: Network Fundamentals

FTP means that uses TCP ports 20 and 21.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes FTP.

34tfcol tfcolTrue / False

Topic: Network Fundamentals

FTP is unrelated to uses TCP ports 20 and 21.

True

False

Correct Answer:

Correct Answer: False

Concept: FTP is directly related to uses TCP ports 20 and 21.

35tfcol tfcolTrue / False

Topic: Network Fundamentals

HTTP means that uses TCP port 80.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HTTP.

36tfcol tfcolTrue / False

Topic: Network Fundamentals

HTTP is unrelated to uses TCP port 80.

True

False

Correct Answer:

Correct Answer: False

Concept: HTTP is directly related to uses TCP port 80.

37tfcol tfcolTrue / False

Topic: Network Fundamentals

HTTPS means that uses TCP port 443.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HTTPS.

38tfcol tfcolTrue / False

Topic: Network Fundamentals

HTTPS is unrelated to uses TCP port 443.

True

False

Correct Answer:

Correct Answer: False

Concept: HTTPS is directly related to uses TCP port 443.

39tfcol tfcolTrue / False

Topic: Network Fundamentals

SSH means that uses TCP port 22.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SSH.

40tfcol tfcolTrue / False

Topic: Network Fundamentals

SSH is unrelated to uses TCP port 22.

True

False

Correct Answer:

Correct Answer: False

Concept: SSH is directly related to uses TCP port 22.

41tfcol tfcolTrue / False

Topic: Network Fundamentals

Telnet means that uses TCP port 23 and is insecure.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Telnet.

42tfcol tfcolTrue / False

Topic: Network Fundamentals

Telnet is unrelated to uses TCP port 23 and is insecure.

True

False

Correct Answer:

Correct Answer: False

Concept: Telnet is directly related to uses TCP port 23 and is insecure.

43tfcol tfcolTrue / False

DNS means that primarily uses UDP port 53.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DNS.

44tfcol tfcolTrue / False

DNS is unrelated to primarily uses UDP port 53.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: DNS is directly related to primarily uses UDP port 53.

45tfcol tfcolTrue / False

SMTP means that uses TCP port 25.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SMTP.

46tfcol tfcolTrue / False

SMTP is unrelated to uses TCP port 25.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: SMTP is directly related to uses TCP port 25.

47tfcol tfcolTrue / False

DHCP means that dynamically assigns IP addresses.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DHCP.

48tfcol tfcolTrue / False

DHCP is unrelated to dynamically assigns IP addresses.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: DHCP is directly related to dynamically assigns IP addresses.

49tfcol tfcolTrue / False

Topic: Network Fundamentals

ARP means that maps IP addresses to MAC addresses.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ARP.

50tfcol tfcolTrue / False

Topic: Network Fundamentals

ARP is unrelated to maps IP addresses to MAC addresses.

True

False

Correct Answer:

Correct Answer: False

Concept: ARP is directly related to maps IP addresses to MAC addresses.

51tfcol tfcolTrue / False

Topic: Network Fundamentals

ICMP means that provides error reporting and diagnostics.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ICMP.

52tfcol tfcolTrue / False

Topic: Network Fundamentals

ICMP is unrelated to provides error reporting and diagnostics.

True

False

Correct Answer:

Correct Answer: False

Concept: ICMP is directly related to provides error reporting and diagnostics.

53tfcol tfcolTrue / False

Topic: Network Fundamentals

IPv4 address means that is 32 bits long.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IPv4 address.

54tfcol tfcolTrue / False

Topic: Network Fundamentals

IPv4 address is unrelated to is 32 bits long.

True

False

Correct Answer:

Correct Answer: False

Concept: IPv4 address is directly related to is 32 bits long.

55tfcol tfcolTrue / False

IPv6 address means that is 128 bits long.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IPv6 address.

56tfcol tfcolTrue / False

IPv6 address is unrelated to is 128 bits long.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: IPv6 address is directly related to is 128 bits long.

57tfcol tfcolTrue / False

MAC address means that is a Layer 2 hardware address.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes MAC address.

58tfcol tfcolTrue / False

MAC address is unrelated to is a Layer 2 hardware address.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: MAC address is directly related to is a Layer 2 hardware address.

59tfcol tfcolTrue / False

Switch means that operates at Layer 2 and forwards frames by MAC.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Switch.

60tfcol tfcolTrue / False

Switch is unrelated to operates at Layer 2 and forwards frames by MAC.

Topic: Network Fundamentals

True

False

Correct Answer:

Correct Answer: False

Concept: Switch is directly related to operates at Layer 2 and forwards frames by MAC.

61tfcot tfcolTrue / False

Topic: Network Fundamentals

Router means that operates at Layer 3 and forwards packets by IP.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Router.

62tfcot tfcolTrue / False

Topic: Network Fundamentals

Router is unrelated to operates at Layer 3 and forwards packets by IP.

True

False

Correct Answer:

Correct Answer: False

Concept: Router is directly related to operates at Layer 3 and forwards packets by IP.

63tfcot tfcolTrue / False

Topic: Network Fundamentals

Hub means that operates at Layer 1 and repeats signals.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Hub.

64tfcot tfcolTrue / False

Topic: Network Fundamentals

Hub is unrelated to operates at Layer 1 and repeats signals.

True

False

Correct Answer:

Correct Answer: False

Concept: Hub is directly related to operates at Layer 1 and repeats signals.

65tfcot tfcolTrue / False

Topic: Network Fundamentals

VLAN means that logically segments a network into broadcast domains.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VLAN.

66tfcot tfcolTrue / False

Topic: Network Fundamentals

VLAN is unrelated to logically segments a network into broadcast domains.

True

False

Correct Answer:

Correct Answer: False

Concept: VLAN is directly related to logically segments a network into broadcast domains.

67tfcot tfcolTrue / False

Topic: Network Fundamentals

STP means that prevents Layer 2 loops.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes STP.

68tfcot tfcolTrue / False

Topic: Network Fundamentals

STP is unrelated to prevents Layer 2 loops.

True

False

Correct Answer:

Correct Answer: False

Concept: STP is directly related to prevents Layer 2 loops.

69tfcot tfcolTrue / False

Topic: Network Fundamentals

RFC 1918 means that defines private IPv4 address ranges.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes RFC 1918.

70tfcot tfcolTrue / False

Topic: Network Fundamentals

RFC 1918 is unrelated to defines private IPv4 address ranges.

True

False

Correct Answer:

Correct Answer: False

Concept: RFC 1918 is directly related to defines private IPv4 address ranges.

79 OSI and TCP/IP Models

1tfcot tfcolTrue / False

Topic: OSI and TCP/IP Models

The TCP/IP model combines the OSI session and presentation layers into the application layer.

True

False

Correct Answer:

Correct Answer: True

Concept: The TCP/IP model does not have separate session and presentation layers; their functions are included in the application layer.

80 Network Protocols

1tfcol tfcolTrue / False

Topic: Network Protocols

UDP guarantees ordered delivery of packets.

True

False

Correct Answer:

Correct Answer: False

Concept: UDP is connectionless and does not guarantee delivery, ordering, or duplicate protection.

81 Network Devices

1tfcol tfcolTrue / False

Topic: Network Devices

A router forwards frames based on MAC addresses.

True

False

Correct Answer:

Correct Answer: False

Concept: Routers forward packets based on IP addresses; switches forward frames based on MAC addresses.

2tfcol tfcolTrue / False

Topic: Network Devices

A switch creates a separate collision domain for each port.

True

False

Correct Answer:

Correct Answer: True

Concept: Each switch port is its own collision domain, reducing collisions compared to a hub.

82 Enterprise Security Threats

1tfcol tfcolTrue / False

Topic: Enterprise Security Threats

A worm requires a host program to spread.

True

False

Correct Answer:

Correct Answer: False

Concept: Worms are standalone malware that can self-replicate; viruses require host programs.

2tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Phishing attacks target human psychology rather than software vulnerabilities.

True

False

Correct Answer:

Correct Answer: True

Concept: Phishing uses deception and social engineering to trick users.

3tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Using HTTP instead of HTTPS improves confidentiality.

True

False

Correct Answer:

Correct Answer: False

Concept: HTTP sends data in plaintext; HTTPS encrypts data to protect confidentiality.

4tfcol tfcolTrue / False

Topic: Enterprise Security Threats

The DMZ zone is typically more trusted than the Untrust zone but less trusted than the Trust zone.

True

False

Correct Answer:

Correct Answer: True

Concept: DMZ has an intermediate trust level for public-facing servers.

5tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Patch management helps eliminate known vulnerabilities on hosts.

True

False

Correct Answer:

Correct Answer: True

Concept: Applying patches closes security holes that attackers could exploit.

6tfcol tfcolTrue / False

Topic: Enterprise Security Threats

All security threats originate from external attackers.

True

False

Correct Answer:

Correct Answer: False

Concept: Threats can be external or internal, accidental or malicious.

7tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Phishing is a type of social engineering.

True

False

Correct Answer:

Correct Answer: True

Concept: Phishing manipulates users into revealing information.

8tfcol tfcolTrue / False

Topic: Enterprise Security Threats

A firewall at the network border can enforce zone-based policies.

True

False

Correct Answer:

Correct Answer: True

Concept: Firewalls enforce policies between security zones.

9tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Least privilege reduces the impact of compromised accounts.

True

False

Correct Answer:

Correct Answer: True

Concept: Limiting access limits damage from compromised credentials.

10tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Passive attack means that monitors traffic without altering it.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Passive attack.

11tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Passive attack is unrelated to monitors traffic without altering it.

True

False

Correct Answer:

Correct Answer: False

Concept: Passive attack is directly related to monitors traffic without altering it.

12tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Active attack means that modifies or disrupts systems or data.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Active attack.

13tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Active attack is unrelated to modifies or disrupts systems or data.

True

False

Correct Answer:

Correct Answer: False

Concept: Active attack is directly related to modifies or disrupts systems or data.

14tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Malware means that includes viruses, worms, Trojans, and ransomware.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Malware.

15tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Malware is unrelated to includes viruses, worms, Trojans, and ransomware.

True

False

Correct Answer:

Correct Answer: False

Concept: Malware is directly related to includes viruses, worms, Trojans, and ransomware.

16tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Virus means that requires a host program to spread.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Virus.

17tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Virus is unrelated to requires a host program to spread.

True

False

Correct Answer:

Correct Answer: False

Concept: Virus is directly related to requires a host program to spread.

18tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Worm means that can self-replicate across networks.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Worm.

19tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Worm is unrelated to can self-replicate across networks.

True

False

Correct Answer:

Correct Answer: False

Concept: Worm is directly related to can self-replicate across networks.

20tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Trojan means that disguises itself as legitimate software.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Trojan.

21tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Trojan is unrelated to disguises itself as legitimate software.

True

False

Correct Answer:

Correct Answer: False

Concept: Trojan is directly related to disguises itself as legitimate software.

22tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Ransomware means that encrypts data and demands payment.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Ransomware.

23tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Ransomware is unrelated to encrypts data and demands payment.

True

False

Correct Answer:

Correct Answer: False

Concept: Ransomware is directly related to encrypts data and demands payment.

24tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Phishing means that uses deception to steal information.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Phishing.

25tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Phishing is unrelated to uses deception to steal information.

True

False

Correct Answer:

Correct Answer: False

Concept: Phishing is directly related to uses deception to steal information.

26tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Social engineering means that manipulates people into breaking security rules.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Social engineering.

27tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Social engineering is unrelated to manipulates people into breaking security rules.

True

False

Correct Answer:

Correct Answer: False

Concept: Social engineering is directly related to manipulates people into breaking security rules.

28tfcot tfcolTrue / False

Topic: Enterprise Security Threats

DoS attack means that aims to make a service unavailable.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DoS attack.

29tfcot tfcolTrue / False

Topic: Enterprise Security Threats

DoS attack is unrelated to aims to make a service unavailable.

True

False

Correct Answer:

Correct Answer: False

Concept: DoS attack is directly related to aims to make a service unavailable.

30tfcot tfcolTrue / False

Topic: Enterprise Security Threats

DDoS attack means that uses many distributed sources to flood a target.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DDoS attack.

31tfcot tfcolTrue / False

Topic: Enterprise Security Threats

DDoS attack is unrelated to uses many distributed sources to flood a target.

True

False

Correct Answer:

Correct Answer: False

Concept: DDoS attack is directly related to uses many distributed sources to flood a target.

32tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Man-in-the-middle attack means that intercepts communication between two parties.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Man-in-the-middle attack.

33tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Man-in-the-middle attack is unrelated to intercepts communication between two parties.

True

False

Correct Answer:

Correct Answer: False

Concept: Man-in-the-middle attack is directly related to intercepts communication between two parties.

34tfcot tfcolTrue / False

Topic: Enterprise Security Threats

ARP spoofing means that forges ARP replies to redirect traffic.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ARP spoofing.

35tfcot tfcolTrue / False

Topic: Enterprise Security Threats

ARP spoofing is unrelated to forges ARP replies to redirect traffic.

True

False

Correct Answer:

Correct Answer: False

Concept: ARP spoofing is directly related to forges ARP replies to redirect traffic.

36tfcot tfcolTrue / False

Topic: Enterprise Security Threats

SYN flood means that sends many half-open TCP connections.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SYN flood.

37tfcot tfcolTrue / False

Topic: Enterprise Security Threats

SYN flood is unrelated to sends many half-open TCP connections.

True

False

Correct Answer:

Correct Answer: False

Concept: SYN flood is directly related to sends many half-open TCP connections.

38tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Ping of death means that sends oversized ICMP packets.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Ping of death.

39tfcot tfcolTrue / False

Topic: Enterprise Security Threats

Ping of death is unrelated to sends oversized ICMP packets.

True

False

Correct Answer:

Correct Answer: False

Concept: Ping of death is directly related to sends oversized ICMP packets.

40tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Smurf attack means that uses broadcast pings to amplify traffic.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Smurf attack.

41tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Smurf attack is unrelated to uses broadcast pings to amplify traffic.

True

False

Correct Answer:

Correct Answer: False

Concept: Smurf attack is directly related to uses broadcast pings to amplify traffic.

42tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Least privilege means that limits user access to the minimum necessary.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Least privilege.

43tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Least privilege is unrelated to limits user access to the minimum necessary.

True

False

Correct Answer:

Correct Answer: False

Concept: Least privilege is directly related to limits user access to the minimum necessary.

44tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Patch management means that eliminates known vulnerabilities.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Patch management.

45tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Patch management is unrelated to eliminates known vulnerabilities.

True

False

Correct Answer:

Correct Answer: False

Concept: Patch management is directly related to eliminates known vulnerabilities.

46tfcol tfcolTrue / False

Topic: Enterprise Security Threats

SIEM means that centralizes log collection and analysis.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SIEM.

47tfcol tfcolTrue / False

Topic: Enterprise Security Threats

SIEM is unrelated to centralizes log collection and analysis.

True

False

Correct Answer:

Correct Answer: False

Concept: SIEM is directly related to centralizes log collection and analysis.

48tfcol tfcolTrue / False

Topic: Enterprise Security Threats

DMZ means that hosts public-facing servers between Internet and internal network.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DMZ.

49tfcol tfcolTrue / False

Topic: Enterprise Security Threats

DMZ is unrelated to hosts public-facing servers between Internet and internal network.

True

False

Correct Answer:

Correct Answer: False

Concept: DMZ is directly related to hosts public-facing servers between Internet and internal network.

50tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Insider threat means that originates from within the organization.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Insider threat.

51tfcol tfcolTrue / False

Topic: Enterprise Security Threats

Insider threat is unrelated to originates from within the organization.

True

False

Correct Answer:

Correct Answer: False

Concept: Insider threat is directly related to originates from within the organization.

83 Communication Network Security

1tfc0l tfcolTrue / False

Topic: Communication Network Security

Encryption is an effective way to protect data confidentiality over public networks.

True

False

Correct Answer:

Correct Answer: True

Concept: Encryption ensures that intercepted data cannot be read without the proper key.

84 Zone Border Security

1tfc0l tfcolTrue / False

Topic: Zone Border Security

A firewall at the zone border can enforce security policies between trusted and untrusted networks.

True

False

Correct Answer:

Correct Answer: True

Concept: Firewalls inspect and control traffic crossing security zone boundaries based on policy.

85 Firewall Basics

1tfc0l tfcolTrue / False

Topic: Firewall Basics

Security policies on Huawei firewalls are matched from bottom to top by default.

True

False

Correct Answer:

Correct Answer: False

Concept: Policies are matched from top to bottom based on configured order/priority.

2tfc0l tfcolTrue / False

Topic: Firewall Basics

A packet-filtering firewall tracks the state of every connection.

True

False

Correct Answer:

Correct Answer: False

Concept: Only stateful firewalls maintain connection state; packet filters inspect packets individually.

3tfc0l tfcolTrue / False

Topic: Firewall Basics

ASPF can generate temporary server-map entries for dynamic protocols.

True

False

Correct Answer:

Correct Answer: True

Concept: ASPF creates server-map entries to permit negotiated secondary connections.

4tfcol tfcolTrue / False

Topic: Firewall Basics

Next-generation firewalls can identify applications regardless of port.

True

False

Correct Answer:

Correct Answer: True

Concept: NGFWs use application-layer inspection to identify applications beyond simple port numbers.

5tfcol tfcolTrue / False

Topic: Firewall Basics

Traffic between interfaces in the same security zone is always allowed.

True

False

Correct Answer:

Correct Answer: False

Concept: Even intra-zone traffic can be controlled by policies, though default behavior may vary by configuration.

6tfcol tfcolTrue / False

Topic: Firewall Basics

A deny action in a security policy can generate logs.

True

False

Correct Answer:

Correct Answer: True

Concept: Both permit and deny actions can be configured to log traffic.

7tfcol tfcolTrue / False

Topic: Firewall Basics

The default Huawei firewall action is permit.

True

False

Correct Answer:

Correct Answer: False

Concept: The default action is deny unless explicitly permitted.

8tfcol tfcolTrue / False

Topic: Firewall Basics

ASPF inspects application-layer negotiation.

True

False

Correct Answer:

Correct Answer: True

Concept: ASPF handles dynamic protocols such as FTP.

9tfcol tfcolTrue / False

Topic: Firewall Basics

Stateful firewalls use session tables to track connections.

True

False

Correct Answer:

Correct Answer: True

Concept: Session tables store connection state for return traffic.

10tfcot tfcolTrue / False

Topic: Firewall Basics

Trust zone priority means that is 85 by default.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Trust zone priority.

11tfcot tfcolTrue / False

Topic: Firewall Basics

Trust zone priority is unrelated to is 85 by default.

True

False

Correct Answer:

Correct Answer: False

Concept: Trust zone priority is directly related to is 85 by default.

12tfcot tfcolTrue / False

Topic: Firewall Basics

DMZ zone priority means that is 50 by default.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DMZ zone priority.

13tfcot tfcolTrue / False

Topic: Firewall Basics

DMZ zone priority is unrelated to is 50 by default.

True

False

Correct Answer:

Correct Answer: False

Concept: DMZ zone priority is directly related to is 50 by default.

14tfcot tfcolTrue / False

Topic: Firewall Basics

Untrust zone priority means that is 5 by default.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Untrust zone priority.

15tfcot tfcolTrue / False

Topic: Firewall Basics

Untrust zone priority is unrelated to is 5 by default.

True

False

Correct Answer:

Correct Answer: False

Concept: Untrust zone priority is directly related to is 5 by default.

16tfcol tfcolTrue / False

Local zone priority means that is 100 by default.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Local zone priority.

17tfcol tfcolTrue / False

Local zone priority is unrelated to is 100 by default.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: False

Concept: Local zone priority is directly related to is 100 by default.

18tfcol tfcolTrue / False

Default security policy action means that is deny.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Default security policy action.

19tfcol tfcolTrue / False

Default security policy action is unrelated to is deny.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: False

Concept: Default security policy action is directly related to is deny.

20tfcol tfcolTrue / False

Stateful firewall means that tracks connections in a session table.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Stateful firewall.

21tfcol tfcolTrue / False

Stateful firewall is unrelated to tracks connections in a session table.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: False

Concept: Stateful firewall is directly related to tracks connections in a session table.

22tfcol tfcolTrue / False

Topic: Firewall Basics

Packet-filtering firewall means that inspects packets individually without state.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Packet-filtering firewall.

23tfcol tfcolTrue / False

Topic: Firewall Basics

Packet-filtering firewall is unrelated to inspects packets individually without state.

True

False

Correct Answer:

Correct Answer: False

Concept: Packet-filtering firewall is directly related to inspects packets individually without state.

24tfcol tfcolTrue / False

Topic: Firewall Basics

ASPF means that inspects multi-channel protocols like FTP.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ASPF.

25tfcol tfcolTrue / False

Topic: Firewall Basics

ASPF is unrelated to inspects multi-channel protocols like FTP.

True

False

Correct Answer:

Correct Answer: False

Concept: ASPF is directly related to inspects multi-channel protocols like FTP.

26tfcol tfcolTrue / False

Topic: Firewall Basics

Next-generation firewall means that includes application awareness and IPS.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Next-generation firewall.

27tfcol tfcolTrue / False

Topic: Firewall Basics

Next-generation firewall is unrelated to includes application awareness and IPS.

True

False

Correct Answer:

Correct Answer: False

Concept: Next-generation firewall is directly related to includes application awareness and IPS.

28tfcot tfcolTrue / False

Topic: Firewall Basics

Security zones means that group interfaces with similar trust levels.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Security zones.

29tfcot tfcolTrue / False

Topic: Firewall Basics

Security zones is unrelated to group interfaces with similar trust levels.

True

False

Correct Answer:

Correct Answer: False

Concept: Security zones is directly related to group interfaces with similar trust levels.

30tfcot tfcolTrue / False

Topic: Firewall Basics

Intra-zone traffic means that can be controlled by policies.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Intra-zone traffic.

31tfcot tfcolTrue / False

Topic: Firewall Basics

Intra-zone traffic is unrelated to can be controlled by policies.

True

False

Correct Answer:

Correct Answer: False

Concept: Intra-zone traffic is directly related to can be controlled by policies.

32tfcot tfcolTrue / False

Topic: Firewall Basics

Policy hit count means that helps verify which rules are used.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Policy hit count.

33tfcot tfcolTrue / False

Topic: Firewall Basics

Policy hit count is unrelated to helps verify which rules are used.

True

False

Correct Answer:

Correct Answer: False

Concept: Policy hit count is directly related to help verify which rules are used.

34tfcot tfcolTrue / False

Topic: Firewall Basics

Security policy logging means that can be enabled for permit and deny actions.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Security policy logging.

35tfcot tfcolTrue / False

Topic: Firewall Basics

Security policy logging is unrelated to can be enabled for permit and deny actions.

True

False

Correct Answer:

Correct Answer: False

Concept: Security policy logging is directly related to can be enabled for permit and deny actions.

36tfcot tfcolTrue / False

Topic: Firewall Basics

Routing mode means that is a firewall deployment mode.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Routing mode.

37tfcot tfcolTrue / False

Topic: Firewall Basics

Routing mode is unrelated to is a firewall deployment mode.

True

False

Correct Answer:

Correct Answer: False

Concept: Routing mode is directly related to is a firewall deployment mode.

38tfcot tfcolTrue / False

Topic: Firewall Basics

Transparent mode means that is a firewall deployment mode.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Transparent mode.

39tfcot tfcolTrue / False

Topic: Firewall Basics

Transparent mode is unrelated to is a firewall deployment mode.

True

False

Correct Answer:

Correct Answer: False

Concept: Transparent mode is directly related to is a firewall deployment mode.

40tfcol tfcolTrue / False

Hybrid mode means that is a firewall deployment mode.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Hybrid mode.

41tfcol tfcolTrue / False

Hybrid mode is unrelated to is a firewall deployment mode.

Topic: Firewall Basics

True

False

Correct Answer:

Correct Answer: False

Concept: Hybrid mode is directly related to is a firewall deployment mode.

86 Security Zones

1tfcol tfcolTrue / False

The Local zone represents traffic destined to or originated from the firewall itself.

Topic: Security Zones

True

False

Correct Answer:

Correct Answer: True

Concept: Local zone traffic involves the firewall services and management interfaces.

87 Security Policies

1tfcol tfcolTrue / False

By default, Huawei firewalls permit all traffic unless a policy explicitly denies it.

Topic: Security Policies

True

False

Correct Answer:

Correct Answer: False

Concept: The default action is to deny traffic unless a security policy explicitly permits it.

88 Stateful Inspection

1tfcol tfcolTrue / False

A stateful firewall can allow return traffic for an established session without a separate policy.

Topic: Stateful Inspection

True

False

Correct Answer:

Correct Answer: True

Concept: Return packets matching an existing session table entry are permitted automatically.

89 ASPF

1tfcot tcolTrue / False

Topic: ASPF

ASPF is used to support protocols that open secondary or dynamic connections.

True

False

Correct Answer:

Correct Answer: True

Concept: ASPF inspects application-layer negotiation to dynamically permit related channels.

90 NAT

1tfcot tcolTrue / False

Topic: NAT

NAT can hide internal network topology from external observers.

True

False

Correct Answer:

Correct Answer: True

Concept: NAT translates private addresses, making internal topology invisible from outside.

2tfcot tcolTrue / False

Topic: NAT

Easy-IP uses a fixed public IP address pool.

True

False

Correct Answer:

Correct Answer: False

Concept: Easy-IP uses the firewall egress interface IP address dynamically.

3tfcot tcolTrue / False

Topic: NAT

NAT ALG is required for HTTP because HTTP embeds IP addresses in payloads.

True

False

Correct Answer:

Correct Answer: False

Concept: HTTP generally does not embed IP addresses in payloads, so ALG is not required.

4tfcot tcolTrue / False

Topic: NAT

NAPT conserves public IPv4 addresses by using port multiplexing.

True

False

Correct Answer:

Correct Answer: True

Concept: NAT maps many private addresses to one public IP using unique port numbers.

5tfcot tfcolTrue / False

Topic: NAT

FTP data channel failure can be caused by disabled NAT ALG.

True

False

Correct Answer:

Correct Answer: True

Concept: Without ALG, the firewall cannot identify and permit the negotiated FTP data channel.

6tfcot tfcolTrue / False

Topic: NAT

NAPT translates both IP addresses and port numbers.

True

False

Correct Answer:

Correct Answer: True

Concept: NAT uses port multiplexing for many-to-one translation.

7tfcot tfcolTrue / False

Topic: NAT

NAT Server is used for outbound Internet access.

True

False

Correct Answer:

Correct Answer: False

Concept: NAT Server publishes inbound services.

8tfcot tfcolTrue / False

Topic: NAT

Static NAT preserves a one-to-one public-private mapping.

True

False

Correct Answer:

Correct Answer: True

Concept: Static NAT maps one public IP to one private IP consistently.

9tfcot tfcolTrue / False

Topic: NAT

NAT means that translates IP addresses between private and public.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NAT.

10tfcot tfcolTrue / False

Topic: NAT

NAT is unrelated to translates IP addresses between private and public.

True

False

Correct Answer:

Correct Answer: False

Concept: NAT is directly related to translates IP addresses between private and public.

11tfcol tfcolTrue / False

Topic: NAT

Source NAT means that translates the source IP of outbound packets.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Source NAT.

12tfcol tfcolTrue / False

Topic: NAT

Source NAT is unrelated to translates the source IP of outbound packets.

True

False

Correct Answer:

Correct Answer: False

Concept: Source NAT is directly related to translates the source IP of outbound packets.

13tfcol tfcolTrue / False

Topic: NAT

Destination NAT means that translates the destination IP of inbound packets.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Destination NAT.

14tfcol tfcolTrue / False

Topic: NAT

Destination NAT is unrelated to translates the destination IP of inbound packets.

True

False

Correct Answer:

Correct Answer: False

Concept: Destination NAT is directly related to translates the destination IP of inbound packets.

15tfcol tfcolTrue / False

Topic: NAT

NAT Server means that publishes internal services to the Internet.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NAT Server.

16tfcol tfcolTrue / False

Topic: NAT

NAT Server is unrelated to publishes internal services to the Internet.

True

False

Correct Answer:

Correct Answer: False

Concept: NAT Server is directly related to publishes internal services to the Internet.

17tfcol tfcolTrue / False

Topic: NAT

Bidirectional NAT means that translates both source and destination addresses.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Bidirectional NAT.

18tfcol tfcolTrue / False

Topic: NAT

Bidirectional NAT is unrelated to translates both source and destination addresses.

True

False

Correct Answer:

Correct Answer: False

Concept: Bidirectional NAT is directly related to translates both source and destination addresses.

19tfcol tfcolTrue / False

Topic: NAT

NAPT means that translates many private addresses to one public IP using ports.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NAPT.

20tfcol tfcolTrue / False

Topic: NAT

NAPT is unrelated to translates many private addresses to one public IP using ports.

True

False

Correct Answer:

Correct Answer: False

Concept: NAPT is directly related to translates many private addresses to one public IP using ports.

21tfcol tfcolTrue / False

Topic: NAT

NAT No-PAT means that performs one-to-one address translation preserving ports.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NAT No-PAT.

22tfcol tfcolTrue / False

Topic: NAT

NAT No-PAT is unrelated to performs one-to-one address translation preserving ports.

True

False

Correct Answer:

Correct Answer: False

Concept: NAT No-PAT is directly related to performs one-to-one address translation preserving ports.

23tfcot tfcolTrue / False

Topic: NAT

Easy-IP means that uses the outbound interface IP for translation.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Easy-IP.

24tfcot tfcolTrue / False

Topic: NAT

Easy-IP is unrelated to uses the outbound interface IP for translation.

True

False

Correct Answer:

Correct Answer: False

Concept: Easy-IP is directly related to uses the outbound interface IP for translation.

25tfcot tfcolTrue / False

Topic: NAT

NAT ALG means that rewrites addresses inside application payloads.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NAT ALG.

26tfcot tfcolTrue / False

Topic: NAT

NAT ALG is unrelated to rewrites addresses inside application payloads.

True

False

Correct Answer:

Correct Answer: False

Concept: NAT ALG is directly related to rewrites addresses inside application payloads.

27tfcot tfcolTrue / False

Topic: NAT

RFC 1918 means that defines private IPv4 address ranges.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes RFC 1918.

28tfcot tfcolTrue / False

Topic: NAT

RFC 1918 is unrelated to defines private IPv4 address ranges.

True

False

Correct Answer:

Correct Answer: False

Concept: RFC 1918 is directly related to defines private IPv4 address ranges.

29tfcol tfcolTrue / False

Topic: NAT

Hairpin NAT means that lets internal users access servers via public NAT address.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Hairpin NAT.

30tfcol tfcolTrue / False

Topic: NAT

Hairpin NAT is unrelated to lets internal users access servers via public NAT address.

True

False

Correct Answer:

Correct Answer: False

Concept: Hairpin NAT is directly related to lets internal users access servers via public NAT address.

31tfcol tfcolTrue / False

Topic: NAT

Static NAT means that provides a persistent one-to-one mapping.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Static NAT.

32tfcol tfcolTrue / False

Topic: NAT

Static NAT is unrelated to provides a persistent one-to-one mapping.

True

False

Correct Answer:

Correct Answer: False

Concept: Static NAT is directly related to provides a persistent one-to-one mapping.

91 NAT Basics

1tfcol tfcolTrue / False

Topic: NAT Basics

NAT can conserve public IPv4 addresses.

True

False

Correct Answer:

Correct Answer: True

Concept: NAT allows many private hosts to share a smaller number of public IP addresses.

92 Source NAT

1tfcol tfcolTrue / False

Topic: Source NAT

NAPT translates both IP addresses and port numbers.

True

False

Correct Answer:

Correct Answer: True

Concept: NAPT multiplexes many private addresses onto one public IP by translating ports.

93 Destination NAT

1tfcol tfcolTrue / False

Topic: Destination NAT

Destination NAT is typically used for outbound Internet access.

True

False

Correct Answer:

Correct Answer: False

Concept: Destination NAT is used for inbound traffic to expose internal services.

94 NAT Server

1tfcol tfcolTrue / False

Topic: NAT Server

NAT Server maps public IP addresses to private server addresses.

True

False

Correct Answer:

Correct Answer: True

Concept: NAT Server publishes internal services by mapping public IPs/ports to private servers.

95 NAT ALG

1tfcol tfcolTrue / False

Topic: NAT ALG

NAT ALG is needed when protocols embed IP addresses inside application payloads.

True

False

Correct Answer:

Correct Answer: True

Concept: ALG inspects application-layer data to update embedded addresses/ports.

96 Firewall Hot Standby

1tfcol tfcolTrue / False

Topic: Firewall Hot Standby

In active/standby mode, both firewalls actively forward the same traffic.

True

False

Correct Answer:

Correct Answer: False

Concept: Only the active firewall forwards traffic; the standby takes over upon failure.

2tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

VRRP provides default gateway redundancy for end hosts.

True

False

Correct Answer:

Correct Answer: True

Concept: VRRP allows multiple routers to act as a single virtual gateway.

3tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

In VRRP, only the router with the physical IP matching the virtual IP can become master.

True

False

Correct Answer:

Correct Answer: False

Concept: Any router with sufficient priority can become master, not just the IP owner.

4tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

VGMP prevents individual VRRP groups from becoming inconsistent during failover.

True

False

Correct Answer:

Correct Answer: True

Concept: VGMP manages multiple VRRP groups as a single entity.

5tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

HRP can synchronize NAT sessions between active and standby firewalls.

True

False

Correct Answer:

Correct Answer: True

Concept: HRP keeps NAT/session state consistent across hot-standby peers.

6tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

Load-sharing mode improves resource utilization compared to active/standby mode.

True

False

Correct Answer:

Correct Answer: True

Concept: Multiple firewalls forward traffic simultaneously in load-sharing mode.

7tfc0l tfcolTrue / False

Topic: Firewall Hot Standby

The HRP heartbeat link can share bandwidth with regular user traffic without risk.

True

False

Correct Answer:

Correct Answer: False

Concept: Sharing heartbeat and user traffic can cause false failover due to congestion.

8tfcol tfcolTrue / False

VRRP provides a virtual gateway IP.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: VRRP allows multiple routers to share a virtual IP.

9tfcol tfcolTrue / False

HRP synchronizes only configurations.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: HRP synchronizes configurations and runtime state.

10tfcol tfcolTrue / False

Load-sharing mode allows multiple firewalls to forward traffic.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: Multiple active firewalls share the traffic load.

11tfcol tfcolTrue / False

VRRP means that provides gateway redundancy with a virtual IP.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VRRP.

12tfcol tfcolTrue / False

VRRP is unrelated to provides gateway redundancy with a virtual IP.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: VRRP is directly related to provides gateway redundancy with a virtual IP.

13tfcol tfcolTrue / False

VRRP master means that forwards traffic for the virtual IP.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VRRP master.

14tfcot tfcolTrue / False

VRRP master is unrelated to forwards traffic for the virtual IP.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: VRRP master is directly related to forwards traffic for the virtual IP.

15tfcot tfcolTrue / False

VRRP backup means that takes over if the master fails.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VRRP backup.

16tfcot tfcolTrue / False

VRRP backup is unrelated to takes over if the master fails.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: VRRP backup is directly related to takes over if the master fails.

17tfcot tfcolTrue / False

VGMP means that manages VRRP group states consistently.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VGMP.

18tfcot tfcolTrue / False

VGMP is unrelated to manages VRRP group states consistently.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: VGMP is directly related to manages VRRP group states consistently.

19tfcot tfcolTrue / False

HRP means that synchronizes sessions and configurations.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HRP.

20tfcot tfcolTrue / False

HRP is unrelated to synchronizes sessions and configurations.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: HRP is directly related to synchronizes sessions and configurations.

21tfcot tfcolTrue / False

Active/standby mode means that has one firewall forwarding traffic.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Active/standby mode.

22tfcot tfcolTrue / False

Active/standby mode is unrelated to has one firewall forwarding traffic.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: Active/standby mode is directly related to has one firewall forwarding traffic.

23tfcot tfcolTrue / False

Load-sharing mode means that has multiple firewalls forwarding traffic.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Load-sharing mode.

24tfcot tfcolTrue / False

Load-sharing mode is unrelated to has multiple firewalls forwarding traffic.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: Load-sharing mode is directly related to has multiple firewalls forwarding traffic.

25tfcot tfcolTrue / False

Heartbeat link means that should be dedicated and reliable.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Heartbeat link.

26tfcot tfcolTrue / False

Heartbeat link is unrelated to should be dedicated and reliable.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: Heartbeat link is directly related to should be dedicated and reliable.

27tfcot tfcolTrue / False

HRP session backup means that ensures connections survive failover.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HRP session backup.

28tfcot tfcolTrue / False

HRP session backup is unrelated to ensures connections survive failover.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: HRP session backup is directly related to ensures connections survive failover.

29tfcot tfcolTrue / False

Preempt delay means that reduces rapid master-backup flapping.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Preempt delay.

30tfcot tfcolTrue / False

Preempt delay is unrelated to reduces rapid master-backup flapping.

Topic: Firewall Hot Standby

True

False

Correct Answer:

Correct Answer: False

Concept: Preempt delay is directly related to reduces rapid master-backup flapping.

97 VRRP

1tfcot tfcolTrue / False

VRRP uses a virtual IP address shared between multiple physical routers.

Topic: VRRP

True

False

Correct Answer:

Correct Answer: True

Concept: VRRP allows multiple routers to present a single virtual gateway IP to hosts.

98 VGMP

1tfcol tfcolTrue / False

Topic: VGMP

VGMP ensures that VRRP groups on a firewall switch state consistently.

True

False

Correct Answer:

Correct Answer: True

Concept: VGMP prevents split-brain scenarios by managing all VRRP groups as a single entity.

99 HRP

1tfcol tfcolTrue / False

Topic: HRP

HRP only synchronizes configuration files, not session tables.

True

False

Correct Answer:

Correct Answer: False

Concept: HRP synchronizes both configurations and runtime state such as session tables.

100 Intrusion Prevention and Antivirus

1tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

IPS is typically deployed inline to block threats.

True

False

Correct Answer:

Correct Answer: True

Concept: Inline deployment allows IPS to actively block malicious traffic.

2tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Anomaly-based detection requires a database of known attack signatures.

True

False

Correct Answer:

Correct Answer: False

Concept: Anomaly detection uses baselines of normal behavior, not attack signatures.

3tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Heuristic analysis can detect previously unknown malware.

True

False

Correct Answer:

Correct Answer: True

Concept: Heuristics identify suspicious behavior even without a known signature.

4tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Signature-based antivirus is effective against all zero-day threats.

True

False

Correct Answer:

Correct Answer: False

Concept: Zero-day threats lack signatures, so signature-only detection cannot catch them.

5tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

An IDS can block malicious traffic inline.

True

False

Correct Answer:

Correct Answer: False

Concept: IDS typically detects and alerts; IPS blocks inline.

6tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Heuristic detection can identify unknown threats.

True

False

Correct Answer:

Correct Answer: True

Concept: Heuristics analyze behavior to find unknown malware.

7tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Sandboxing executes files in an isolated environment.

True

False

Correct Answer:

Correct Answer: True

Concept: Sandboxing observes behavior without risking production systems.

8tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

IDS means that detects intrusions and generates alerts.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IDS.

9tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

IDS is unrelated to detects intrusions and generates alerts.

True

False

Correct Answer:

Correct Answer: False

Concept: IDS is directly related to detects intrusions and generates alerts.

10tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

IPS means that detects and blocks intrusions inline.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IPS.

11tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

IPS is unrelated to detects and blocks intrusions inline.

True

False

Correct Answer:

Correct Answer: False

Concept: IPS is directly related to detects and blocks intrusions inline.

12tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Signature detection means that matches known attack patterns.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Signature detection.

13tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Signature detection is unrelated to matches known attack patterns.

True

False

Correct Answer:

Correct Answer: False

Concept: Signature detection is directly related to matches known attack patterns.

14tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Anomaly detection means that flags deviations from normal behavior.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Anomaly detection.

15tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Anomaly detection is unrelated to flags deviations from normal behavior.

True

False

Correct Answer:

Correct Answer: False

Concept: Anomaly detection is directly related to flags deviations from normal behavior.

16tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

False positive means that incorrectly blocks legitimate traffic.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes False positive.

17tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

False positive is unrelated to incorrectly blocks legitimate traffic.

True

False

Correct Answer:

Correct Answer: False

Concept: False positive is directly related to incorrectly blocks legitimate traffic.

18tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

False negative means that misses an actual attack.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes False negative.

19tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

False negative is unrelated to misses an actual attack.

True

False

Correct Answer:

Correct Answer: False

Concept: False negative is directly related to misses an actual attack.

20tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Sandboxing means that observes file behavior in an isolated environment.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Sandboxing.

21tfcol tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Sandboxing is unrelated to observes file behavior in an isolated environment.

True

False

Correct Answer:

Correct Answer: False

Concept: Sandboxing is directly related to observes file behavior in an isolated environment.

22tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Heuristic analysis means that detects malware by examining behavior.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Heuristic analysis.

23tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Heuristic analysis is unrelated to detects malware by examining behavior.

True

False

Correct Answer:

Correct Answer: False

Concept: Heuristic analysis is directly related to detects malware by examining behavior.

24tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Whitelisting means that allows only approved applications.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Whitelisting.

25tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Whitelisting is unrelated to allows only approved applications.

True

False

Correct Answer:

Correct Answer: False

Concept: Whitelisting is directly related to allows only approved applications.

26tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Antivirus signatures means that must be updated regularly.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Antivirus signatures.

27tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Antivirus signatures is unrelated to must be updated regularly.

True

False

Correct Answer:

Correct Answer: False

Concept: Antivirus signatures is directly related to must be updated regularly.

28tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

NIDS means that monitors network traffic for intrusions.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes NIDS.

29tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

NIDS is unrelated to monitors network traffic for intrusions.

True

False

Correct Answer:

Correct Answer: False

Concept: NIDS is directly related to monitors network traffic for intrusions.

30tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

HIDS means that monitors host activity for intrusions.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HIDS.

31tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

HIDS is unrelated to monitors host activity for intrusions.

True

False

Correct Answer:

Correct Answer: False

Concept: HIDS is directly related to monitors host activity for intrusions.

32tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Reconnaissance means that is an early attack phase.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Reconnaissance.

33tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Reconnaissance is unrelated to is an early attack phase.

True

False

Correct Answer:

Correct Answer: False

Concept: Reconnaissance is directly related to is an early attack phase.

34tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Exploitation means that gains unauthorized access using vulnerabilities.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Exploitation.

35tfcot tfcolTrue / False

Topic: Intrusion Prevention and Antivirus

Exploitation is unrelated to gains unauthorized access using vulnerabilities.

True

False

Correct Answer:

Correct Answer: False

Concept: Exploitation is directly related to gains unauthorized access using vulnerabilities.

101 Intrusion Prevention

1tfcot tfcolTrue / False

Topic: Intrusion Prevention

An IDS is typically deployed inline and can block attacks.

True

False

Correct Answer:

Correct Answer: False

Concept: IDS is usually passive/out-of-band and only detects and alerts; IPS blocks inline.

2tfcot tfcolTrue / False

Topic: Intrusion Prevention

Signature-based IPS can detect zero-day attacks effectively.

True

False

Correct Answer:

Correct Answer: False

Concept: Signatures require known attack patterns; zero-day attacks are unknown and often missed.

102 Antivirus

1tfcot tfcolTrue / False

Topic: Antivirus

Antivirus signature databases must be updated regularly to detect new malware.

True

False

Correct Answer:**Correct Answer: True****Concept:** New malware variants require updated signatures and detection engines.

2tfc0l tfcolTrue / False

Topic: Antivirus

Sandboxing can detect unknown malware by observing its runtime behavior.

True

False

Correct Answer:**Correct Answer: True****Concept:** Sandboxing reveals malicious behavior of unknown samples in an isolated environment.

103 AAA and User Authentication

1tfc0l tfcolTrue / False

Topic: AAA and User Authentication

Authentication verifies who a user is.

True

False

Correct Answer:**Correct Answer: True****Concept:** Authentication confirms identity before granting access.

2tfc0l tfcolTrue / False

Topic: AAA and User Authentication

TACACS+ separates authentication, authorization, and accounting.

True

False

Correct Answer:**Correct Answer: True****Concept:** TACACS+ provides independent AAA functions.

3tfc0l tfcolTrue / False

Topic: AAA and User Authentication

RADIUS encrypts the entire authentication request.

True

False

Correct Answer:**Correct Answer: False****Concept:** RADIUS encrypts only the password attribute.

4tfc0l tfcolTrue / False

Topic: AAA and User Authentication

Local authentication does not require an external server.

True

False

Correct Answer:**Correct Answer: True****Concept:** Local authentication uses the firewall own user database.

5tfc0l tfcolTrue / False

Topic: AAA and User Authentication

Authorization happens after authentication.

True

False

Correct Answer:

Correct Answer: True

Concept: Authorization defines what an authenticated user may do.

6tfcol tfcolTrue / False
TACACS+ uses UDP.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: False

Concept: TACACS+ uses TCP.

7tfcol tfcolTrue / False
Portal authentication requires a web browser.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: True

Concept: Portal presents a web login page to users.

8tfcol tfcolTrue / False
Authentication means that verifies user identity.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Authentication.

9tfcol tfcolTrue / False
Authentication is unrelated to verifies user identity.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: False

Concept: Authentication is directly related to verifies user identity.

10tfcol tfcolTrue / False
Authorization means that determines what a user can access.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Authorization.

11tfcol tfcolTrue / False
Authorization is unrelated to determines what a user can access.

Topic: AAA and User Authentication

True

False

Correct Answer:

Correct Answer: False

Concept: Authorization is directly related to determines what a user can access.

12tfcol tfcolTrue / False

Topic: AAA and User Authentication

Accounting means that records user activities and resource usage.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Accounting.

13tfcol tfcolTrue / False

Topic: AAA and User Authentication

Accounting is unrelated to records user activities and resource usage.

True

False

Correct Answer:

Correct Answer: False

Concept: Accounting is directly related to records user activities and resource usage.

14tfcol tfcolTrue / False

Topic: AAA and User Authentication

RADIUS means that uses UDP ports 1812 and 1813.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes RADIUS.

15tfcol tfcolTrue / False

Topic: AAA and User Authentication

RADIUS is unrelated to uses UDP ports 1812 and 1813.

True

False

Correct Answer:

Correct Answer: False

Concept: RADIUS is directly related to uses UDP ports 1812 and 1813.

16tfcol tfcolTrue / False

Topic: AAA and User Authentication

TACACS+ means that uses TCP port 49.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes TACACS+.

17tfcol tfcolTrue / False

Topic: AAA and User Authentication

TACACS+ is unrelated to uses TCP port 49.

True

False

Correct Answer:

Correct Answer: False

Concept: TACACS+ is directly related to uses TCP port 49.

18tfcot tfcolTrue / False

Topic: AAA and User Authentication

Portal authentication means that uses a web login page.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Portal authentication.

19tfcot tfcolTrue / False

Topic: AAA and User Authentication

Portal authentication is unrelated to uses a web login page.

True

False

Correct Answer:

Correct Answer: False

Concept: Portal authentication is directly related to uses a web login page.

20tfcot tfcolTrue / False

Topic: AAA and User Authentication

802.1X means that provides port-based network access control.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes 802.1X.

21tfcot tfcolTrue / False

Topic: AAA and User Authentication

802.1X is unrelated to provides port-based network access control.

True

False

Correct Answer:

Correct Answer: False

Concept: 802.1X is directly related to provides port-based network access control.

22tfcot tfcolTrue / False

Topic: AAA and User Authentication

Local authentication means that uses the device user database.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Local authentication.

23tfcot tfcolTrue / False

Topic: AAA and User Authentication

Local authentication is unrelated to uses the device user database.

True

False

Correct Answer:

Correct Answer: False

Concept: Local authentication is directly related to uses the device user database.

24tfcot tfcolTrue / False

Topic: AAA and User Authentication

LDAP means that queries a directory server for authentication.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes LDAP.

25tfcot tfcolTrue / False

Topic: AAA and User Authentication

LDAP is unrelated to queries a directory server for authentication.

True

False

Correct Answer:

Correct Answer: False

Concept: LDAP is directly related to queries a directory server for authentication.

26tfcot tfcolTrue / False

Topic: AAA and User Authentication

User group means that applies policies to multiple users collectively.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes User group.

27tfcot tfcolTrue / False

Topic: AAA and User Authentication

User group is unrelated to applies policies to multiple users collectively.

True

False

Correct Answer:

Correct Answer: False

Concept: User group is directly related to applies policies to multiple users collectively.

28tfcot tfcolTrue / False

Topic: AAA and User Authentication

SSO means that allows one authentication for multiple services.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SSO.

29tfcot tfcolTrue / False

Topic: AAA and User Authentication

SSO is unrelated to allows one authentication for multiple services.

True

False

Correct Answer:

Correct Answer: False

Concept: SSO is directly related to allows one authentication for multiple services.

104 AAA Basics

1tfcol tfcolTrue / False

Topic: AAA Basics

Authorization determines whether an authenticated user is allowed to perform an action.

True

False

Correct Answer:

Correct Answer: True

Concept: Authorization enforces access control decisions based on identity and policy.

2tfcol tfcolTrue / False

Topic: AAA Basics

Accounting in AAA records user activities and resource usage.

True

False

Correct Answer:

Correct Answer: True

Concept: Accounting provides logs for auditing, billing, and forensic analysis.

105 Firewall User Authentication

1tfcol tfcolTrue / False

Topic: Firewall User Authentication

Portal authentication is commonly used for guest access and web-based login.

True

False

Correct Answer:

Correct Answer: True

Concept: Portal authentication provides a convenient web-based login experience.

2tfcol tfcolTrue / False

Topic: Firewall User Authentication

RADIUS encrypts the entire authentication packet including the username.

True

False

Correct Answer:

Correct Answer: False

Concept: RADIUS encrypts only the password field; the rest of the packet is not encrypted.

106 Cryptography and PKI

1tfcol tfcolTrue / False

Topic: Cryptography and PKI

AES is a symmetric encryption algorithm.

True

False

Correct Answer:

Correct Answer: True

Concept: AES uses the same key for encryption and decryption.

2tfcol tfcolTrue / False

RSA is faster than AES for bulk data encryption.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: RSA is slower and typically used for key exchange or signatures, not bulk encryption.

3tfcol tfcolTrue / False

SHA-256 produces a 256-bit digest.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: SHA-256 generates a fixed 256-bit output.

4tfcol tfcolTrue / False

Hash functions are reversible.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Hash functions are designed to be one-way.

5tfcol tfcolTrue / False

A digital signature provides non-repudiation.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: Digital signatures prove the signer created the message and cannot easily deny it.

6tfcol tfcolTrue / False

A root CA certificate is typically signed by another CA.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Root CA certificates are self-signed and act as trust anchors.

7tfcol tfcolTrue / False

OCSP provides real-time certificate revocation checking.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: OCSP allows clients to query revocation status online.

8tfcol tfcolTrue / False

Topic: Cryptography and PKI

A hash function output can be reversed to find the input.

True

False

Correct Answer:

Correct Answer: False

Concept: Hash functions are one-way.

9tfcol tfcolTrue / False

Topic: Cryptography and PKI

A digital signature verifies integrity and authenticity.

True

False

Correct Answer:

Correct Answer: True

Concept: Signatures prove the message came from the signer and was not altered.

10tfcol tfcolTrue / False

Topic: Cryptography and PKI

RSA is a symmetric encryption algorithm.

True

False

Correct Answer:

Correct Answer: False

Concept: RSA is asymmetric.

11tfcol tfcolTrue / False

Topic: Cryptography and PKI

Symmetric encryption means that uses the same key for encryption and decryption.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Symmetric encryption.

12tfcol tfcolTrue / False

Topic: Cryptography and PKI

Symmetric encryption is unrelated to uses the same key for encryption and decryption.

True

False

Correct Answer:

Correct Answer: False

Concept: Symmetric encryption is directly related to uses the same key for encryption and decryption.

13tfcol tfcolTrue / False

Topic: Cryptography and PKI

Asymmetric encryption means that uses public and private key pairs.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Asymmetric encryption.

14tfcol tfcolTrue / False

Topic: Cryptography and PKI

Asymmetric encryption is unrelated to uses public and private key pairs.

True

False

Correct Answer:

Correct Answer: False

Concept: Asymmetric encryption is directly related to uses public and private key pairs.

15tfcol tfcolTrue / False

Topic: Cryptography and PKI

AES means that is a symmetric encryption standard.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes AES.

16tfcol tfcolTrue / False

Topic: Cryptography and PKI

AES is unrelated to is a symmetric encryption standard.

True

False

Correct Answer:

Correct Answer: False

Concept: AES is directly related to is a symmetric encryption standard.

17tfcol tfcolTrue / False

Topic: Cryptography and PKI

RSA means that is an asymmetric encryption algorithm.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes RSA.

18tfcol tfcolTrue / False

Topic: Cryptography and PKI

RSA is unrelated to is an asymmetric encryption algorithm.

True

False

Correct Answer:

Correct Answer: False

Concept: RSA is directly related to is an asymmetric encryption algorithm.

19tfcol tfcolTrue / False

Topic: Cryptography and PKI

DES means that uses a 56-bit effective key.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes DES.

20tfcol tfcolTrue / False

DES is unrelated to uses a 56-bit effective key.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: DES is directly related to uses a 56-bit effective key.

21tfcol tfcolTrue / False

Hash function means that produces a fixed-length digest.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Hash function.

22tfcol tfcolTrue / False

Hash function is unrelated to produces a fixed-length digest.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Hash function is directly related to produces a fixed-length digest.

23tfcol tfcolTrue / False

MD5 means that is considered broken due to collision attacks.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes MD5.

24tfcol tfcolTrue / False

MD5 is unrelated to is considered broken due to collision attacks.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: MD5 is directly related to is considered broken due to collision attacks.

25tfcol tfcolTrue / False

SHA-256 means that produces a 256-bit digest.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SHA-256.

26tfcol tfcolTrue / False

SHA-256 is unrelated to produces a 256-bit digest.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: SHA-256 is directly related to produces a 256-bit digest.

27tfcol tfcolTrue / False

HMAC means that provides message authentication using a hash and key.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes HMAC.

28tfcol tfcolTrue / False

HMAC is unrelated to provides message authentication using a hash and key.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: HMAC is directly related to provides message authentication using a hash and key.

29tfcol tfcolTrue / False

Digital signature means that verifies authenticity and integrity.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Digital signature.

30tfcol tfcolTrue / False

Digital signature is unrelated to verifies authenticity and integrity.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Digital signature is directly related to verifies authenticity and integrity.

31tfcol tfcolTrue / False

Public key means that is used to verify a digital signature.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Public key.

32tfcot tfcolTrue / False

Public key is unrelated to is used to verify a digital signature.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Public key is directly related to is used to verify a digital signature.

33tfcot tfcolTrue / False

Private key means that is used to create a digital signature.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Private key.

34tfcot tfcolTrue / False

Private key is unrelated to is used to create a digital signature.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Private key is directly related to is used to create a digital signature.

35tfcot tfcolTrue / False

PKI means that manages public keys and certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes PKI.

36tfcot tfcolTrue / False

PKI is unrelated to manages public keys and certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: PKI is directly related to manages public keys and certificates.

37tfcot tfcolTrue / False

CA means that issues and signs digital certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes CA.

38tfcot tfcolTrue / False

CA is unrelated to issues and signs digital certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: CA is directly related to issues and signs digital certificates.

39tfcot tfcolTrue / False

RA means that validates identity before certificate issuance.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes RA.

40tfcot tfcolTrue / False

RA is unrelated to validates identity before certificate issuance.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: RA is directly related to validates identity before certificate issuance.

41tfcot tfcolTrue / False

CRL means that lists revoked certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes CRL.

42tfcot tfcolTrue / False

CRL is unrelated to lists revoked certificates.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: CRL is directly related to lists revoked certificates.

43tfcot tfcolTrue / False

OCSP means that provides online certificate status checking.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes OCSP.

44tfcot tfcolTrue / False

OCSP is unrelated to provides online certificate status checking.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: OCSP is directly related to provides online certificate status checking.

45tfcot tfcolTrue / False

X.509 means that is the standard certificate format.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes X.509.

46tfcot tfcolTrue / False

X.509 is unrelated to is the standard certificate format.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: X.509 is directly related to is the standard certificate format.

47tfcot tfcolTrue / False

Root CA certificate means that is self-signed and acts as a trust anchor.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Root CA certificate.

48tfcot tfcolTrue / False

Root CA certificate is unrelated to is self-signed and acts as a trust anchor.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: Root CA certificate is directly related to is self-signed and acts as a trust anchor.

49tfcot tfcolTrue / False

GCM means that provides authenticated encryption.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes GCM.

50tfcol tfcolTrue / False

GCM is unrelated to provides authenticated encryption.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: GCM is directly related to provides authenticated encryption.

51tfcol tfcolTrue / False

ECB means that should be avoided because it reveals plaintext patterns.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ECB.

52tfcol tfcolTrue / False

ECB is unrelated to should be avoided because it reveals plaintext patterns.

Topic: Cryptography and PKI

True

False

Correct Answer:

Correct Answer: False

Concept: ECB is directly related to should be avoided because it reveals plaintext patterns.

107 Cryptography Basics

1tfcol tfcolTrue / False

Symmetric encryption is generally faster than asymmetric encryption.

Topic: Cryptography Basics

True

False

Correct Answer:

Correct Answer: True

Concept: Symmetric algorithms are faster and suitable for bulk data encryption.

2tfcol tfcolTrue / False

Asymmetric encryption is ideal for encrypting large amounts of data directly.

Topic: Cryptography Basics

True

False

Correct Answer:

Correct Answer: False

Concept: Asymmetric encryption is slower; it is often used for key exchange and signatures, while symmetric encryption handles bulk data.

108 Hash Algorithms

1tfcol tfcolTrue / False

Topic: Hash Algorithms

A hash function can be reversed to recover the original input.

True

False

Correct Answer:

Correct Answer: False

Concept: Hash functions are one-way; the original input cannot be derived from the digest.

109 PKI

1tfcol tfcolTrue / False

Topic: PKI

A digital certificate binds a public key to an identity.

True

False

Correct Answer:

Correct Answer: True

Concept: Certificates contain a public key and identity information, signed by a trusted CA.

2tfcol tfcolTrue / False

Topic: PKI

A CRL lists certificates that are still valid.

True

False

Correct Answer:

Correct Answer: False

Concept: A CRL lists certificates that have been revoked and should no longer be trusted.

110 VPN

1tfcol tfcolTrue / False

Topic: VPN

A VPN allows secure communication over an untrusted public network.

True

False

Correct Answer:

Correct Answer: True

Concept: VPNs create encrypted tunnels over public networks.

2tfcol tfcolTrue / False

Topic: VPN

GRE can tunnel multicast traffic.

True

False

Correct Answer:

Correct Answer: True

Concept: GRE supports multicast tunneling, which is useful for routing protocols.

3tfcol tfcolTrue / False

Topic: VPN

IPsec transport mode encrypts the entire original IP packet.

True

False

Correct Answer:

Correct Answer: False

Concept: Transport mode encrypts only the payload; tunnel mode encrypts the entire packet.

4tfcol tfcolTrue / False

Topic: VPN

IKE is responsible for negotiating IPsec keys and security associations.

True

False

Correct Answer:

Correct Answer: True

Concept: IKE automates key exchange and SA negotiation.

5tfcol tfcolTrue / False

Topic: VPN

L2TP provides encryption by itself.

True

False

Correct Answer:

Correct Answer: False

Concept: L2TP does not encrypt traffic; it is usually combined with IPsec.

6tfcol tfcolTrue / False

Topic: VPN

SSL VPN can be accessed through a standard web browser.

True

False

Correct Answer:

Correct Answer: True

Concept: SSL VPN supports browser-based access via HTTPS.

7tfcol tfcolTrue / False

Topic: VPN

GRE provides encryption by default.

True

False

Correct Answer:

Correct Answer: False

Concept: GRE encapsulates but does not encrypt traffic.

8tfcol tfcolTrue / False

Topic: VPN

SSL VPN commonly uses TCP port 443.

True

False

Correct Answer:

Correct Answer: True

Concept: SSL VPN uses HTTPS/TCP 443.

9tfcol tfcolTrue / False

Topic: VPN

IPsec tunnel mode encrypts the entire original IP packet.

True

False

Correct Answer:

Correct Answer: True

Concept: Tunnel mode encapsulates the whole original packet.

10tfcot tfcolTrue / False

Topic: VPN

VPN means that provides secure communication over public networks.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes VPN.

11tfcot tfcolTrue / False

Topic: VPN

VPN is unrelated to provides secure communication over public networks.

True

False

Correct Answer:

Correct Answer: False

Concept: VPN is directly related to provides secure communication over public networks.

12tfcot tfcolTrue / False

Topic: VPN

Site-to-site VPN means that connects entire networks at different locations.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Site-to-site VPN.

13tfcot tfcolTrue / False

Topic: VPN

Site-to-site VPN is unrelated to connects entire networks at different locations.

True

False

Correct Answer:

Correct Answer: False

Concept: Site-to-site VPN is directly related to connects entire networks at different locations.

14tfcot tfcolTrue / False

Topic: VPN

Remote-access VPN means that connects individual users to a corporate network.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Remote-access VPN.

15tfcot tfcolTrue / False

Topic: VPN

Remote-access VPN is unrelated to connects individual users to a corporate network.

True

False

Correct Answer:

Correct Answer: False

Concept: Remote-access VPN is directly related to connects individual users to a corporate network.

16tfcol tfcolTrue / False

Topic: VPN

GRE means that is IP protocol 47 and lacks built-in encryption.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes GRE.

17tfcol tfcolTrue / False

Topic: VPN

GRE is unrelated to is IP protocol 47 and lacks built-in encryption.

True

False

Correct Answer:

Correct Answer: False

Concept: GRE is directly related to is IP protocol 47 and lacks built-in encryption.

18tfcol tfcolTrue / False

Topic: VPN

GRE means that can tunnel multicast and non-IP protocols.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes GRE.

19tfcol tfcolTrue / False

Topic: VPN

GRE is unrelated to can tunnel multicast and non-IP protocols.

True

False

Correct Answer:

Correct Answer: False

Concept: GRE is directly related to can tunnel multicast and non-IP protocols.

20tfcol tfcolTrue / False

Topic: VPN

IPsec means that provides confidentiality, integrity, and authentication.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IPsec.

21tfcol tfcolTrue / False

Topic: VPN

IPsec is unrelated to provides confidentiality, integrity, and authentication.

True

False

Correct Answer:

Correct Answer: False

Concept: IPsec is directly related to provides confidentiality, integrity, and authentication.

22tfcot tfcolTrue / False

Topic: VPN

ESP means that provides encryption and authentication.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes ESP.

23tfcot tfcolTrue / False

Topic: VPN

ESP is unrelated to provides encryption and authentication.

True

False

Correct Answer:

Correct Answer: False

Concept: ESP is directly related to provides encryption and authentication.

24tfcot tfcolTrue / False

Topic: VPN

AH means that provides authentication without encryption.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes AH.

25tfcot tfcolTrue / False

Topic: VPN

AH is unrelated to provides authentication without encryption.

True

False

Correct Answer:

Correct Answer: False

Concept: AH is directly related to provides authentication without encryption.

26tfcot tfcolTrue / False

Topic: VPN

IKE means that negotiates IPsec keys and security associations.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IKE.

27tfcot tfcolTrue / False

Topic: VPN

IKE is unrelated to negotiates IPsec keys and security associations.

True

False

Correct Answer:

Correct Answer: False

Concept: IKE is directly related to negotiates IPsec keys and security associations.

28tfcol tfcolTrue / False

Topic: VPN

Tunnel mode means that encrypts the entire original IP packet.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Tunnel mode.

29tfcol tfcolTrue / False

Topic: VPN

Tunnel mode is unrelated to encrypts the entire original IP packet.

True

False

Correct Answer:

Correct Answer: False

Concept: Tunnel mode is directly related to encrypts the entire original IP packet.

30tfcol tfcolTrue / False

Topic: VPN

Transport mode means that encrypts only the IP payload.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Transport mode.

31tfcol tfcolTrue / False

Topic: VPN

Transport mode is unrelated to encrypts only the IP payload.

True

False

Correct Answer:

Correct Answer: False

Concept: Transport mode is directly related to encrypts only the IP payload.

32tfcol tfcolTrue / False

Topic: VPN

L2TP means that uses UDP port 1701 and lacks encryption.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes L2TP.

33tfcol tfcolTrue / False

Topic: VPN

L2TP is unrelated to uses UDP port 1701 and lacks encryption.

True

False

Correct Answer:

Correct Answer: False

Concept: L2TP is directly related to uses UDP port 1701 and lacks encryption.

34tfcol tfcolTrue / False

Topic: VPN

L2TP over IPsec means that combines L2TP tunneling with IPsec encryption.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes L2TP over IPsec.

35tfcol tfcolTrue / False

Topic: VPN

L2TP over IPsec is unrelated to combines L2TP tunneling with IPsec encryption.

True

False

Correct Answer:

Correct Answer: False

Concept: L2TP over IPsec is directly related to combines L2TP tunneling with IPsec encryption.

36tfcol tfcolTrue / False

Topic: VPN

SSL VPN means that commonly uses TCP port 443.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes SSL VPN.

37tfcol tfcolTrue / False

Topic: VPN

SSL VPN is unrelated to commonly uses TCP port 443.

True

False

Correct Answer:

Correct Answer: False

Concept: SSL VPN is directly related to commonly uses TCP port 443.

38tfcol tfcolTrue / False

Topic: VPN

Web proxy mode means that provides browser-based access to web apps.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Web proxy mode.

39tfcol tfcolTrue / False

Topic: VPN

Web proxy mode is unrelated to provides browser-based access to web apps.

True

False

Correct Answer:

Correct Answer: False

Concept: Web proxy mode is directly related to provides browser-based access to web apps.

40tfcol tfcolTrue / False

Topic: VPN

Network extension mode means that assigns a virtual IP to the remote host.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes Network extension mode.

41tfcol tfcolTrue / False

Topic: VPN

Network extension mode is unrelated to assigns a virtual IP to the remote host.

True

False

Correct Answer:

Correct Answer: False

Concept: Network extension mode is directly related to assigns a virtual IP to the remote host.

42tfcol tfcolTrue / False

Topic: VPN

IKEv2 means that is faster and more reliable than IKEv1.

True

False

Correct Answer:

Correct Answer: True

Concept: This statement correctly describes IKEv2.

43tfcol tfcolTrue / False

Topic: VPN

IKEv2 is unrelated to is faster and more reliable than IKEv1.

True

False

Correct Answer:

Correct Answer: False

Concept: IKEv2 is directly related to is faster and more reliable than IKEv1.

111 GRE VPN

1tfcol tfcolTrue / False

Topic: GRE VPN

GRE provides strong encryption by default.

True

False

Correct Answer:

Correct Answer: False

Concept: GRE only encapsulates traffic; encryption must be added separately, often with IPsec.

112 IPsec VPN

1tfcot tfcolTrue / False

Topic: IPsec VPN

ESP provides both encryption and authentication.

True

False

Correct Answer:

Correct Answer: True

Concept: ESP can encrypt payloads and authenticate packets.

2tfcot tfcolTrue / False

Topic: IPsec VPN

AH provides payload encryption.

True

False

Correct Answer:

Correct Answer: False

Concept: AH provides authentication and integrity but does not encrypt the payload.

113 L2TP VPN

1tfcot tfcolTrue / False

Topic: L2TP VPN

L2TP alone does not provide encryption.

True

False

Correct Answer:

Correct Answer: True

Concept: L2TP provides tunneling but relies on IPsec or other mechanisms for encryption.

114 SSL VPN

1tfcot tfcolTrue / False

Topic: SSL VPN

SSL VPN typically uses TCP port 443.

True

False

Correct Answer:

Correct Answer: True

Concept: SSL/TLS VPNs commonly use HTTPS/TCP 443, which is usually allowed through firewalls.